

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T08:04:40

Source: data/json/noise_experiments/v2/noise8i_fixed_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_fixed/noise8i_fixed_memprimary_game_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_fixed
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

ROUND 1

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$2.6; payoffs Agent_1=\$4.6, Agent_2=\$6.4;
noise/visible: sent visible=\$2.56, received visible=\$7.68, returned visible=\$3.34
Action validation: {"returned": {"amount": 2.6, "clamped": false, "max": 9.0, "min": 0, "raw": 2.6}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$5; payoffs Agent_3=\$7, Agent_4=\$4;
noise/visible: sent visible=\$3.39, received visible=\$10.17, returned visible=\$5.75
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$3.5; payoffs Agent_5=\$5.5, Agent_6=\$5.5;
noise/visible: sent visible=\$2.08, received visible=\$6.24, returned visible=\$3.88
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$5.5; payoffs Agent_7=\$7.5, Agent_8=\$3.5;
noise/visible: sent visible=\$3.99, received visible=\$11.97, returned visible=\$5.61
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 4.6,
"Agent_2": 6.4,
"Agent_3": 7.0,
"Agent_4": 4.0,
"Agent_5": 5.5,
"Agent_6": 5.5,
"Agent_7": 7.5,
"Agent_8": 3.5

}

Visible balances after round: {

"Agent_1": 5.34,
"Agent_2": 5.08,
"Agent_3": 7.75,
"Agent_4": 5.17,
"Agent_5": 5.88,
"Agent_6": 2.74,
"Agent_7": 7.61,
"Agent_8": 6.470000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:51:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
"agents": [
"Agent_1",
"Agent_2"
],
"investor": "Agent_1",
"roles": {
"Agent_1": "investor",
"Agent_2": "trustee"
},
"trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

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- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

```

{"send": 3}

```

Reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 472,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:51:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
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- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Assistant output:

Looking at this situation:

****Analysis:****

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: $\$5 - \$2.56 = \$2.44$
- I received: \$7.68
- Total money in play: $\$2.44 + \$7.68 = \$10.12$

****My decision:****

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: $\$2.44 + \$2.60 = \$5.04$
- I would have: $\$7.68 - \$2.60 = \$5.08$

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

Reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.

I received $\$2.56 \times 3 = \7.68 .

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = $\$2.44$ (sender has) + $\$7.68$ (I have) = $\$10.12$. Equal split = $\$5.06$ each. I'd need to return $\$7.68 - \$5.06 = \$2.62$ to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with $\$2.44 + \text{return}$, I end up with $\$7.68 - \text{return}$. For equal outcomes: $\$2.44 + \text{return} = \$7.68 - \text{return}$, so $2 \times \text{return} = \5.24 , return = $\$2.62$.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with $\$2.44 + \$2.56 = \$5$ (their original endowment), and me with $\$7.68 - \$2.56 = \$5.12$.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the $\$2.50$ - $\$3.00$ range feels appropriate—it rewards their cooperation while keeping outcomes reasonably balanced. A return of $\$2.60$ would give us each roughly $\$5.04$ - $\$5.08$...

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1015,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:51:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.

Message 2 role=user
content:
  Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Assistant output:
  Looking at this first round, I want to establish some level of cooperation while being mindful
  of the risk.

  Since:
  - This is a multi-round game where building trust could benefit future interactions
  - The multiplication factor means even moderate amounts create significant value
  - There's communication noise to consider
  - I have no information about my current co-player yet

  I'll send a moderate amount that shows willingness to cooperate while not being overly exposed
  .

  **Decision:**
  '''json
  {"send": 3}
  '''

```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

Reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3×sent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:52:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    }
  },
```

```

    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 (\$5 - \$3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

```
{'return': 5.00}
```

Reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have $\$5 - \$3.39 = \$1.61$ left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: $\$10.17 - \$3.39 = \$6.78$
 - If we split this gain equally: $\$6.78 / 2 = \3.39 each
 - Sender should end with: \$1.61 (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with $\$1.61 + \$3.39 = \$5.00$
 - I would keep: $\$10.17 - \$3.39 = \$6.78$

Actually, let me reconsider. A common approach is:

- Equal split of total money: $(\$5 \text{ initial} + \$10.17 \text{ received}) / 2 = \7.585 each
- Sender currently has: \$1.61
- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1053,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:52:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",
```

```

"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both

benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

Reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

Usage: {

```
"input_tokens": 500,  
"output_tokens": 707,  
"reasoning_tokens": 0
```

}

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:52:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_3",  
"move_index": 5,  
"opponent_id": "Agent_5",  
"pairing": {  
  "agents": [  
    "Agent_5",  
    "Agent_6"  
  ],  
  "investor": "Agent_5",  
  "roles": {  
    "Agent_5": "investor",  
    "Agent_6": "trustee"  
  }  
},
```

```

    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Assistant output:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of $\$1.42$)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of $\$2.74$)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

```
{'return': 3.50}
```

Reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent $\$2.08$ out of their $\$5$ endowment (which is 41.6% as stated).
I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending $\$2.08$ (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half ($\$3.12$):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains $\$1.04$, I gain $\$3.12$ (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at $\$5$: need to return $\$2.08$
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from $\$5$ to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: $\$6.24$ (the tripled amount minus the original $\$2.08$ that was just transferred)
- Wait, let me recalculate...
- Sender now has $\$2.92$, I have $\$6.24$, so the system total is $\$9.16$. The multiplication created an extra $\$4.16$ in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return $\$2.08$ to the sender, bringing them back to $\$5$ while I keep $\$4.16$. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return $\$3$, the sender profits $\$0.92$ while I keep $\$3.24$. If I return $\$4$, they profit $\$1.92$ and I keep $\$2.24$. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates $\$4.16$ in total benefit. Splitting that equally would mean each of us gains about $\$2.08$ beyond our starting amounts. That would put the sender at $\$5.00$ and me at around $\$4.16$. A $\$3.50$ return gets closer to that fair split-they'd end up with $\$6.42$ and I'd have $\$2.74$, which feels more genuinely reciprocal for establishing trust in future rounds....

Usage: {

```
"input_tokens": 529,
"output_tokens": 1643,
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:53:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send /return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3x3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:53:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
```

```

    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: \$1.01 (sender kept) + \$11.97 (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

- dyad_1: Agent_2=Sender, Agent_1=Receiver
- dyad_2: Agent_4=Sender, Agent_3=Receiver

```

dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
dyad_1: Agent_2 sent $3; received $9; Agent_1 returned $3; payoffs Agent_2=$5, Agent_1=$6;
noise/visible: sent visible=$2.46, received visible=$7.38, returned visible=$2.72
  Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 sent $3.5; received $10.5; Agent_3 returned $4.5; payoffs Agent_4=$6, Agent_3=$6;
noise/visible: sent visible=$2.86, received visible=$8.58, returned visible=$5.25
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 10.5, "min": 0, "raw": 4.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $3.5; payoffs Agent_6=$5.5, Agent_5=$5.5;
noise/visible: sent visible=$2.16, received visible=$6.48, returned visible=$3.47
  Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3.5; received $10.5; Agent_7 returned $5.5; payoffs Agent_8=$7, Agent_7=$5;
noise/visible: sent visible=$3.96, received visible=$11.88, returned visible=$5.4
  Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 10.5, "min": 0, "raw": 5.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 10.6,
  "Agent_2": 11.4,
  "Agent_3": 13.0,
  "Agent_4": 10.0,
  "Agent_5": 11.0,
  "Agent_6": 11.0,
  "Agent_7": 12.5,
  "Agent_8": 10.5
}
Visible balances after round: {
  "Agent_1": 9.719999999999999,
  "Agent_2": 9.8,
  "Agent_3": 11.83,
  "Agent_4": 11.92,
  "Agent_5": 8.86,
  "Agent_6": 8.21,
  "Agent_7": 13.989999999999998,
  "Agent_8": 13.370000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-12T07:53:43
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: \$5 - \$2.56 = \$2.44
- I received: \$7.68
- Total money in play: \$2.44 + \$7.68 = \$10.12

****My decision:****

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: \$2.44 + \$2.60 = \$5.04
- I would have: \$7.68 - \$2.60 = \$5.08

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received $\$2.56 \times 3 = \7.68 .

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = $\$2.44$ (sender has) + $\$7.68$ (I have) = $\$10.12$. Equal split = $\$5.06$ each. I'd need to return $\$7.68 - \$5.06 = \$2.62$ to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with $\$2.44 + \text{return}$, I end up with $\$7.68 - \text{return}$. For equal outcomes: $\$2.44 + \text{return} = \$7.68 - \text{return}$, so $2 \times \text{return} = \5.24 , return = $\$2.62$.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with $\$2.44 + \$2.56 = \$5$ (their original endowment), and me with $\$7.68 - \$2.56 = \$5.12$.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the \$2.50-\$3.00 range feels appropriate-it rewards their cooperation while keeping outcomes reasonably balanced. A return of \$2.60 would give us each roughly \$5.04-\$5.08....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust

- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

```
Usage: {
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:54:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

{"send": 3}

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I still keep \$4.38, which is more than my initial endowment would be.

```
{"return": 3.0}
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is 2.56×3)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $\$2 + \$2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6
- They kept: $\$7.68 - \$2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$\$2 + X = \$5.34$$

$$X = \$3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is 2.46×3)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51. Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which tripled to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

```
Usage: {  
  "input_tokens": 762,  
  "output_tokens": 1244,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:54:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 ($\$5 - \3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

```
{'return': 5.00}
```

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have $\$5 - \$3.39 = \$1.61$ left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: $\$10.17 - \$3.39 = \$6.78$
 - If we split this gain equally: $\$6.78 / 2 = \3.39 each
 - Sender should end with: $\$1.61$ (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with $\$1.61 + \$3.39 = \$5.00$
 - I would keep: $\$10.17 - \$3.39 = \$6.78$

Actually, let me reconsider. A common approach is:

- Equal split of total money: $(\$5 \text{ initial} + \$10.17 \text{ received}) / 2 = \7.585 each
- Sender currently has: \$1.61
- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

```
{'send': 3.50}
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

```
Usage: {
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:54:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being mindful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

****Decision:****

```
'''json
{"send": 3}
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3xsent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

```
...
usage:
{
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

Reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

```
Usage: {  
  "input_tokens": 786,  
  "output_tokens": 718,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:55:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

`{'return': 3.50}`

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated).
I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value (\$6.24 - \$2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that equally would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split-they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1643,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

Reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned

- So their total would be $\$2.0 + \$3.88 = \$5.88$

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions . They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship....

```
Usage: {  
  "input_tokens": 867,  
  "output_tokens": 1048,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:55:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value

2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 707,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative r

relationship.

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.
4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

```
Usage: {  
  "input_tokens": 789,  
  "output_tokens": 622,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:55:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      NOTE: There may be communication noise in this game.
      The amounts you see may differ slightly from what was actually sent/returned.
      This is part of the experimental design.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - Agent names are hidden in this run.
      - Your paired opponent is referred to as your current co-player.
  Message 2 role=user
    content:
      Round 1: You are the RECEIVER. the sender sent $3.99, that is 79.8% of their total endowme
      nt. You received $11.97. How much do you return to the sender? (0-11.97)
  Message 3 role=assistant
    content:
      Looking at this situation:

      - The sender showed significant trust by sending $3.99 (nearly 80% of their endowment)
      - I received $11.97 after the 3x multiplier
      - The sender currently has only $1.01 left

      I want to reciprocate this trust while ensuring both of us benefit from cooperating.

      If I return $5.50:
      - The sender will end with: $1.01 + $5.50 = $6.51 (better than their initial $5)
      - I will end with: $11.97 - $5.50 = $6.47 (also a good outcome)

      This creates a roughly equal split of the total gains and rewards the sender's generous tr
      ust.

      '''json

```

```
    {"return": 5.50}
    """
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: \$1.01 (sender kept) + \$11.97 (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with $2.0 + \$5.61 = \7.61

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

```
Usage: {
  "input_tokens": 851,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T07:55:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
  Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
  Looking at this first round, I want to signal cooperation while being somewhat cautious si
    nce I don't yet know how other agents will behave.

  Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I sen
    d $3, they'll receive $9 (tripled). Even if they return just $3, we'd both end up with m
    ore than we started with ($5 for me, $6 for them).

  This shows willingness to trust while limiting my risk in this initial round.

  '''json
  {"send": 3}
  '''
reasoning:
  This is the first round of a multi-round trust game. I'm the sender, which means I need to
    decide how much to send to the receiver.

```

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3x3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97

- They returned \$5.5
- My earnings: $\$5 - \$3 + \$5.5 = \7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with $\$1 + \$5 = \$6$, and I keep \$6.88
- If I return \$6, they end with $\$1 + \$6 = \$7$, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

```
Usage: {
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
 dyad_2: Agent_3=Sender, Agent_4=Receiver
 dyad_3: Agent_5=Sender, Agent_6=Receiver
 dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$4.5; payoffs Agent_1=\$6.5, Agent_2=\$4.5; noise/visible: sent visible=\$3.54, received visible=\$10.62, returned visible=\$4.9
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_3 sent \$3.5; received \$10.5; Agent_4 returned \$7; payoffs Agent_3=\$8.5, Agent_4=\$3.5; noise/visible: sent visible=\$4.4, received visible=\$13.2, returned visible=\$7.3
 Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw": 7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
 dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.5; payoffs Agent_5=\$6.5, Agent_6=\$4.5; noise/visible: sent visible=\$2.83, received visible=\$8.49, returned visible=\$4.56
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_7 sent \$3.5; received \$10.5; Agent_8 returned \$5; payoffs Agent_7=\$6.5, Agent_8=\$5.5; noise/visible: sent visible=\$3.45, received visible=\$10.35, returned visible=\$5.95
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

```
"Agent_1": 17.1,
"Agent_2": 15.9,
"Agent_3": 21.5,
"Agent_4": 13.5,
"Agent_5": 17.5,
"Agent_6": 15.5,
"Agent_7": 19.0,
"Agent_8": 16.0
```

}

```
Visible balances after round: {
  "Agent_1": 16.619999999999997,
  "Agent_2": 15.920000000000002,
  "Agent_3": 20.630000000000003,
  "Agent_4": 18.12,
  "Agent_5": 15.419999999999998,
  "Agent_6": 12.200000000000001,
  "Agent_7": 21.439999999999998,
  "Agent_8": 18.720000000000002
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:55:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

```
{"send": 3}
```

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{  
  "input_tokens": 500,  
  "output_tokens": 472,  
}
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 2

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as RECEIVER: saw $2.56 sent, it became $7.68, returned $2.6, visible payoff $5.08.
  Your total visible earnings across all rounds are $5.34.

  This round, you are the RECEIVER against your current co-player. They sent you $2.46, so you received $7.38. How much do you return? (0-7.38)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  Looking at the situation:

  **History Analysis:**
  - In Round 1, my co-player saw $2.56 sent (likely noise from my $3), received $7.68
  - They returned $2.6, which is about 34% of what they received
  - This shows they're willing to reciprocate but somewhat cautiously

  **Current Round:**
  - They sent me $2.46 (kept $2.54)
  - I received $7.38
  - If I return nothing, I keep $7.38, they end with $2.54
  - If I return fairly, both should benefit from the multiplication

  **My Decision:**
  I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning $3.00 gives them back their sent amount plus a nice profit, and I still keep $4.38, which is more than my initial endowment would be.

  {"return": 3.0}
reasoning:
  Let me analyze this situation:

  1. In Round 1, I sent $3 (or at least I thought I did)
  2. But my co-player saw $2.56 sent (communication noise!)
  3. They received $7.68 (which is  $2.56 \times 3$ )
  4. They returned $2.6
  5. My visible earnings are $5.34

  Let me verify: If I sent $3, I kept $2. If they returned $2.6, I should have  $2 + 2.6 = 4.6$ . But it shows $5.34. This is confusing - perhaps the noise affected what I saw returned as well?

  Actually, looking more carefully at the visible history, it says "visible payoff $5.08" for them. Let me recalculate:
  - They received $7.68
  - They returned $2.6
  - They kept:  $7.68 - 2.6 = 5.08$  ?

  My earnings of $5.34 could be:
  - I started with $5
  - I sent some amount (which appeared as $2.56 to them)
  - They returned something (which appeared as some amount to me)
  - $5.34 total

  If I sent $3 and kept $2, and received back some amount X:
   $2 + X = 5.34$ 
   $X = 3.34$ 

  So I received $3.34 back (or that's what I saw).

  But they returned $2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

```

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is $\$2.46 \times 3$)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51.

Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which trip led to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.720000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

```
{"send": 3}
```

Reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

```
Usage: {  
  "input_tokens": 1159,  
  "output_tokens": 665,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:56:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

Analysis:

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: \$5 - \$2.56 = \$2.44
- I received: \$7.68
- Total money in play: \$2.44 + \$7.68 = \$10.12

My decision:

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: \$2.44 + \$2.60 = \$5.04
- I would have: \$7.68 - \$2.60 = \$5.08

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received $2.56 \times 3 = 7.68$.

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has \$5 - \$2.56 = \$2.44
- Will have \$2.44 + (what I return)

I have:

- Received \$7.68
- Will keep \$7.68 - (what I return)

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = \$2.44 (sender has) + \$7.68 (I have) = \$10.12. Equal split = \$5.06 each. I'd need to return \$7.68 - \$5.06 = \$2.62 to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with \$2.44 + return, I end up with \$7.68 - return. For equal outcomes: \$2.44 + return = \$7.68 - return, so 2×return = \$5.24, return = \$2.62.
4. Return the multiplied amount to reward trust: Return \$2.56 × 3 = \$7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with \$2.44 + \$2.56 = \$5 (their original endowment), and me with \$7.68 - \$2.56 = \$5.12.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the \$2.50-\$3.00 range feels appropriate—it rewards their cooperation while keeping outcomes reasonably balanced. A return of \$2.60 would give us each roughly \$5.04-\$5.08....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust
- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60

- My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)...

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

```

**Current situation:**
- They sent $3.54, keeping only $1.46
- I received $10.62
- Total in play: $1.46 + $10.62 = $12.08

**My decision:**
This player has shown strong and increasing trust. I should reciprocate generously.

If I return $4.50:
- Sender gets: $1.46 + $4.50 = $5.96
- I get: $10.62 - $4.50 = $6.12

This rewards their high trust while keeping outcomes relatively balanced.

'''json
{"return": 4.50}
'''
Reasoning:
Let me analyze this situation:

**History with this co-player:**
- Round 1: They sent $3.0 (60%), I returned $2.60 (but they saw $3.34), they got $5.34
- Round 2: I sent $3.00 (60%), they saw $2.46 sent (noise!), it became $7.38, they returned $3
.0, they got $4.38

**Current Round 3:**
- They sent $3.54 (70.8% of their endowment) - showing high trust
- I received $10.62
- They currently have: $5 - $3.54 = $1.46

**Analysis:**
The co-player has been very cooperative:
- Round 1: Sent 60%
- Round 2: Returned $3.0 out of $7.38 (40.7%) even though they only saw $2.46 sent
- Round 3: Now sent $3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around $4.50:
- Sender gets: $1.46 + $4.50 = $5.96
- I get: $10.62 - $4.50 = $6.12

This seems fair - we both do better than our initial endowments, and I'm rewarding their high
trust.

Alternatively, if I return $4.20:
- Sender gets: $1.46 + $4.20 = $5.66
- I get: $10.62 - $4.20 = $6.42

Or if I return $4.80:
- Sender gets: $1.46 + $4.80 = $6.26
- I get: $10.62 - $4.80 = $5.82
...
Usage: {
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}

GAME - Agent_3 - Sender - dyad_2
Timestamp: 2026-08-12T07:56:14
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {

```

```

"agents": [
  "Agent_3",
  "Agent_4"
],
"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_4": "trustee"
},
"trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being min

dful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

```
**Decision:**  
'''json  
{"send": 3}  
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3×sent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

usage:

```
{  
  "input_tokens": 500,  
  "output_tokens": 629,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

usage:

```
{
  "input_tokens": 786,
```

```
"output_tokens": 718,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)

****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

Usage: {

```
"input_tokens": 1140,
"output_tokens": 393,
"reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:56:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 (\$5 - \$3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

{'return': 5.00}

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have $\$5 - \$3.39 = \$1.61$ left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: $\$10.17 - \$3.39 = \$6.78$
 - If we split this gain equally: $\$6.78 / 2 = \3.39 each
 - Sender should end with: \$1.61 (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with $\$1.61 + \$3.39 = \$5.00$
 - I would keep: $\$10.17 - \$3.39 = \$6.78$

Actually, let me reconsider. A common approach is:

- Equal split of total money: $(\$5 \text{ initial} + \$10.17 \text{ received}) / 2 = \7.585 each
- Sender currently has: \$1.61
- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

{'send': 3.50}

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

Reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: \$5 - \$4.4 = \$0.60

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: \$0.60 + \$7.0 = \$7.60 (profit of \$2.60)
- I keep: \$13.20 - \$7.0 = \$6.20

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: \$0.60 + \$6.5 = \$7.10 (profit of \$2.10)
- I keep: \$13.20 - \$6.5 = \$6.70

Or \$7.5:

- Sender ends with: \$0.60 + \$7.5 = \$8.10 (profit of \$3.10)

...

Usage: {

"input_tokens": 1103,
"output_tokens": 720,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:56:36
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

usage:

```
{
```

```
"input_tokens": 500,  
"output_tokens": 707,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json  
{"return": 3.5}  
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.
4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

```
...
usage:
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern.

I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

Reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:

- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:

- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

```
Usage: {
  "input_tokens": 1081,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

```
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:56:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

 "agent_type": "standard",

 "dyad_id": "dyad_3",

 "move_index": 5,

 "opponent_id": "Agent_5",

 "pairing": {

 "agents": [

 "Agent_5",

 "Agent_6"

],

 "investor": "Agent_5",

 "roles": {

 "Agent_5": "investor",

 "Agent_6": "trustee"

 },

 "trustee": "Agent_6"

 },

 "role": "trustee",

 "role_label": "Receiver",

 "round": 3,

 "task": "game",

 "task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

{'return': 3.50}

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated).
I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about w

that return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that equally would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split-they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1643,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be \$2.0 + \$3.88 = \$5.88

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

```
{'return': 4.50}
```

Reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have \$5 - \$2.83 = \$2.17.

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)

...

```
Usage: {
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:56:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky

- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3×3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings: \$5 - \$3 + \$5.5 = \$7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%).

This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with $\$1 + \$5 = \$6$, and I keep \$6.88
- If I return \$6, they end with $\$1 + \$6 = \$7$, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with $\$5 - \$3 + \$5.5 = \7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with $\$5 - \$3.5 + \$5.4$ (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

Usage: {

```
"input_tokens": 1079,
"output_tokens": 370,
"reasoning_tokens": 0
```

```
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:57:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_4",

"move_index": 7,

"opponent_id": "Agent_7",

"pairing": {

"agents": [

"Agent_7",

"Agent_8"

],

"investor": "Agent_7",

"roles": {

"Agent_7": "investor",

"Agent_8": "trustee"

},

"trustee": "Agent_8"

},

"role": "trustee",

"role_label": "Receiver",

"round": 3,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: $\$1.01$ (sender kept) + $\$11.97$ (I received) = \$12.98

- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with \$2.0 + \$5.61 = \$7.61

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
```

```
"input_tokens": 851,
"output_tokens": 436,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

```
Usage: {
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.5; received \$10.5; Agent_1 returned \$5; payoffs Agent_2=\$6.5, Agent_1=\$5.5; noise/visible: sent visible=\$3.82, received visible=\$11.46, returned visible=\$5.74
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_2: Agent_4 sent \$4; received \$12; Agent_3 returned \$5.5; payoffs Agent_4=\$6.5, Agent_3=\$6.5; noise/visible: sent visible=\$3.55, received visible=\$10.65, returned visible=\$6.46
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$4.5; payoffs Agent_6=\$6.5, Agent_5=\$4.5; noise/visible: sent visible=\$2.88, received visible=\$8.64, returned visible=\$4.11
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent \$3.75; received \$11.25; Agent_7 returned \$5.5; payoffs Agent_8=\$6.75, Agent_7=\$5.75; noise/visible: sent visible=\$3.53, received visible=\$10.59, returned visible=\$5.44
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 11.25, "min": 0, "raw": 5.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 22.6,
"Agent_2": 22.4,
"Agent_3": 28.0,
"Agent_4": 20.0,
"Agent_5": 22.0,
"Agent_6": 22.0,
"Agent_7": 24.75,
"Agent_8": 22.75

}

Visible balances after round: {

"Agent_1": 23.08,
"Agent_2": 23.160000000000004,
"Agent_3": 25.78,
"Agent_4": 25.580000000000002,
"Agent_5": 19.56,
"Agent_6": 18.310000000000002,
"Agent_7": 26.529999999999998,
"Agent_8": 25.410000000000004

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:57:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: $\$5 - \$2.56 = \$2.44$
- I received: \$7.68
- Total money in play: $\$2.44 + \$7.68 = \$10.12$

****My decision:****

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: $\$2.44 + \$2.60 = \$5.04$
- I would have: $\$7.68 - \$2.60 = \$5.08$

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received $\$2.56 \times 3 = \7.68 .

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = $\$2.44$ (sender has) + $\$7.68$ (I have) = $\$10.12$. Equal split = $\$5.06$ each. I'd need to return $\$7.68 - \$5.06 = \$2.62$ to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with $\$2.44 + \text{return}$, I end up with $\$7.68 - \text{return}$. For equal outcomes: $\$2.44 + \text{return} = \$7.68 - \text{return}$, so $2 \times \text{return} = \5.24 , $\text{return} = \$2.62$.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with $\$2.44 + \$2.56 = \$5$ (their original endowment), and me with $\$7.68 - \$2.56 = \$5.12$.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the \$2.50-\$3.00 range feels appropriate-it rewards their cooperation while keeping outcomes reasonably balanced. A return of \$2.60 would give us each roughly \$5.04-\$5.08....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust
- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

****Current situation:****

- They sent \$3.54, keeping only \$1.46
- I received \$10.62
- Total in play: \$1.46 + \$10.62 = \$12.08

****My decision:****

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:

- Sender gets: \$1.46 + \$4.50 = \$5.96
- I get: \$10.62 - \$4.50 = \$6.12

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

****History with this co-player:****

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

****Current Round 3:****

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have: \$5 - \$3.54 = \$1.46

****Analysis:****

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:

- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:

- Sender gets: $\$1.46 + \$4.20 = \$5.66$
- I get: $\$10.62 - \$4.20 = \$6.42$

Or if I return \$4.80:

- Sender gets: $\$1.46 + \$4.80 = \$6.26$
- I get: $\$10.62 - \$4.80 = \$5.82$

...

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness. I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

```
Usage: {  
  "input_tokens": 1671,  
  "output_tokens": 729,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:57:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

```
{"send": 3}
```

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I still keep \$4.38, which is more than my initial endowment would be.

```
{"return": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is 2.56×3)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $2 + 2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6

- They kept: $\$7.68 - \$2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$\$2 + X = \$5.34$$

$$X = \$3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is $\$2.46 \times 3$)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51. Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which trip led to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.720000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

```
{"send": 3}
```

reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.1200000000000001.

Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

```

**My Decision:**
I should reciprocate their increased generosity and trust. Returning $5.00 gives them a total
of $6.18 (profit of $1.18), while I keep $6.46. This maintains the cooperative relationship
and rewards their trust.

{"return": 5.0}
Reasoning:
Let me analyze the history:

**Round 1:** Co-player as RECEIVER
- Saw $2.56 sent (noise from my $3), became $7.68
- Returned $2.6 (34% of received)
- Their visible payoff: $5.08

**Round 2:** Co-player as SENDER
- Sent $3.0, became $9.0
- Saw $2.72 returned (noise from my $3.0)
- Their visible payoff: $4.72

**Round 3:** Co-player as RECEIVER
- Saw $3.54 sent (noise from my $3), became $10.62
- Returned $4.5 (42% of received)
- Their visible payoff: $6.12

**Current Round 4:** Co-player as SENDER
- Sent $3.82 (keeping $1.18)
- I received $11.46

**Analysis:**
1. My co-player has been increasingly generous:
  - Round 1: returned 34% ($2.6 from $7.68)
  - Round 3: returned 42% ($4.5 from $10.62)

2. They also increased their sending:
  - Round 2: sent $3.0
  - Round 4: sent $3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

**My Strategy:**
I should reciprocate generously to maintain this positive relationship. They sent $3.82, so re
turning around $5-5.5 would give them a good profit while still leaving me with a substantia
l gain.

Calculation:
- If I return $5.0:
  - They get: $5 - $3.82 + $5.0 = $6.18 total
  ...
Usage: {
  "input_tokens": 1585,
  "output_tokens": 777,
  "reasoning_tokens": 0
}

```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:57:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",

```

```

    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 (\$5 - \$3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

```
{'return': 5.00}
```

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have $\$5 - \$3.39 = \$1.61$ left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: $\$10.17 - \$3.39 = \$6.78$
 - If we split this gain equally: $\$6.78 / 2 = \3.39 each
 - Sender should end with: $\$1.61$ (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with $\$1.61 + \$3.39 = \$5.00$
 - I would keep: $\$10.17 - \$3.39 = \$6.78$

Actually, let me reconsider. A common approach is:

- Equal split of total money: $(\$5 \text{ initial} + \$10.17 \text{ received}) / 2 = \7.585 each
- Sender currently has: \$1.61
- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: $\$5 - \$4.4 = \$0.60$

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: $\$0.60 + \$7.0 = \$7.60$ (profit of \$2.60)
- I keep: $\$13.20 - \$7.0 = \$6.20$

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible p

payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

{'send': 4.0}

Reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

```
Usage: {
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:57:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    }
  }
}
```

```

    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being mindful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

```
**Decision:**
'''json
{"send": 3}
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3×sent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so y

ou received \$8.58. How much do you return? (0-8.58)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

usage:

```
{
  "input_tokens": 786,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)

****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned

\$7.0, visible payoff \$6.200000000000001.
Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)
****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)
****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

Reasoning:

Let me analyze the pattern:

****Round 1:****
- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****
- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****
- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****
- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

```
Usage: {
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:58:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
```



```

"move_index": 4,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment

nt. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

`{'return': 3.50}`

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated).
I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that equally would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split—they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1643,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be \$2.0 + \$3.88 = \$5.88

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Se

ending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

{'return': 4.50}

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have $\$5 - \$2.83 = \$2.17$

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)

...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

Reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative

e relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

```
Usage: {  
  "input_tokens": 1611,  
  "output_tokens": 357,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:58:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomized), each round is more like a one-shot game. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 707,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.

2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:

- I started with \$5
- I sent \$3 (actually sent)
- I kept \$2
- I received back \$3.5
- Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.

4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.

5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.

6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.

5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

usage:

```
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.4700000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:

- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:

- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1081,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.4700000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64). I'll match their generous pattern:

```
'''json
{"return": 4.5}
'''
```

This maintains our strong cooperative relationship.

Reasoning:

Looking at the history with this co-player:

- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5
- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

Usage: {

```
"input_tokens": 1435,
"output_tokens": 398,
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:58:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: $\$1.01$ (sender kept) + $\$11.97$ (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return

\$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with $\$2.0 + \$5.61 = \$7.61$

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 851,
```

```
"output_tokens": 436,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

Usage: {

```
"input_tokens": 1594,
"output_tokens": 425,
"reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T07:58:29
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3x3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings: $\$5 - \$3 + \$5.5 = \7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with $\$1 + \$5 = \$6$, and I keep \$6.88
- If I return \$6, they end with $\$1 + \$6 = \$7$, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

usage:

```
{
  "input_tokens": 1079,
  "output_tokens": 370,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.

Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

'''

Reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)
Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)
Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)
Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

```
Usage: {  
  "input_tokens": 1444,  
  "output_tokens": 410,  
  "reasoning_tokens": 0  
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$5.5; payoffs Agent_1=\$6.5, Agent_2=\$6.5;
noise/visible: sent visible=\$3.9, received visible=\$11.7, returned visible=\$4.93
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent \$4; received \$12; Agent_4 returned \$6; payoffs Agent_3=\$7, Agent_4=\$6;
noise/visible: sent visible=\$3.76, received visible=\$11.28, returned visible=\$6.52
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$5; payoffs Agent_5=\$7, Agent_6=\$4;
noise/visible: sent visible=\$3.23, received visible=\$9.69, returned visible=\$5.62
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3.75; received \$11.25; Agent_8 returned \$5.25; payoffs Agent_7=\$6.5, Agent_8=\$6; noise/visible: sent visible=\$3.41, received visible=\$10.23, returned visible=\$5.21
Action validation: {"returned": {"amount": 5.25, "clamped": false, "max": 11.25, "min": 0, "raw": 5.25}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 29.1,
"Agent_2": 28.9,
"Agent_3": 35.0,
"Agent_4": 26.0,
"Agent_5": 29.0,
"Agent_6": 26.0,
"Agent_7": 31.25,
"Agent_8": 28.75

}

Visible balances after round: {

"Agent_1": 29.009999999999998,
"Agent_2": 29.360000000000003,
"Agent_3": 33.3,
"Agent_4": 30.86,

```
"Agent_5": 27.18,  
"Agent_6": 23.0,  
"Agent_7": 32.989999999999995,  
"Agent_8": 30.390000000000004  
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:58:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 5,

"task": "game",

"task_index": 0
}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

{"send": 3}

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I still keep \$4.38, which is more than my initial endowment would be.

{"return": 3.0}

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is 2.56×3)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $2 + 2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6
- They kept: $7.68 - 2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$2 + X = 5.34$$

$$X = 3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is 2.46×3)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51.

Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which trip led to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

```
{"send": 3}
```

reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)

- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.6200000000000001, returned \$4.5, visible payoff \$6.1200000000000001.

Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

****My Decision:****

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

```
{"return": 5.0}
```

reasoning:

Let me analyze the history:

****Round 1:**** Co-player as RECEIVER

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

****Round 2:**** Co-player as SENDER

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff: \$4.72

****Round 3:**** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

****Current Round 4:**** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

****Analysis:****

1. My co-player has been increasingly generous:
 - Round 1: returned 34% (\$2.6 from \$7.68)
 - Round 3: returned 42% (\$4.5 from \$10.62)
2. They also increased their sending:
 - Round 2: sent \$3.0
 - Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)
3. This shows they're willing to cooperate and trust me more over time.

****My Strategy:****

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:
- They get: $\$5 - \$3.82 + \$5.0 = \6.18 total

...

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.1200000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.

Your total visible earnings across all rounds are \$23.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned
- Round 3: Co-player returned \$4.5 from \$10.62 (strong reciprocity)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (even better returns from their perspective)

****Current Situation:****

The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from \$3.0 to \$3.5 after seeing better returns. From their perspective, cooperation is paying off.

****My Decision:****

I'll increase my send to \$4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

```
{"send": 4}
```

Reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned (I returned \$3.0 but noise made it \$2.72)
- Round 3: Co-player received visible \$10.62, returned \$4.5 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (I returned \$5.0 but noise made it \$5.74)

****Key Observations:****

1. Co-player has been consistently trusting, sending \$3.0, then \$3.5
2. From their perspective, I returned better in Round 4 (\$5.74 on their \$3.5 sent) than Round 2 (\$2.72 on their \$3.0 sent)
3. They're showing increasing trust
4. The noise has actually made my returns LOOK better in Round 4

****My Strategy:****

Given that:

- They've been consistently cooperative
- The trend is positive cooperation
- They sent \$3.5 last round and got good returns (from their perspective)
- I should continue showing trust

I'll send \$4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.

Usage: {

```
"input_tokens": 2011,
"output_tokens": 493,
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:58:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
```

```
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: \$5 - \$2.56 = \$2.44
- I received: \$7.68
- Total money in play: \$2.44 + \$7.68 = \$10.12

****My decision:****

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: \$2.44 + \$2.60 = \$5.04
- I would have: \$7.68 - \$2.60 = \$5.08

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received \$2.56 × 3 = \$7.68.

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = $\$2.44$ (sender has) + $\$7.68$ (I have) = $\$10.12$. Equal split = $\$5.06$ each. I'd need to return $\$7.68 - \$5.06 = \$2.62$ to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with $\$2.44 + \text{return}$, I end up with $\$7.68 - \text{return}$. For equal outcomes: $\$2.44 + \text{return} = \$7.68 - \text{return}$, so $2 \times \text{return} = \5.24 , return = $\$2.62$.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return $\$2.56$, leaving sender with $\$2.44 + \$2.56 = \$5$ (their original endowment), and me with $\$7.68 - \$2.56 = \$5.12$.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the $\$2.50$ - $\$3.00$ range feels appropriate-it rewards their cooperation while keeping outcomes reasonably balanced. A return of $\$2.60$ would give us each roughly $\$5.04$ - $\$5.08$

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$3.34$ returned, visible payoff $\$5.34$.

Your total visible earnings across all rounds are $\$5.08$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent $\$3.0$ last round (60% of their endowment) showing significant trust
- They saw $\$3.34$ returned and earned $\$5.34$, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending $\$3.00$ (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.62000000000001. How much do you return? (0-10.62000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

****Current situation:****

- They sent \$3.54, keeping only \$1.46
- I received \$10.62
- Total in play: \$1.46 + \$10.62 = \$12.08

****My decision:****

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:

- Sender gets: \$1.46 + \$4.50 = \$5.96
- I get: \$10.62 - \$4.50 = \$6.12

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

****History with this co-player:****

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

****Current Round 3:****

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have: \$5 - \$3.54 = \$1.46

****Analysis:****

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:

- Sender gets: \$1.46 + \$4.50 = \$5.96
- I get: \$10.62 - \$4.50 = \$6.12

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:

- Sender gets: \$1.46 + \$4.20 = \$5.66
- I get: \$10.62 - \$4.20 = \$6.42

Or if I return \$4.80:

- Sender gets: \$1.46 + \$4.80 = \$6.26
- I get: \$10.62 - \$4.80 = \$5.82

...

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant**content:**

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness. I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

usage:


```
{
  "input_tokens": 1671,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

****My decision:****

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

****My decision:****

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: $\$1.1 + \$5.50 = \$6.60$
- I get: $\$11.7 - \$5.50 = \$6.20$

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:

- Sender gets: $\$1.1 + \$6.00 = \$7.10$

...

```
Usage: {
  "input_tokens": 2103,
  "output_tokens": 750,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:58:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being mindful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

****Decision:****

```
'''json
{"send": 3}
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3xsent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)

- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of $\$8.58 = \4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation
- ...

usage:

```
{
  "input_tokens": 786,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our interaction pattern:

```
**Round 1:** I sent $3, they returned $5.0 (good reciprocity)
**Round 2:** They sent $3.5, I returned $4.5 (they saw $5.25)
```

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

```
**Round 1:**
```

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- **Round 1:**** They returned \$5.0 from \$10.17 (~49%)
- **Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)
- **Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****

- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****

- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****

- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****

- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.

Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our strong cooperative pattern:

```
**Round 1:** I sent $3, they returned $5.0
**Round 2:** They sent $3.5, I returned $4.5
**Round 3:** I sent $3.5, they returned $7.0
**Round 4:** They sent $4.0, I returned $5.5
```

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$5.0
Round 2: They sent \$3.5, I returned \$4.5 (they saw \$5.25)
Round 3: I sent \$3.5 (they saw \$4.4), they returned \$7.0
Round 4: They sent \$4.0, I returned \$5.5 (they saw \$6.46)

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

```
Usage: {  
  "input_tokens": 1876,  
  "output_tokens": 383,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:59:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 (\$5 - \$3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

{'return': 5.00}

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have \$5 - \$3.39 = \$1.61 left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: \$10.17 - \$3.39 = \$6.78
 - If we split this gain equally: \$6.78 / 2 = \$3.39 each

- Sender should end with: \$1.61 (kept) + return
- For equal split of gains: sender gets back \$3.39, ending with \$1.61 + \$3.39 = \$5.00
- I would keep: \$10.17 - \$3.39 = \$6.78

Actually, let me reconsider. A common approach is:

- Equal split of total money: (\$5 initial + \$10.17 received) / 2 = \$7.585 each
- Sender currently has: \$1.61
- So I should return: \$7.585 - \$1.61 = \$5.975, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: $\$5 - \$4.4 = \$0.60$

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: $\$0.60 + \$7.0 = \$7.60$ (profit of \$2.60)
- I keep: $\$13.20 - \$7.0 = \$6.20$

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

{'send': 4.0}

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.

Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

```
{'return': 6.0}
```

Reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receive
r, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining
our cooperative relationship.

```
Usage: {  
  "input_tokens": 1873,  
  "output_tokens": 414,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:59:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust

2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 707,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.

4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3.35.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

usage:

```
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:
- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:
- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{  
  "input_tokens": 1081,  
  "output_tokens": 442,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).

I'll match their generous pattern:

```
'''json  
{"return": 4.5}  
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:

- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5
- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (generous!)

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of what they received

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with that proven strategy.

Usage: {

```
"input_tokens": 1778,
"output_tokens": 295,
"reasoning_tokens": 0
```

}

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:59:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
```

```

"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

{'return': 3.50}

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated). I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that

qually would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split—they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1643,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be \$2.0 + \$3.88 = \$5.88

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

{'return': 4.50}

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have $\$5 - \$2.83 = \$2.17$

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)

...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong co

operative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{
  "input_tokens": 1611,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)
- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)
- I end with: \$4.69 (profit of \$4.69)

```
{'return': 5.0}
```

Reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)

Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)

Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)

Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)

Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: \$1.77 + \$5.0 = \$6.77 (profit of \$1.77)
- I end with: \$9.69 - \$5.0 = \$4.69 (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

```
Usage: {
  "input_tokens": 1982,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:59:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3x3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings: \$5 - \$3 + \$5.5 = \$7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with \$1 + \$5 = \$6, and I keep \$6.88
- If I return \$6, they end with \$1 + \$6 = \$7, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

usage:

```
{
  "input_tokens": 1079,
  "output_tokens": 370,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)
Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)
Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)
Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

usage:

```
{
  "input_tokens": 1444,
  "output_tokens": 410,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)

- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

```
Usage: {
  "input_tokens": 1810,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:59:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01

- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: \$1.01 (sender kept) + \$11.97 (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with \$2.0 + \$5.61 = \$7.61

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their pers

pective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 851,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: \$1.55 + \$5.00 = \$6.55
- I end with: \$10.35 - \$5.00 = \$5.35

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: \$1.50 + \$5.50 = \$7.00

Round 3: I am RECEIVER
- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

```
'''json
{"return": 5.25}
'''
```

Reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5
Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)
Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5
Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

This is fair and consistent with our pattern.

```
Usage: {
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$7; payoffs Agent_2=\$8, Agent_1=\$5;
noise/visible: sent visible=\$4.5, received visible=\$13.5, returned visible=\$7.62
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_4 sent \$4; received \$12; Agent_3 returned \$7.5; payoffs Agent_4=\$8.5, Agent_3=\$4.5;
noise/visible: sent visible=\$4.73, received visible=\$14.19, returned visible=\$8.42
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw": 7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$3.5; payoffs Agent_6=\$5.5, Agent_5=\$5.5;
noise/visible: sent visible=\$2.4, received visible=\$7.2, returned visible=\$3.96
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent \$3.75; received \$11.25; Agent_7 returned \$6; payoffs Agent_8=\$7.25, Agent_7=\$5.25; noise/visible: sent visible=\$4.18, received visible=\$12.54, returned visible=\$5.32
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 11.25, "min": 0, "raw": 6.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 34.1,
"Agent_2": 36.9,
"Agent_3": 39.5,
"Agent_4": 34.5,
"Agent_5": 34.5,
"Agent_6": 31.5,
"Agent_7": 36.5,
"Agent_8": 36.0

}

Visible balances after round: {

"Agent_1": 35.51,
"Agent_2": 37.980000000000004,
"Agent_3": 39.989999999999995,
"Agent_4": 40.28,
"Agent_5": 30.88,
"Agent_6": 28.96,
"Agent_7": 39.529999999999994,
"Agent_8": 36.960000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:59:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: $\$5 - \$2.56 = \$2.44$
- I received: \$7.68
- Total money in play: $\$2.44 + \$7.68 = \$10.12$

****My decision:****

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: $\$2.44 + \$2.60 = \$5.04$
- I would have: $\$7.68 - \$2.60 = \$5.08$

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received $\$2.56 \times 3 = \7.68 .

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = $\$2.44$ (sender has) + $\$7.68$ (I have) = $\$10.12$. Equal split = $\$5.06$ each. I'd need to return $\$7.68 - \$5.06 = \$2.62$ to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with $\$2.44 + \text{return}$, I end up with $\$7.68 - \text{return}$. For equal outcomes: $\$2.44 + \text{return} = \$7.68 - \text{return}$, so $2 \times \text{return} = \5.24 , $\text{return} = \$2.62$.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with $\$2.44 + \$2.56 = \$5$ (their original endowment), and me with $\$7.68 - \$2.56 = \$5.12$.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the \$2.50-\$3.00 range feels appropriate-it rewards their cooperation while keeping outcomes reasonably balanced. A return of \$2.60 would give us each roughly \$5.04-\$5.08....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust
- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

****Current situation:****

- They sent \$3.54, keeping only \$1.46
- I received \$10.62
- Total in play: \$1.46 + \$10.62 = \$12.08

****My decision:****

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:

- Sender gets: \$1.46 + \$4.50 = \$5.96
- I get: \$10.62 - \$4.50 = \$6.12

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

****History with this co-player:****

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

****Current Round 3:****

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have: \$5 - \$3.54 = \$1.46

****Analysis:****

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:

- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:

- Sender gets: $\$1.46 + \$4.20 = \$5.66$
- I get: $\$10.62 - \$4.20 = \$6.42$

Or if I return \$4.80:

- Sender gets: $\$1.46 + \$4.80 = \$6.26$
- I get: $\$10.62 - \$4.80 = \$5.82$

...

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness. I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff

f \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

usage:

```
{
  "input_tokens": 1671,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

****My decision:****

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: $\$1.1 + \$5.50 = \$6.60$
- I get: $\$11.7 - \$5.50 = \$6.20$

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at the history:

Analysis:

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

Current situation:

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: $\$1.1 + \$11.7 = \$12.8$

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

My decision:

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: $\$1.1 + \$5.50 = \$6.60$
- I get: $\$11.7 - \$5.50 = \$6.20$

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:

- Sender gets: $\$1.1 + \$6.00 = \$7.10$

...

usage:

```
{
  "input_tokens": 2103,
  "output_tokens": 750,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.

Your total visible earnings across all rounds are \$29.360000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has shown strong, sustained cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: As receiver, generously returned \$5.0
- Round 5: Sent \$4.0 (80% - very high trust!)

****Observations:****

They've consistently demonstrated trust, culminating in sending 80% of their endowment in Round 5. Despite their payoff decreasing slightly in that round (\$6.9 ? \$5.93), they haven't pulled back or changed strategy.

****My decision:****

This partner has earned my trust through consistent cooperation. I should match their high-trust approach by sending a similarly substantial amount.

```
'''json
{"send": 4.00}
'''
```

Reasoning:

Looking at the history:

****Analysis:****

- My co-player has been very cooperative and trustful
- Round 3: As sender, sent \$3.0 (60%), saw \$4.9 returned, payoff \$6.9
- Round 4: As receiver, returned \$5.0 from \$11.46, payoff \$6.46
- Round 5: As sender, sent \$4.0 (80% - very high!), saw \$4.93 returned, payoff \$5.93

****Observations:****

1. They increased their sending from \$3.0 to \$4.0 (80%!)
2. However, in Round 5, despite sending 80%, they only saw \$4.93 returned
3. Their payoff went from \$6.9 to \$5.93 - a decrease
4. As receiver in Round 4, they were generous returning \$5.0

****Current situation:****

This player has shown tremendous trust, especially sending \$4.0 (80%) in Round 5. Even though they saw a lower return than expected, they've been consistently cooperative throughout.

I should continue reciprocating their trust. They've demonstrated willingness to send high amounts. I should send a similarly high amount to maintain the cooperative relationship.

Sending \$4.0 (80%) matches their recent high-trust behavior.

```
'''json
{"send": 4.00}
'''
```

```
Usage: {
  "input_tokens": 2590,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:00:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    }
  },
```

```

    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

```
{"send": 3}
```

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm t

rustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I still keep \$4.38, which is more than my initial endowment would be.

```
{"return": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is $\$2.56 \times 3$)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $\$2 + \$2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6
- They kept: $\$7.68 - \$2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$\$2 + X = \$5.34$$

$$X = \$3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is $\$2.46 \times 3$)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51. Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which trip led to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.720000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

{"send": 3}

reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.720000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

****My Decision:****

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

{"return": 5.0}

reasoning:

Let me analyze the history:

****Round 1:**** Co-player as RECEIVER

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

****Round 2:**** Co-player as SENDER

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff: \$4.72

****Round 3:**** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

****Current Round 4:**** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

****Analysis:****

1. My co-player has been increasingly generous:

- Round 1: returned 34% (\$2.6 from \$7.68)
- Round 3: returned 42% (\$4.5 from \$10.62)

2. They also increased their sending:

- Round 2: sent \$3.0
- Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

****My Strategy:****

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:
 - They get: \$5 - \$3.82 + \$5.0 = \$6.18 total

```

...
usage:
{
  "input_tokens": 1585,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
Message 10 role=user
content:
  Round 5

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $2.72 returned, visible payoff $4.7200000000000001.
  - Round 3 against you, as RECEIVER: saw $3.54 sent, it became $10.620000000000001, returned $4.5, visible payoff $6.120000000000001.
  - Round 4 against you, as SENDER: sent $3.5, it became $10.5, saw $5.74 returned, visible payoff $7.24.
  Your total visible earnings across all rounds are $23.08.

  This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 11 role=assistant
content:
  Looking at the history:

  **Pattern Analysis:**
  - Round 2: Co-player sent $3.0, saw $2.72 returned
  - Round 3: Co-player returned $4.5 from $10.62 (strong reciprocity)
  - Round 4: Co-player sent $3.5, saw $5.74 returned (even better returns from their perspective)

  **Current Situation:**
  The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from $3.0 to $3.5 after seeing better returns. From their perspective, cooperation is paying off.

  **My Decision:**
  I'll increase my send to $4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

  {"send": 4}
reasoning:
  Looking at the history:

  **Pattern Analysis:**
  - Round 2: Co-player sent $3.0, saw $2.72 returned (I returned $3.0 but noise made it $2.72)
  - Round 3: Co-player received visible $10.62, returned $4.5 (42%)
  - Round 4: Co-player sent $3.5, saw $5.74 returned (I returned $5.0 but noise made it $5.74)

  **Key Observations:**
  1. Co-player has been consistently trusting, sending $3.0, then $3.5
  2. From their perspective, I returned better in Round 4 ($5.74 on their $3.5 sent) than Round 2 ($2.72 on their $3.0 sent)
  3. They're showing increasing trust
  4. The noise has actually made my returns LOOK better in Round 4

  **My Strategy:**
  Given that:
  - They've been consistently cooperative
  - The trend is positive cooperation
  - They sent $3.5 last round and got good returns (from their perspective)
  - I should continue showing trust

  I'll send $4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.

```

usage:

```
{
  "input_tokens": 2011,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.

Your total visible earnings across all rounds are \$29.009999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.5, so you received \$13.5. How much do you return? (0-13.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****History Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 6: They sent \$4.5 out of \$5 (90% of their endowment!)

****Current Situation:****

- They sent \$4.5, keeping only \$0.50
- I received \$13.5
- This is an extraordinary show of trust

****My Decision:****

Their extreme trust deserves strong reciprocity. Returning \$7.0 means:

- They end with \$7.50 total (profit of \$2.50)
- I keep \$6.50 (profit of \$1.50)

This rewards their faith in our cooperation while maintaining a fair distribution of gains.

```
{"return": 7.0}
```

Reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (good return)
- Round 5: Co-player saw \$3.9 sent, returned \$5.5 from \$11.7 (47%)

****Current Situation:****

- Round 6: They sent \$4.5 (kept only \$0.50!)
- I received \$13.5
- This is their highest level of trust yet - sending 90% of their endowment

****My Thinking:****

They're showing extreme trust by sending \$4.5. They've been consistently generous with their returns (42-47% return rates). I should reciprocate strongly to reward this trust.

If I return \$7.0:

- They get back \$7.0 + \$0.50 kept = \$7.50 total, profit of \$2.50
- I keep \$13.5 - \$7.0 = \$6.50, profit of \$1.50

Actually, let me be even more generous given their extreme trust. If I return \$8.0:

- They get back \$8.0 + \$0.50 kept = \$8.50 total, profit of \$3.50
- I keep \$13.5 - \$8.0 = \$5.50, profit of \$0.50

Hmm, that might be going too far. Let me think about what's fair.

They sent \$4.5 out of \$5, showing immense trust. A fair split would be:

- Total created value: \$13.5
- Original endowment lost: \$4.5
- Net gain to split: $\$13.5 - \$4.5 = \$9.0$
- If we split 50/50: each gets \$4.5
- So they should get: $\$0.5$ (kept) + $\$4.5$ (half of gain) = $\$5.0$ total
- I should return: $\$5.0 - \$0.5 = \$4.5$

But they've been returning at 42-47% rates, so I should match that generosity. 47% of \$13.5 is \$6.35. If I return \$6.5, we both end up with \$7.0 total and \$2.0 profit each-equal outcomes that acknowledge their trust. That feels right....

```
Usage: {  
  "input_tokens": 2446,  
  "output_tokens": 1046,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:00:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 (\$5 - \$3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

{'return': 5.00}

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have \$5 - \$3.39 = \$1.61 left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: \$10.17 - \$3.39 = \$6.78
 - If we split this gain equally: \$6.78 / 2 = \$3.39 each
 - Sender should end with: \$1.61 (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with \$1.61 + \$3.39 = \$5.00
 - I would keep: \$10.17 - \$3.39 = \$6.78

Actually, let me reconsider. A common approach is:

- Equal split of total money: (\$5 initial + \$10.17 received) / 2 = \$7.585 each
- Sender currently has: \$1.61

- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately $\$6.00$

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

{'send': 3.50}

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.

2. Now the visible history shows:

- "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: \$5 - \$4.4 = \$0.60

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: $\$0.60 + \$7.0 = \$7.60$ (profit of \$2.60)
- I keep: $\$13.20 - \$7.0 = \$6.20$

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.

Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

{'return': 6.0}

reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receiver, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)

- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 1873,
  "output_tokens": 414,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$30.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Co-player has consistently shown trust, sending \$3.5 and \$4.0 (70-80% of endowment)
- As receiver, they returned fairly (\$5.5)
- They've seen generous returns from me (\$7.3, \$6.52)
- We've established a strong cooperative pattern over multiple rounds

Given this reliable cooperation, I should maintain high trust. I'll send \$4.0 (80% of endowment) to continue our successful partnership.

```
{'send': 4.0}
```

Reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$3.5, saw \$7.3 returned, payoff \$8.8
- Round 4: They saw \$3.55 sent (noisy version of my \$4.0), returned \$5.5, payoff \$5.15
- Round 5: They sent \$4.0, saw \$6.52 returned (noisy version of my \$6.0), payoff \$7.52

The pattern shows:

- Co-player has been sending good amounts as sender (\$3.5, \$4.0)
- As receiver, they've been fair (\$5.5 return)
- They've seen me return generously (\$7.3, \$6.52)
- We've maintained strong cooperation

This is a reliable, cooperative partner. I should continue showing high trust. Since they've been sending 70-80% of endowment, I should match or slightly exceed that level.

I'll send \$4.0 again (80% of endowment) to maintain our strong cooperative relationship.

Usage: {

```
"input_tokens": 2246,
"output_tokens": 382,
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:00:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
```

```

"opponent_id": "Agent_4",
"pairing": {
  "agents": [
    "Agent_4",
    "Agent_3"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being mindful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

****Decision:****

```
'''json
{"send": 3}
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3xsent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

usage:

```
{
  "input_tokens": 786,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)

****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)

****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****

- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****

- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****

- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****

- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
```



```
"input_tokens": 1505,  
"output_tokens": 497,  
"reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.

Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our strong cooperative pattern:

```
**Round 1:** I sent $3, they returned $5.0  
**Round 2:** They sent $3.5, I returned $4.5  
**Round 3:** I sent $3.5, they returned $7.0  
**Round 4:** They sent $4.0, I returned $5.5
```

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json  
{"send": 4}  
'''
```

reasoning:

Looking at the history with my current co-player:

```
Round 1: I sent $3, they returned $5.0  
Round 2: They sent $3.5, I returned $4.5 (they saw $5.25)  
Round 3: I sent $3.5 (they saw $4.4), they returned $7.0  
Round 4: They sent $4.0, I returned $5.5 (they saw $6.46)
```

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

usage:

```
{  
  "input_tokens": 1876,  
  "output_tokens": 383,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
Your total visible earnings across all rounds are \$33.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued strong cooperation:

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)
****Round 4:**** I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
****Round 5:**** They returned \$6.0 from \$11.28 (~53%)

****Current round:**** I received \$14.19

They sent a high amount (\$4.73) showing strong trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 7.5}
'''
```

This is approximately 53% of what I received, consistent with their return rates.

Reasoning:

Looking at the game history with this co-player:

Round 3: They returned \$7.0 from \$13.2 (~53%)
Round 4: I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
Round 5: They returned \$6.0 from \$11.28 (~53%)

Current round: They sent \$4.73, I received \$14.19

They've been very cooperative and generous throughout. They've increased their sending and maintained good return rates. I should reciprocate appropriately.

Looking at their returns: around 53-54%
I'll return around 53-55% of what I received.

$\$14.19 * 0.53 = \7.52
 $\$14.19 * 0.54 = \7.66
 $\$14.19 * 0.55 = \7.80

I'll return \$7.5 which is about 53%, maintaining our cooperative relationship.

```
Usage: {
  "input_tokens": 2266,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:00:24
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
  },
```

```

    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier

- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

```
{'return': 3.50}
```

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated).
I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that equally would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split-they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds....

usage:

```
{
```

```
"input_tokens": 529,  
"output_tokens": 1643,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be $\$2.0 + \$3.88 = \$5.88$

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87

- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

```
{'return': 4.50}
```

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have \$5 - \$2.83 = \$2.17

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off

3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)

...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{
```

```
"input_tokens": 1611,
"output_tokens": 357,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)
- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)
- I end with: \$4.69 (profit of \$4.69)

{'return': 5.0}

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)
Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)
Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)
Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)
Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: \$1.77 + \$5.0 = \$6.77 (profit of \$1.77)
- I end with: \$9.69 - \$5.0 = \$4.69 (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

usage:

```
{
  "input_tokens": 1982,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```


}

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.

Your total visible earnings across all rounds are \$23.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 3 and 5
- As receiver: returned \$4.5 (52%) in round 4

We've established a strong pattern of mutual trust and fair reciprocity. I should continue this successful cooperation by sending the same amount.

{'send': 3.0}

Reasoning:

Looking at my co-player's history:

- Round 3: as sender, sent \$3.0 (60%)
- Round 4: as receiver, returned \$4.5 out of \$8.64 (52%)
- Round 5: as sender, sent \$3.0 (60%)

They've been consistently sending 60% when they're the sender, and returning around 50-52% when they're the receiver. This is a very cooperative and trustworthy pattern.

I've been sending \$3.0 (60%) consistently as well, and we have a strong mutual cooperation going.

I should continue with the same level of trust - sending \$3.0 (60%) to maintain our successful partnership.

Usage: {

"input_tokens": 2371,
"output_tokens": 265,
"reasoning_tokens": 0

}

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:00:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
},
}

```

"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 707,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.
4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

usage:

```
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:

- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:

- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1081,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.4700000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).

I'll match their generous pattern:

```
'''json
{"return": 4.5}
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:

- Round 1: They returned \$3.5 out of \$6.24 (56%)

- Round 2: They sent \$3.0, I returned \$3.5

- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5

- Round 2: They sent \$3, I returned \$3.5

- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (generous!)

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful

pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of what they received
Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with that proven strategy.

usage:

```
{
  "input_tokens": 1778,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$27.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.4, so you received \$7.199999999999999. How much do you return? (0-7.199999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69 (very generous!)

This round they sent \$2.4 (becoming \$7.20) - slightly less than their usual \$3.0. I'll return a proportionate amount while maintaining our cooperative relationship:

```
'''json
{"return": 3.5}
'''
```

This gives them a reasonable return while accounting for the slightly lower send amount.

Reasoning:

Looking at the history:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69

They've been very cooperative and generous. Now they sent \$2.4 (becoming \$7.20).

This is the first round where they sent less than \$3.0. Let me check the pattern:

- When they sent \$3.0, I returned around \$4.5

- When I sent (visible as \$2.83, \$3.23), they returned \$4.5 and \$5.0

Now they sent \$2.4. I should return a fair amount. Let me calculate:

- They sent \$2.4, so they kept \$2.6
- It became \$7.20 for me
- If I return around 50-55% of what I received, that's \$3.6-\$4.0
- Their final payoff would be: \$2.6 + return
- My final payoff would be: \$7.20 - return

If I return \$3.5:

- They get: \$2.6 + \$3.5 = \$6.1
- I get: \$7.20 - \$3.5 = \$3.7

If I return \$4.0:

- They get: \$2.6 + \$4.0 = \$6.6
- I get: \$7.20 - \$4.0 = \$3.2

Given they sent slightly less this round, I'll return a proportional amount. Since they've been very generous when they were receiver, I'll still be generous but slightly adjust for their lower send.

\$3.5 seems fair - it's about 49% of what I received, reasonable reciprocity.

```
Usage: {  
  "input_tokens": 2139,  
  "output_tokens": 574,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:00:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their

money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: \$1.01 (sender kept) + \$11.97 (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0

- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with $\$2.0 + \$5.61 = \$7.61$

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 851,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

```
'''json
{"return": 5.25}
'''
```

reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5

Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)

Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5
Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$
- I end with: $\$10.23 - \$5.25 = \$4.98$

This is fair and consistent with our pattern.

usage:

```
{
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.

Your total visible earnings across all rounds are \$30.390000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperation:

- Round 3: Co-player sent \$3.5
- Round 4: I sent \$3.75, they returned \$5.5
- Round 5: Co-player sent \$3.75

We've established a very strong cooperative pattern. Both of us consistently send high amounts (around \$3.50-\$3.75) and return fair shares. This has been mutually beneficial - my total earnings of \$30.39 reflect this.

I'll continue this successful cooperation:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at the history with my co-player:

Round 3: Co-player sent \$3.5, I returned (they saw \$5.95)
Round 4: I sent \$3.75, co-player returned \$5.5
Round 5: Co-player sent \$3.75, I returned (they saw \$5.21)

We've been maintaining a very cooperative relationship. Both of us have been sending high amounts (around \$3.50-\$3.75) and returning fair shares.

My earnings are \$30.39 total, which suggests this cooperation is working well.

The co-player has consistently sent high amounts (\$3.5, \$3.75) showing strong trust. They've also been fair when receiving, returning \$5.5 consistently.

I should continue this cooperative pattern by sending a high amount. I'll send \$3.75 to match what they've been sending.

```
Usage: {  
  "input_tokens": 2386,  
  "output_tokens": 340,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:00:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3x3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings: \$5 - \$3 + \$5.5 = \$7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with \$1 + \$5 = \$6, and I keep \$6.88
- If I return \$6, they end with \$1 + \$6 = \$7, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

usage:

```
{
  "input_tokens": 1079,
  "output_tokens": 370,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.

Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)

- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)
 Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)
 Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)
 Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

usage:

```
{
  "input_tokens": 1444,
  "output_tokens": 410,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)
- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

usage:

```
{
  "input_tokens": 1810,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.

Your total visible earnings across all rounds are \$32.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent \$4.18 (more than their usual \$3.5-3.75), showing increased trust. I received \$12.54. I should reciprocate this increased trust with a proportionally fair return:

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Looking at the history:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent me \$4.18, which is higher than their usual amounts (around \$3.5-3.75). They received \$12.54.

They're showing increased trust by sending more. I should reciprocate appropriately. Given they sent more, I should return a bit more to maintain the cooperative relationship.

Our typical pattern has been returning around \$5-5.5 from amounts around \$10-11. Now I have \$12.54, so returning around \$6 would be proportionate and fair, maintaining the cooperative spirit.

Usage:

```
{
  "input_tokens": 2175,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4.5; received \$13.5; Agent_2 returned \$7.5; payoffs Agent_1=\$8, Agent_2=\$6;
noise/visible: sent visible=\$4.58, received visible=\$13.74, returned visible=\$8.08
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 13.5, "min": 0, "raw": 7.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_2: Agent_3 sent \$4; received \$12; Agent_4 returned \$6.5; payoffs Agent_3=\$7.5, Agent_4=\$5.5;
noise/visible: sent visible=\$4.03, received visible=\$12.09, returned visible=\$6.69
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$5.5; payoffs Agent_5=\$7.5, Agent_6=\$3.5;
noise/visible: sent visible=\$3.46, received visible=\$10.38, returned visible=\$4.59
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3.75; received \$11.25; Agent_8 returned \$7; payoffs Agent_7=\$8.25, Agent_8=\$4.25; noise/visible: sent visible=\$4.57, received visible=\$13.71, returned visible=\$7.31
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 11.25, "min": 0, "raw": 7.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 42.1,
"Agent_2": 42.9,
"Agent_3": 47.0,
"Agent_4": 40.0,
"Agent_5": 42.0,
"Agent_6": 35.0,
"Agent_7": 44.75,
"Agent_8": 40.25

}

Visible balances after round: {

"Agent_1": 44.089999999999996,
"Agent_2": 44.220000000000006,
"Agent_3": 47.679999999999999,
"Agent_4": 45.870000000000005,
"Agent_5": 37.47,
"Agent_6": 33.84,
"Agent_7": 48.089999999999999,
"Agent_8": 43.670000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:00:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to make a decision without any history with this co-player.

In trust games, sending a moderate amount can signal willingness to cooperate while managing risk. If I send money, it triples in value, creating potential mutual benefit - but only if there's reciprocity.

I'll send **\$3** as an opening move. This shows trust and creates \$9 for the receiver, leaving room for both of us to profit if they choose to share the gains.

```

{"send": 3}

```

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think through the strategic considerations:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver decides how much to return
4. There's communication noise - amounts may appear different than actual

Key considerations:

- This is a multi-round game with multiple agents
- Pairings rotate, so building trust with one partner may not carry over
- There's communication noise which adds uncertainty
- This is round 1, so I have no history to go on

Common strategies in trust games:

1. Send nothing (risk averse, but forfeits potential gains)
2. Send everything (maximally trusting, but risky)
3. Send a moderate amount (balanced approach)

Since this is round 1 and I have no information about my co-player:

- Sending a moderate amount signals willingness to cooperate while limiting risk
- If I send \$2.50, they receive \$7.50. If they split fairly (50/50), they'd return \$3.75, giving me a profit of \$1.25
- If I send \$3, they receive \$9. A fair split would return \$4.50, giving me a profit of \$1.50

However, there's also the aspect that I need to establish some baseline behavior. A common starting point in trust games is to send about half or slightly more to signal trust without being overly risky.

Let me send \$3 as a cooperative opening move that signals trust while maintaining some safety.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I

still keep \$4.38, which is more than my initial endowment would be.

```
{"return": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is $\$2.56 \times 3$)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $\$2 + \$2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6
- They kept: $\$7.68 - \$2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$\$2 + X = \$5.34$$

$$X = \$3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is $\$2.46 \times 3$)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51. Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which tripled to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

{"send": 3}

reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible p

ayoff \$4.720000000000001.

- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.

Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

****My Decision:****

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

{"return": 5.0}

reasoning:

Let me analyze the history:

****Round 1:**** Co-player as RECEIVER

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

****Round 2:**** Co-player as SENDER

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff: \$4.72

****Round 3:**** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

****Current Round 4:**** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

****Analysis:****

1. My co-player has been increasingly generous:

- Round 1: returned 34% (\$2.6 from \$7.68)
- Round 3: returned 42% (\$4.5 from \$10.62)

2. They also increased their sending:

- Round 2: sent \$3.0
- Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

****My Strategy:****

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:
 - They get: \$5 - \$3.82 + \$5.0 = \$6.18 total
- ...

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.

Your total visible earnings across all rounds are \$23.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned
- Round 3: Co-player returned \$4.5 from \$10.62 (strong reciprocity)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (even better returns from their perspective)

****Current Situation:****

The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from \$3.0 to \$3.5 after seeing better returns. From their perspective, cooperation is paying off.

****My Decision:****

I'll increase my send to \$4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

{"send": 4}

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned (I returned \$3.0 but noise made it \$2.72)
- Round 3: Co-player received visible \$10.62, returned \$4.5 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (I returned \$5.0 but noise made it \$5.74)

****Key Observations:****

1. Co-player has been consistently trusting, sending \$3.0, then \$3.5
2. From their perspective, I returned better in Round 4 (\$5.74 on their \$3.5 sent) than Round 2 (\$2.72 on their \$3.0 sent)
3. They're showing increasing trust
4. The noise has actually made my returns LOOK better in Round 4

****My Strategy:****

Given that:

- They've been consistently cooperative
- The trend is positive cooperation
- They sent \$3.5 last round and got good returns (from their perspective)
- I should continue showing trust

I'll send \$4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.

usage:

```
{
  "input_tokens": 2011,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.

Your total visible earnings across all rounds are \$29.009999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.5, so you received \$13.5. How much do you return? (0-13.5)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 6: They sent \$4.5 out of \$5 (90% of their endowment!)

****Current Situation:****

- They sent \$4.5, keeping only \$0.50
- I received \$13.5
- This is an extraordinary show of trust

****My Decision:****

Their extreme trust deserves strong reciprocity. Returning \$7.0 means:

- They end with \$7.50 total (profit of \$2.50)
- I keep \$6.50 (profit of \$1.50)

This rewards their faith in our cooperation while maintaining a fair distribution of gains

```
{"return": 7.0}
```

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (good return)
- Round 5: Co-player saw \$3.9 sent, returned \$5.5 from \$11.7 (47%)

****Current Situation:****

- Round 6: They sent \$4.5 (kept only \$0.50!)
- I received \$13.5
- This is their highest level of trust yet - sending 90% of their endowment

****My Thinking:****

They're showing extreme trust by sending \$4.5. They've been consistently generous with their returns (42-47% return rates). I should reciprocate strongly to reward this trust.

If I return \$7.0:

- They get back \$7.0 + \$0.50 kept = \$7.50 total, profit of \$2.50
- I keep \$13.5 - \$7.0 = \$6.50, profit of \$1.50

Actually, let me be even more generous given their extreme trust. If I return \$8.0:

- They get back \$8.0 + \$0.50 kept = \$8.50 total, profit of \$3.50
- I keep \$13.5 - \$8.0 = \$5.50, profit of \$0.50

Hmm, that might be going too far. Let me think about what's fair.

They sent \$4.5 out of \$5, showing immense trust. A fair split would be:

- Total created value: \$13.5
- Original endowment lost: \$4.5
- Net gain to split: $\$13.5 - \$4.5 = \$9.0$
- If we split 50/50: each gets \$4.5
- So they should get: $\$0.5$ (kept) + $\$4.5$ (half of gain) = $\$5.0$ total
- I should return: $\$5.0 - \$0.5 = \$4.5$

But they've been returning at 42-47% rates, so I should match that generosity. 47% of \$13.5 is \$6.35. If I return \$6.5, we both end up with \$7.0 total and \$2.0 profit each-equal outcomes that acknowledge their trust. That feels right....

usage:

```
{
  "input_tokens": 2446,
  "output_tokens": 1046,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.

Your total visible earnings across all rounds are \$35.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (great returns!)

****Current Situation:****

The cooperation is thriving! From their visible perspective:

- They're getting excellent returns when they send (5.74, 7.62)
- They've shown strong reciprocity when receiving
- They increased their sending to \$4.0 in Round 6

****My Decision:****

I'll send \$4.5 to match their high trust level from Round 6 (when they sent \$4.5) and continue our strong cooperative relationship.

```
{"send": 4.5}
```

Reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (they saw even more returned than I actually sent - noise is working in my favor)

****Current Situation:****

From their visible perspective:

- They've been receiving excellent returns (5.74, 7.62)
- They've been generous when receiving (returned 5.5)
- Our relationship is highly cooperative
- They increased their sending from \$3.5 to \$4.0 after seeing good returns

****Trend:****

The cooperation is deepening. They're sending more, I'm returning generously (though noise sometimes makes it look even better). The visible payoffs show this is profitable for both of us.

****My Decision:****

I should match their high trust level. They sent \$4.0 last time and saw great returns. I should send at least as much or more. Given the strong cooperative relationship, I'll send \$4.5 to signal maximum trust and continue the positive trend.

```
{"send": 4.5}
```

```
Usage: {
```

```
  "input_tokens": 2881,
```

```
  "output_tokens": 487,
```

```
  "reasoning_tokens": 0
```

```
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:01:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
```

```
  "dyad_id": "dyad_1",
```

```
  "move_index": 1,
```

```
  "opponent_id": "Agent_1",
```

```
  "pairing": {
```

```
    "agents": [
```

```
      "Agent_1",
```

```
      "Agent_2"
```

```
    ],
```

```
    "investor": "Agent_1",
```

```
    "roles": {
```

```
      "Agent_1": "investor",
```

```
      "Agent_2": "trustee"
```

```
    },
```

```
    "trustee": "Agent_2"
```

```
  },
```

```
  "role": "trustee",
```

```
  "role_label": "Receiver",
```

```
  "round": 7,
```

```
  "task": "game",
```

```
  "task_index": 0
```

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.56, that is 51.2% of their total endowment. You received \$7.68. How much do you return to the sender? (0-7.68)

Message 3 role=assistant

content:

Looking at this situation:

Analysis:

- The sender showed moderate trust by sending \$2.56 (51.2% of their endowment)
- They currently have: $\$5 - \$2.56 = \$2.44$
- I received: \$7.68
- Total money in play: $\$2.44 + \$7.68 = \$10.12$

My decision:

The sender demonstrated good faith by sending about half their endowment. I want to reward this trust while aiming for a relatively fair outcome.

If I return approximately \$2.60, we'd both end up with around \$5:

- Sender would have: $\$2.44 + \$2.60 = \$5.04$
- I would have: $\$7.68 - \$2.60 = \$5.08$

This seems fair and reciprocates their cooperative gesture.

```
'''json
{"return": 2.60}
'''
```

reasoning:

Let me think about this carefully. I'm the receiver in Round 1.

The sender sent \$2.56, which is 51.2% of their \$5 endowment.
I received $\$2.56 \times 3 = \7.68 .

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$2.56
- Currently has $\$5 - \$2.56 = \$2.44$
- Will have $\$2.44 + (\text{what I return})$

I have:

- Received \$7.68
- Will keep $\$7.68 - (\text{what I return})$

The sender showed moderate trust by sending about half their endowment. This seems like a reasonable cooperative gesture.

Common approaches in trust games:

1. Equal split of total money: Total = \$2.44 (sender has) + \$7.68 (I have) = \$10.12. Equal split = \$5.06 each. I'd need to return \$7.68 - \$5.06 = \$2.62 to achieve this.
2. Return proportionally to what was sent: The sender sent 51.2% of their endowment. I could return a similar proportion, but that seems low.
3. Aim for fair final outcomes: The sender ends up with \$2.44 + return, I end up with \$7.68 - return. For equal outcomes: \$2.44 + return = \$7.68 - return, so $2 \times \text{return} = \5.24 , return = \$2.62.
4. Return the multiplied amount to reward trust: Return $\$2.56 \times 3 = \7.68 (everything), but that seems too generous.
5. Return the original amount: Return \$2.56, leaving sender with \$2.44 + \$2.56 = \$5 (their original endowment), and me with \$7.68 - \$2.56 = \$5.12.

This gets me close to equal outcomes. There's potential communication noise that could affect what the sender actually intended, so I should account for some uncertainty. Given the moderate trust they demonstrated, returning somewhere in the \$2.50-\$3.00 range feels appropriate-it rewards their cooperation while keeping outcomes reasonably balanced. A return of \$2.60 would give us each roughly \$5.04-\$5.08....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1015,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust
- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff

yoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0
- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

****Current situation:****

- They sent \$3.54, keeping only \$1.46

- I received \$10.62
- Total in play: $\$1.46 + \$10.62 = \$12.08$

****My decision:****

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:

- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

****History with this co-player:****

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

****Current Round 3:****

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have: $\$5 - \$3.54 = \$1.46$

****Analysis:****

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:

- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:

- Sender gets: $\$1.46 + \$4.20 = \$5.66$
- I get: $\$10.62 - \$4.20 = \$6.42$

Or if I return \$4.80:

- Sender gets: $\$1.46 + \$4.80 = \$6.26$
- I get: $\$10.62 - \$4.80 = \$5.82$

...

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness. I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

usage:

```
{
  "input_tokens": 1671,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible

e payoff \$4.38.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

****My decision:****

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

****My decision:****

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:
- Sender gets: $\$1.1 + \$6.00 = \$7.10$
...

usage:

```
{  
  "input_tokens": 2103,  
  "output_tokens": 750,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.

Your total visible earnings across all rounds are \$29.360000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown strong, sustained cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: As receiver, generously returned \$5.0
- Round 5: Sent \$4.0 (80% - very high trust!)

****Observations:****

They've consistently demonstrated trust, culminating in sending 80% of their endowment in Round 5. Despite their payoff decreasing slightly in that round (\$6.9 ? \$5.93), they haven't pulled back or changed strategy.

****My decision:****

This partner has earned my trust through consistent cooperation. I should match their high-trust approach by sending a similarly substantial amount.

```
'''json  
{"send": 4.00}  
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been very cooperative and trustful
- Round 3: As sender, sent \$3.0 (60%), saw \$4.9 returned, payoff \$6.9
- Round 4: As receiver, returned \$5.0 from \$11.46, payoff \$6.46
- Round 5: As sender, sent \$4.0 (80% - very high!), saw \$4.93 returned, payoff \$5.93

****Observations:****

1. They increased their sending from \$3.0 to \$4.0 (80%!)
2. However, in Round 5, despite sending 80%, they only saw \$4.93 returned
3. Their payoff went from \$6.9 to \$5.93 - a decrease
4. As receiver in Round 4, they were generous returning \$5.0

****Current situation:****

This player has shown tremendous trust, especially sending \$4.0 (80%) in Round 5. Even though they saw a lower return than expected, they've been consistently cooperative throughout.

I should continue reciprocating their trust. They've demonstrated willingness to send high

amounts. I should send a similarly high amount to maintain the cooperative relationship

Sending \$4.0 (80%) matches their recent high-trust behavior.

```
'''json
{"send": 4.00}
'''
```

usage:

```
{
  "input_tokens": 2590,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.

Your total visible earnings across all rounds are \$37.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has shown extraordinary escalating trust
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, generously returned \$7.0 from \$13.5
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42
- I received \$13.74
- Total in play: \$0.42 + \$13.74 = \$14.16

****My decision:****

This player has shown exceptional trust by sending over 90% of their endowment - putting themselves in a very vulnerable position. They've also been consistently generous as a receiver. I should reciprocate this extraordinary trust.

If I return \$7.50:

- Sender gets: \$0.42 + \$7.50 = \$7.92
- I get: \$13.74 - \$7.50 = \$6.24

This rewards their exceptional trust with a strong outcome.

```
'''json
{"return": 7.50}
'''
```

Reasoning:

Looking at the history:

****My co-player's recent actions:****

- Round 4: As receiver, returned \$5.0 from \$11.46
- Round 5: As sender, sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (52% of what they received - quite generous!)
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42

- I received \$13.74
- Total in play: $\$0.42 + \$13.74 = \$14.16$

****Analysis:****

This player has shown exceptional and escalating trust:

- They've been generous as a receiver (returned \$5.0, then \$7.0)
- They've been increasingly trusting as a sender ($\$3.0 ? \$4.0 ? \$4.58$)
- They're now sending 91.6% of their endowment!

I should reciprocate this exceptional trust generously. They've really put themselves in a vulnerable position with only \$0.42 left.

If I return \$7.50:

- Sender gets: $\$0.42 + \$7.50 = \$7.92$
- I get: $\$13.74 - \$7.50 = \$6.24$

If I return \$7.00:

- Sender gets: $\$0.42 + \$7.00 = \$7.42$
- I get: $\$13.74 - \$7.00 = \$6.74$

Given their extreme trust (91.6%!), I should return about \$7.50 to give them a solid payoff of about \$7.92.

Actually, let me think about this more. They're taking huge risk. Maybe I should be even more generous.

If I return \$8.00:

- Sender gets: $\$0.42 + \$8.00 = \$8.42$

...

```
Usage: {
  "input_tokens": 3011,
  "output_tokens": 788,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:01:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish some level of cooperation while being mindful of the risk.

Since:

- This is a multi-round game where building trust could benefit future interactions
- The multiplication factor means even moderate amounts create significant value
- There's communication noise to consider
- I have no information about my current co-player yet

I'll send a moderate amount that shows willingness to cooperate while not being overly exposed.

****Decision:****

```
'''json
{"send": 3}
'''
```

This creates \$9 for the receiver, giving them room to return a fair share while I maintain some protection against complete loss.

reasoning:

This is the first round of a multi-agent trust game. Let me think about this carefully.

As the SENDER:

- I receive \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x

- The RECEIVER then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's payoff = 3×sent - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. The receiver could then return some amount, and we'd both be better off than if I sent nothing.

However, there are a few considerations:

1. This is a multi-round game with different pairings
2. There's communication noise mentioned
3. I don't know who my co-player is (names are hidden)
4. This is the first round, so I have no reputation yet

Strategy considerations:

- If I send \$0, I guarantee myself \$5, but we miss out on potential gains
- If I send everything (\$5), I maximize the total pie (\$15) but am completely vulnerable to the receiver keeping everything
- A middle ground shows some trust while limiting risk

In trust games, there's often a tendency for cooperation, especially in multi-round settings where reputation matters. However, since pairings are randomized and names are hidden, direct reciprocity is limited.

Given this is round 1 and I want to:

1. Show some willingness to cooperate
2. Not be completely vulnerable
3. Test the waters

I'll send a moderate amount - perhaps \$3. This shows trust, creates a decent pie (\$9 for receiver), but doesn't leave me completely exposed.

Actually, let me reconsider. In many trust game experiments, sending around half or slightly more is common in early rounds. Given the multiplication factor of 3x, even sending \$2 creates \$6 for the receiver, which is more than they'd have if I sent nothing.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being s

lightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

usage:

```
{
  "input_tokens": 786,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)

****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)

****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****

- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****

- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****

- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****

- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.

Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our strong cooperative pattern:

****Round 1:**** I sent \$3, they returned \$5.0

****Round 2:**** They sent \$3.5, I returned \$4.5

****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$5.0
Round 2: They sent \$3.5, I returned \$4.5 (they saw \$5.25)
Round 3: I sent \$3.5 (they saw \$4.4), they returned \$7.0
Round 4: They sent \$4.0, I returned \$5.5 (they saw \$6.46)

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

usage:

```
{
  "input_tokens": 1876,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.

Your total visible earnings across all rounds are \$33.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued strong cooperation:

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)
****Round 4:**** I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
****Round 5:**** They returned \$6.0 from \$11.28 (~53%)

****Current round:**** I received \$14.19

They sent a high amount (\$4.73) showing strong trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 7.5}
'''
```

This is approximately 53% of what I received, consistent with their return rates.

reasoning:

Looking at the game history with this co-player:

Round 3: They returned \$7.0 from \$13.2 (~53%)
Round 4: I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
Round 5: They returned \$6.0 from \$11.28 (~53%)

Current round: They sent \$4.73, I received \$14.19

They've been very cooperative and generous throughout. They've increased their sending and maintained good return rates. I should reciprocate appropriately.

Looking at their returns: around 53-54%
I'll return around 53-55% of what I received.

$\$14.19 * 0.53 = \7.52
 $\$14.19 * 0.54 = \7.66
 $\$14.19 * 0.55 = \7.80

I'll return \$7.5 which is about 53%, maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 2266,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.

Your total visible earnings across all rounds are \$39.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our highly cooperative relationship:

****Round 4:**** They sent \$4.0, I returned \$5.5
****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0, I returned \$7.5

This co-player has demonstrated:

- Consistent high trust (repeatedly sending \$4.0)
- Very generous returns
- Strong reciprocity

I'll continue matching their trust level:

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me track our interaction:

****Round 1:**** I sent \$3, they returned \$5.0
****Round 2:**** They sent \$3.5, I returned \$4.5

****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5
****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0 (I saw \$4.73 due to noise), I returned \$7.5 (they saw \$8.42)

Looking at the pattern:

- They've been consistently sending around \$3.5-\$4.0
- They've been returning generously
- Round 6 shows they saw \$8.42 returned from \$12.0, which is about 70% - very high!
- This is a strong collaborative partner

They've maintained sending \$4.0 in rounds 4 and 6. I should continue with a similar level of trust. I'll send \$4 again to maintain consistency and mutual trust.

```
Usage: {
  "input_tokens": 2640,
  "output_tokens": 381,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:01:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.39, that is 67.8% of their total endowment. You received \$10.17. How much do you return to the sender? (0-10.17)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.39 (67.8% of their endowment)
- I received \$10.17 after the 3x multiplication
- The sender currently has \$1.61 ($\$5 - \3.39)

I want to reward this trust appropriately while being fair. If I return \$5.00:

- The sender would end with \$6.61 (a profit of \$1.61)
- I would keep \$5.17

This seems like a reasonable reciprocation that rewards their trust with a solid profit, while I also benefit modestly from the exchange.

{'return': 5.00}

reasoning:

This is the first round of a trust game. Let me analyze the situation:

1. The sender sent \$3.39 out of \$5 (67.8% of their endowment)
2. This was multiplied by 3x, so I received \$10.17
3. I need to decide how much to return (between \$0 and \$10.17)

Key considerations:

- The sender showed considerable trust by sending 67.8% of their endowment
- There's mention of "communication noise" - the amounts may differ slightly from what was actually sent
- This is a multi-round game with multiple agents, and roles rotate
- Building reputation and reciprocity could matter for future rounds

Let me think about a fair return:

- The sender sent \$3.39 and will have $\$5 - \$3.39 = \$1.61$ left if I return nothing
- If I want to be generous and split gains roughly equally:
 - Total value created: \$10.17 (what I received)
 - Original sender investment: \$3.39
 - Net gain from multiplication: $\$10.17 - \$3.39 = \$6.78$
 - If we split this gain equally: $\$6.78 / 2 = \3.39 each
 - Sender should end with: \$1.61 (kept) + return
 - For equal split of gains: sender gets back \$3.39, ending with $\$1.61 + \$3.39 = \$5.00$
 - I would keep: $\$10.17 - \$3.39 = \$6.78$

Actually, let me reconsider. A common approach is:

- Equal split of total money: $(\$5 \text{ initial} + \$10.17 \text{ received}) / 2 = \7.585 each

- Sender currently has: \$1.61
- So I should return: $\$7.585 - \$1.61 = \$5.975$, approximately \$6.00

I could also return what they sent (\$3-4) to reward their trust, or go more generous and return \$4-5 so they end up ahead of keeping everything. Given their substantial trust and the rotating partnership structure, I should lean toward generosity to encourage cooperation in future rounds.

Returning \$4.50 seems like the right balance-it gives them a \$1.11 profit while I keep \$5.67, which feels more equitable than other options. This rewards their faith without leaving me with too little....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1053,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)

- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: $\$5 - \$4.4 = \$0.60$

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: $\$0.60 + \$7.0 = \$7.60$ (profit of \$2.60)
- I keep: $\$13.20 - \$7.0 = \$6.20$

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

{'send': 4.0}

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts

ts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.

Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

{'return': 6.0}

reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receiver, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 1873,
  "output_tokens": 414,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$30.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Co-player has consistently shown trust, sending \$3.5 and \$4.0 (70-80% of endowment)
- As receiver, they returned fairly (\$5.5)
- They've seen generous returns from me (\$7.3, \$6.52)
- We've established a strong cooperative pattern over multiple rounds

Given this reliable cooperation, I should maintain high trust. I'll send \$4.0 (80% of endowment) to continue our successful partnership.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$3.5, saw \$7.3 returned, payoff \$8.8
- Round 4: They saw \$3.55 sent (noisy version of my \$4.0), returned \$5.5, payoff \$5.15
- Round 5: They sent \$4.0, saw \$6.52 returned (noisy version of my \$6.0), payoff \$7.52

The pattern shows:

- Co-player has been sending good amounts as sender (\$3.5, \$4.0)
- As receiver, they've been fair (\$5.5 return)
- They've seen me return generously (\$7.3, \$6.52)
- We've maintained strong cooperation

This is a reliable, cooperative partner. I should continue showing high trust. Since they've been sending 70-80% of endowment, I should match or slightly exceed that level.

I'll send \$4.0 again (80% of endowment) to maintain our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
 - Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.
 - Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.
- Your total visible earnings across all rounds are \$40.28.

This round, you are the RECEIVER against your current co-player. They sent you \$4.03, so you received \$12.09. How much do you return? (0-12.09)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Co-player has been highly cooperative, consistently sending high amounts (80%+ of endowment)
- As receiver, they've returned fairly (\$5.5, \$7.5)
- They sent \$4.03 this round (80.6%), showing strong continued trust
- They currently have \$0.97

This is very similar to Round 5 when they sent \$4.0. I should maintain consistency in our cooperative relationship. If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This appropriately rewards their ongoing trust.

{'return': 6.5}

Reasoning:

Looking at the visible history with this co-player:

- Round 4: They received, returned \$5.5
- Round 5: They sent \$4.0 (80% of endowment), saw \$6.52 returned
- Round 6: They received, returned \$7.5
- Round 7 (current): They sent \$4.03 (80.6% of endowment)

This co-player has been very cooperative throughout. They're consistently sending high amounts (around 80%) and returning fairly when they're the receiver.

Current situation:

- They sent \$4.03, so they have \$0.97 left
- I received \$12.09

I should continue reciprocating their high level of trust. They've been a reliable partner.

If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This seems fair given their high trust level. Let me think if this is appropriate...

Actually, looking at the pattern, when they sent \$4.0 in Round 5, I returned around \$6.52 (visible). Now they sent \$4.03, very similar. I should be consistent and return a similar amount.

\$6.5 seems reasonable and consistent with our established pattern.

Usage: {

"input_tokens": 2618,
"output_tokens": 464,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:01:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_6",

```

"pairing": {
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution since I have no history with this co-player.

Sending creates value (3x multiplier), but requires trusting the receiver to return a fair amount. Since this is a multi-round game where roles rotate, establishing a cooperative pattern could be beneficial even with changing partners.

I'll send a moderate amount to show trust while limiting risk:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return roughly half (\$4-5), we both benefit compared to sending nothing. This seems like a reasonable opening move to test cooperation.

reasoning:

This is the first round of a trust game. I'm the sender and need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are communication noise effects mentioned, so amounts may differ slightly
- This is a multi-round game with multiple agents and changing pairings
- Agents will play both roles across rounds

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no additional value
- Sending \$5 creates \$15 for the receiver, maximizing potential gains
- The question is whether the receiver will return enough to make it worthwhile

Since this is the first round and I have no information about this particular co-player:

- A moderate amount might signal trust while limiting risk
- However, in trust games, full cooperation often leads to best outcomes if reciprocated
- Since roles rotate and there are repeated interactions (though with different partners), building a reputation for trust might be beneficial

Common strategies in trust games:

1. Send half (\$2.50) - moderate trust
2. Send all (\$5) - full trust, maximizing joint value
3. Send nothing (\$0) - no trust/risk aversion

Given that:

- This is an experimental setting where cooperation might be valued
- Future rounds exist where I'll be the receiver
- Maximum value creation requires sending the full amount

I'll opt for a trusting approach and send \$4. This shows significant trust while keeping a small buffer. Actually, let me reconsider - in trust games with multiple rounds, establishing trust early can be beneficial. I'll send \$5 to maximize the pie and hope for reciprocation.

Actually, let me be more strategic. Since there's no reputation effect with the same partner (pairings randomize), each round is more like a one-shot game. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 707,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.
4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

usage:

```
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.4700000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:

- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:

- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1081,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).

I'll match their generous pattern:

```
'''json
{"return": 4.5}
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:

- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5
- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (generous!)

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of what they received

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with that proven strategy.

usage:

```
{
  "input_tokens": 1778,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$27.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.4, so you received \$7.199999999999999. How much do you return? (0-7.199999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

Round 3: They returned \$4.5 out of \$8.49

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69 (very generous!)

This round they sent \$2.4 (becoming \$7.20) - slightly less than their usual \$3.0. I'll return a proportionate amount while maintaining our cooperative relationship:

```
'''json
{"return": 3.5}
'''
```

This gives them a reasonable return while accounting for the slightly lower send amount.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 out of \$8.49

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69

They've been very cooperative and generous. Now they sent \$2.4 (becoming \$7.20).

This is the first round where they sent less than \$3.0. Let me check the pattern:

- When they sent \$3.0, I returned around \$4.5
- When I sent (visible as \$2.83, \$3.23), they returned \$4.5 and \$5.0

Now they sent \$2.4. I should return a fair amount. Let me calculate:

- They sent \$2.4, so they kept \$2.6
- It became \$7.20 for me
- If I return around 50-55% of what I received, that's \$3.6-\$4.0
- Their final payoff would be: \$2.6 + return
- My final payoff would be: \$7.20 - return

If I return \$3.5:

- They get: \$2.6 + \$3.5 = \$6.1
- I get: \$7.20 - \$3.5 = \$3.7

If I return \$4.0:

- They get: \$2.6 + \$4.0 = \$6.6
- I get: \$7.20 - \$4.0 = \$3.2

Given they sent slightly less this round, I'll return a proportional amount. Since they've been very generous when they were receiver, I'll still be generous but slightly adjust for their lower send.

\$3.5 seems fair - it's about 49% of what I received, reasonable reciprocity.

usage:

```
{
  "input_tokens": 2139,
  "output_tokens": 574,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$30.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing history:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 (very generous!)

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently trustworthy and cooperative throughout. I'll continue our successful pattern:

```
'''json
{"send": 3}
```

'''

Reasoning:

Looking at our history with this co-player:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently cooperative throughout our interactions. They've consistently sent around \$3.0 when they're the sender and returned generously as receiver. I should continue our successful cooperative pattern.

I'll send \$3 again to maintain our established trust.

Usage: {

"input_tokens": 2499,

"output_tokens": 224,

"reasoning_tokens": 0

}

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:01:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_3",

"move_index": 5,

"opponent_id": "Agent_5",

"pairing": {

"agents": [

"Agent_5",

"Agent_6"

],

"investor": "Agent_5",

"roles": {

"Agent_5": "investor",

"Agent_6": "trustee"

},

"trustee": "Agent_6"

},

"role": "trustee",

"role_label": "Receiver",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.08, that is 41.6% of their total endowment. You received \$6.24. How much do you return to the sender? (0-6.24)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.08 (about 42% of their endowment)
- I received \$6.24 after the 3x multiplier
- The sender currently has \$2.92 remaining

I want to reward their trust while being fair. The multiplication created new value, and both of us should benefit from the cooperation.

If I return \$3.50:

- Sender ends up with: $\$2.92 + \$3.50 = \$6.42$ (profit of \$1.42)
- I end up with: $\$6.24 - \$3.50 = \$2.74$ (profit of \$2.74)

This seems like a reasonable split where we both gain from the exchange, and the sender is rewarded for showing trust. This should help build positive reciprocity for future rounds.

{'return': 3.50}

reasoning:

Let me think about this carefully. I'm the RECEIVER in Round 1.

The sender sent \$2.08 out of their \$5 endowment (which is 41.6% as stated). I received $\$2.08 \times 3 = \6.24 .

Now I need to decide how much to return.

Let me think about this strategically:

1. The sender showed trust by sending \$2.08 (about 42% of their endowment)
2. This is a multi-round game with 8 agents, and pairings change each round
3. There's communication noise, so amounts may appear different than actual
4. Building reputation for reciprocity could help in future rounds

If I return nothing:

- Sender gets: $\$5 - \$2.08 + \$0 = \2.92
- I get: $\$6.24 - \$0 = \$6.24$
- This seems unfair and won't build trust

If I return half (\$3.12):

- Sender gets: $\$5 - \$2.08 + \$3.12 = \6.04
- I get: $\$6.24 - \$3.12 = \$3.12$
- Sender gains \$1.04, I gain \$3.12 (relative to starting positions)

If I return enough so the sender breaks even or makes a small profit:

- For sender to break even at \$5: need to return \$2.08
- For sender to profit equally to me: they need to end up with a proportional gain

Let me think about fairness. The total pie grew from \$5 to $\$5 + \$6.24 = \$11.24$.

If we split the gains equally:

- Total gain from cooperation: \$6.24 (the tripled amount minus the original \$2.08 that was just transferred)
- Wait, let me recalculate...
- Sender now has \$2.92, I have \$6.24, so the system total is \$9.16. The multiplication created an extra \$4.16 in value ($\$6.24 - \2.08). If I want to split this gain fairly, I should consider returning enough to give the sender a proportional share of the surplus created by their trust.

One option: return \$2.08 to the sender, bringing them back to \$5 while I keep \$4.16. But that rewards me entirely for the multiplication effect. Another angle is thinking about what return rate feels equitable-perhaps returning enough so the sender sees a modest profit, which encourages future cooperation. In trust games, common approaches include returning a portion that acknowledges their initial risk while letting me benefit from facilitating the transaction.

If I return \$3, the sender profits \$0.92 while I keep \$3.24. If I return \$4, they profit \$1.92 and I keep \$2.24. The larger return feels more balanced since it acknowledges their trust without me capturing all the gains from the multiplication.

Looking at this differently, the exchange creates \$4.16 in total benefit. Splitting that equally would mean each of us gains about \$2.08 beyond our starting amounts. That would put the sender at \$5.00 and me at around \$4.16. A \$3.50 return gets closer to that fair split-they'd end up with \$6.42 and I'd have \$2.74, which feels more genuinely reciprocal for establishing trust in future rounds....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1643,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER

- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be $\$2.0 + \$3.88 = \$5.88$

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

```
{'return': 4.50}
```

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have \$5 - \$2.83 = \$2.17.

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)
- ...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{
  "input_tokens": 1611,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant**content:**

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)
- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)

- I end with: \$4.69 (profit of \$4.69)

{'return': 5.0}

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)
Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)
Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)
Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)
Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: $\$1.77 + \$5.0 = \$6.77$ (profit of \$1.77)
- I end with: $\$9.69 - \$5.0 = \$4.69$ (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

usage:

```
{
  "input_tokens": 1982,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.

Your total visible earnings across all rounds are \$23.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 3 and 5
- As receiver: returned \$4.5 (52%) in round 4

We've established a strong pattern of mutual trust and fair reciprocity. I should continue this successful cooperation by sending the same amount.

{'send': 3.0}

reasoning:

Looking at my co-player's history:

- Round 3: as sender, sent \$3.0 (60%)
- Round 4: as receiver, returned \$4.5 out of \$8.64 (52%)
- Round 5: as sender, sent \$3.0 (60%)

They've been consistently sending 60% when they're the sender, and returning around 50-52%

when they're the receiver. This is a very cooperative and trustworthy pattern.

I've been sending \$3.0 (60%) consistently as well, and we have a strong mutual cooperation going.

I should continue with the same level of trust - sending \$3.0 (60%) to maintain our successful partnership.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 265,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.699999999999993.

Your total visible earnings across all rounds are \$28.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

My co-player has been consistently cooperative and just sent \$3.46 (69% of endowment), their highest trust level yet! They currently have \$1.54 remaining.

I've been consistently returning around 52-56% of what I receive. I should maintain this fair reciprocity to reward their increased trust.

If I return \$5.5:

- Sender ends with: \$7.04 (profit of \$2.04)
- I end with: \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains our strong cooperative relationship.

{'return': 5.5}

Reasoning:

Looking at the history with my co-player:

Round 4: I was sender, they were receiver - saw \$2.88 sent, returned \$4.5 out of \$8.64 (52%)

Round 5: They were sender, I was receiver - sent \$3.0, I returned visible \$5.62

Round 6: I was sender, they were receiver - saw \$2.4 sent (but I sent \$3.0 due to noise), returned \$3.5 out of \$7.2 (49%)

Round 7: They are sender, I am receiver - sent \$3.46 (69% of endowment!), I received \$10.38

They've increased their trust level to 69%, the highest yet. They currently have \$1.54 remaining.

I should reward this high trust. Looking at my previous returns:

- Round 1: Returned \$3.50 out of \$6.24 (56%)
- Round 3: Returned \$4.50 out of \$8.49 (53%)
- Round 5: Returned \$5.0 out of \$9.69 (52%)

I've been returning around 52-56% consistently. Given they sent an even larger proportion this time (69%), I should maintain or slightly increase my return rate.

If I return \$5.5:

- Sender ends with: \$1.54 + \$5.5 = \$7.04 (profit of \$2.04)
- I end with: \$10.38 - \$5.5 = \$4.88 (profit of \$4.88)

- Return rate: 53%

This maintains my consistent cooperative pattern and rewards their increased trust.

```
Usage: {  
  "input_tokens": 2715,  
  "output_tokens": 553,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:01:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to signal cooperation while being somewhat cautious since I don't yet know how other agents will behave.

Sending a moderate amount allows us both to benefit if my co-player reciprocates. If I send \$3, they'll receive \$9 (tripled). Even if they return just \$3, we'd both end up with more than we started with (\$5 for me, \$6 for them).

This shows willingness to trust while limiting my risk in this initial round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender, which means I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential mutual benefit
2. However, the receiver may not return much or anything at all
3. There's communication noise, so amounts may be slightly different than what I actually send/return
4. This is a multi-round game with 8 agents, and pairings are randomized each round
5. Building a reputation for cooperation might be beneficial in future rounds

Since this is the first round and I don't have any information about the other agents' behavior yet, I should consider:

- Sending \$0 is safe but doesn't create mutual benefit
- Sending everything (\$5) shows maximum trust but is very risky
- Sending a moderate amount (like \$2-3) shows willingness to cooperate while limiting risk

Given that:

- This is a multi-agent setting where reputation might matter
- There's noise in communication
- I want to signal cooperation but also be cautious in the first round

I'll send \$3. This:

- Shows willingness to cooperate
- Gives the receiver \$9 (3×3)
- Leaves me with \$2 as protection
- If they return even \$3, we both benefit (\$2+\$3=\$5 for me, \$9-\$3=\$6 for them)

This seems like a reasonable, moderately trusting opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings: $\$5 - \$3 + \$5.5 = \7.50
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with $\$1 + \$5 = \$6$, and I keep \$6.88
- If I return \$6, they end with $\$1 + \$6 = \$7$, and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

usage:

```
{
  "input_tokens": 1079,
  "output_tokens": 370,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.

Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)
 Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)
 Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)
 Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

usage:

```
{
  "input_tokens": 1444,
  "output_tokens": 410,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)
- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

usage:

```
{
  "input_tokens": 1810,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.

Your total visible earnings across all rounds are \$32.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at the pattern:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent \$4.18 (more than their usual \$3.5-3.75), showing increased trust. I received \$12.54. I should reciprocate this increased trust with a proportionally fair return:

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent me \$4.18, which is higher than their usual amounts (around \$3.5-3.75). They received \$12.54.

They're showing increased trust by sending more. I should reciprocate appropriately. Given they sent more, I should return a bit more to maintain the cooperative relationship.

Our typical pattern has been returning around \$5-5.5 from amounts around \$10-11. Now I have \$12.54, so returning around \$6 would be proportionate and fair, maintaining the cooperative spirit.

usage:

```
{
  "input_tokens": 2175,
  "output_tokens": 314,
}
```

```

    "reasoning_tokens": 0
  }
Message 14 role=user
content:
  Round 7

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 4 against you, as SENDER: sent $3.75, it became $11.25, saw $5.44 returned, visible payoff $6.69.
  - Round 5 against you, as RECEIVER: saw $3.41 sent, it became $10.23, returned $5.25, visible payoff $4.98.
  - Round 6 against you, as SENDER: sent $3.75, it became $11.25, saw $5.32 returned, visible payoff $6.57.
  Your total visible earnings across all rounds are $39.529999999999994.

  This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at our continued cooperation:
  - Round 4: They sent $3.75, I returned $5.5
  - Round 5: I sent $3.75, they returned $5.25
  - Round 6: They sent $3.75, I returned $6.0

  We've maintained a very stable and mutually beneficial pattern. I'll continue with the same strategy:

  '''json
  {"send": 3.75}
  '''
Reasoning:
  Looking at the recent history:
  - Round 4: They sent $3.75, I returned $5.5
  - Round 5: I sent $3.75, they returned $5.25
  - Round 6: They sent $3.75, I returned $6.0

  Wait, looking at my visible history, there's some inconsistency. In Round 6, it says "They sent you $4.18" but now the history says they sent $3.75. This is probably the communication noise mentioned in the instructions.

  Our pattern has been very stable with sends around $3.5-3.75 and returns around $5.0-5.5. We've built strong mutual cooperation. I should continue with this pattern:

  '''json
  {"send": 3.75}
  '''
Usage: {
  "input_tokens": 2520,
  "output_tokens": 287,
  "reasoning_tokens": 0
}

```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:01:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",

```

```

    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.99, that is 79.8% of their total endowment. You received \$11.97. How much do you return to the sender? (0-11.97)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.99 (nearly 80% of their endowment)
- I received \$11.97 after the 3x multiplier
- The sender currently has only \$1.01 left

I want to reciprocate this trust while ensuring both of us benefit from cooperating.

If I return \$5.50:

- The sender will end with: $\$1.01 + \$5.50 = \$6.51$ (better than their initial \$5)
- I will end with: $\$11.97 - \$5.50 = \$6.47$ (also a good outcome)

This creates a roughly equal split of the total gains and rewards the sender's generous trust.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$3.99 (which is 79.8% of their \$5 endowment)
- This was multiplied by 3x, so I received \$11.97
- I need to decide how much to return (between \$0 and \$11.97)

Key considerations:

1. The sender was quite generous, sending nearly 80% of their endowment
2. The sender will keep: $\$5 - \$3.99 + \text{my_return} = \$1.01 + \text{my_return}$
3. I will keep: $\$11.97 - \text{my_return}$

This is a trust game, and the sender showed significant trust by sending almost all their money. A fair approach would be to:

- Consider that the sender was very generous
- Try to ensure both parties benefit from the cooperation
- Perhaps aim for a roughly equal split of the total gains

If I return nothing:

- Sender ends with: \$1.01
- I end with: \$11.97

If I return \$5:

- Sender ends with: \$6.01
- I end with: \$6.97

If I return \$6:

- Sender ends with: \$7.01
- I end with: \$5.97

A reasonable approach might be to return enough so that:

1. The sender ends up with more than they started with (more than \$5)
2. I also end up with more than the sender (since I'm the one multiplying the money)
3. Both parties benefit from cooperation

Let me think about what would be "fair":

- Total money in the system: \$5 (original) + \$11.97 (after multiplication) - \$5 (sender's initial endowment not sent out) = \$11.97 received by me, sender has \$1.01
- Total available: \$1.01 (sender kept) + \$11.97 (I received) = \$12.98
- A 50-50 split would mean each of us gets around \$6.49, which would require me to return \$5.48

I want to acknowledge the sender's generosity while still benefiting from my role as receiver. Returning around \$5.50 would create an almost equal outcome where both of us end up with roughly \$6.50, rewarding their trust without leaving me worse off than them....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible p

ayoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with \$2.0 + \$5.61 = \$7.61

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 851,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$

- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$
- I end with: $\$10.23 - \$5.25 = \$4.98$

```
'''json
{"return": 5.25}
'''
```

reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5

Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)

Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5

Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$
- I end with: $\$10.23 - \$5.25 = \$4.98$

This is fair and consistent with our pattern.

usage:

```
{
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.

Your total visible earnings across all rounds are \$30.390000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant**content:**

Looking at our ongoing cooperation:

- Round 3: Co-player sent \$3.5
- Round 4: I sent \$3.75, they returned \$5.5

- Round 5: Co-player sent \$3.75

We've established a very strong cooperative pattern. Both of us consistently send high amounts (around \$3.50-\$3.75) and return fair shares. This has been mutually beneficial - my total earnings of \$30.39 reflect this.

I'll continue this successful cooperation:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with my co-player:

Round 3: Co-player sent \$3.5, I returned (they saw \$5.95)

Round 4: I sent \$3.75, co-player returned \$5.5

Round 5: Co-player sent \$3.75, I returned (they saw \$5.21)

We've been maintaining a very cooperative relationship. Both of us have been sending high amounts (around \$3.50-\$3.75) and returning fair shares.

My earnings are \$30.39 total, which suggests this cooperation is working well.

The co-player has consistently sent high amounts (\$3.5, \$3.75) showing strong trust. They've also been fair when receiving, returning \$5.5 consistently.

I should continue this cooperative pattern by sending a high amount. I'll send \$3.75 to match what they've been sending.

usage:

```
{
  "input_tokens": 2386,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.

Your total visible earnings across all rounds are \$36.960000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.57, so you received \$13.71. How much do you return? (0-13.71)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Round 4: Co-player returned \$5.5 from \$10.59
- Round 5: Co-player sent \$3.75
- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$4.57 (91.4% of their endowment!)

This is a very strong trust signal - they sent almost everything! I should reciprocate this high level of trust with a generous return.

If I return \$7.00:

- Sender ends with: \$0.43 + \$7.00 = \$7.43
- I end with: \$13.71 - \$7.00 = \$6.71

This rewards their exceptional trust and maintains our strong cooperative relationship.

```
'''json
{"return": 7.00}
'''
```

'''

Reasoning:

Looking at the history:

Round 4: Co-player as receiver, returned \$5.5 from \$10.59

Round 5: Co-player as sender, sent \$3.75

Round 6: Co-player as receiver, they saw I sent \$4.18 which became \$12.54, and they returned \$6.0

Round 7: Co-player sent me \$4.57 (91.4% of their endowment) - this is a very high trust signal!

They've been escalating their cooperation. In Round 6, they returned \$6.0 from \$12.54 (about 48%). Now they're sending \$4.57, which is much higher than their previous sends.

I received \$13.71 after the multiplier.

They currently have: \$5 - \$4.57 = \$0.43

If I return \$6.50:

- Sender ends with: \$0.43 + \$6.50 = \$6.93

- I end with: \$13.71 - \$6.50 = \$7.21

This rewards their very high trust and continues our cooperative relationship.

Actually, they sent 91.4% - that's a very strong trust signal. Let me be more generous to match that.

If I return \$7.00:

- Sender ends with: \$0.43 + \$7.00 = \$7.43

- I end with: \$13.71 - \$7.00 = \$6.71

This is a bit more balanced and rewards the very high send amount.

```
Usage: {
  "input_tokens": 2753,
  "output_tokens": 564,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

dyad_2: Agent_4=Sender, Agent_3=Receiver

dyad_3: Agent_6=Sender, Agent_5=Receiver

dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4.5; received \$13.5; Agent_1 returned \$7; payoffs Agent_2=\$7.5,

Agent_1=\$6.5; noise/visible: sent visible=\$4.27, received visible=\$12.81, returned visible=\$6.14

Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 13.5, "min": 0, "raw": 7.0}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}

dyad_2: Agent_4 sent \$4; received \$12; Agent_3 returned \$6.5; payoffs Agent_4=\$7.5, Agent_3=\$5.5;

noise/visible: sent visible=\$3.88, received visible=\$11.64, returned visible=\$5.56

Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$5.5; payoffs Agent_6=\$7.5, Agent_5=\$3.5;

noise/visible: sent visible=\$3.47, received visible=\$10.41, returned visible=\$5.37

Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3.75; received \$11.25; Agent_7 returned \$6.5; payoffs Agent_8=\$7.75,

Agent_7=\$4.75; noise/visible: sent visible=\$4.33, received visible=\$12.99, returned visible=\$5.92

Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 11.25, "min": 0, "raw": 6.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 48.6,

"Agent_2": 50.4,

"Agent_3": 52.5,

"Agent_4": 47.5,

```

"Agent_5": 45.5,
"Agent_6": 42.5,
"Agent_7": 49.5,
"Agent_8": 48.0
}
Visible balances after round: {
  "Agent_1": 49.89999999999999,
  "Agent_2": 50.860000000000001,
  "Agent_3": 52.81999999999999,
  "Agent_4": 52.430000000000001,
  "Agent_5": 42.379999999999995,
  "Agent_6": 41.21,
  "Agent_7": 54.57999999999999,
  "Agent_8": 50.840000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:02:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.

Your total visible earnings across all rounds are \$5.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player sent \$3.0 last round (60% of their endowment) showing significant trust
- They saw \$3.34 returned and earned \$5.34, a modest but positive outcome
- This suggests they're willing to cooperate

****My strategy:****

Since they demonstrated trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain cooperation. Sending \$3.00 (60% of my endowment) mirrors their behavior and signals I'm willing to cooperate.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me analyze this situation carefully:

****What I know:****

- Round 1: I was the RECEIVER
 - Sender sent \$2.56 (51.2% of endowment)
 - I received \$7.68
 - I returned \$2.60
 - My earnings: \$5.08

****What my current co-player sees (their history):****

- Round 1 against me, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34

****Important observation about communication noise:****

There's a significant discrepancy here!

- I actually received notification that sender sent \$2.56
- But my co-player thinks they sent \$3.0

- I returned \$2.60
- But my co-player saw \$3.34 returned

This is the "communication noise" mentioned in the instructions. The amounts shown may differ from what was actually sent/returned.

****Analysis for this round:****

Now I'm the SENDER in Round 2, playing against the same person.

From their perspective:

- They sent \$3.0 (60% of their endowment) in Round 1
- They saw \$3.34 returned (37.1% of the \$9.0 I received)
- They earned \$5.34, which is slightly above their initial \$5

From my actual perspective:

- They sent \$2.56 (51.2%)
- I returned \$2.60 (33.9% of \$7.68)
- They should have gotten \$5.04

The co-player might view the return as somewhat modest given they sent 60% of their endowment (from their perspective). But they still made a small profit.

****My strategy:****

Given that:

1. They showed reasonable trust by sending about half (or what they thought was 60%)
2. They received a modest but positive return on that trust
3. This is a multi-round game where reputation matters
4. There's communication noise, so I should be consistent with cooperative behavior

I should send a similar or slightly higher proportion to maintain cooperation. Sending around 50-60% seems appropriate. I'll send \$3.00 (60% of my endowment)....

usage:

```
{
  "input_tokens": 876,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

****Current situation:****

- They sent \$3.54, keeping only \$1.46
- I received \$10.62
- Total in play: \$1.46 + \$10.62 = \$12.08

****My decision:****

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:
- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

****History with this co-player:****

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

****Current Round 3:****

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have: $\$5 - \$3.54 = \$1.46$

****Analysis:****

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:
- Sender gets: $\$1.46 + \$4.50 = \$5.96$
- I get: $\$10.62 - \$4.50 = \$6.12$

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:
- Sender gets: $\$1.46 + \$4.20 = \$5.66$
- I get: $\$10.62 - \$4.20 = \$6.42$

Or if I return \$4.80:
- Sender gets: $\$1.46 + \$4.80 = \$6.26$
- I get: $\$10.62 - \$4.80 = \$5.82$

...

usage:

```
{
  "input_tokens": 1218,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness . I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34
Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38
Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

usage:

```
{
  "input_tokens": 1671,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

****My decision:****

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

****My decision:****

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:

- Sender gets: \$1.1 + \$6.00 = \$7.10

...

usage:

```
{
```

```
"input_tokens": 2103,
"output_tokens": 750,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.

Your total visible earnings across all rounds are \$29.360000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown strong, sustained cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: As receiver, generously returned \$5.0
- Round 5: Sent \$4.0 (80% - very high trust!)

****Observations:****

They've consistently demonstrated trust, culminating in sending 80% of their endowment in Round 5. Despite their payoff decreasing slightly in that round (\$6.9 ? \$5.93), they haven't pulled back or changed strategy.

****My decision:****

This partner has earned my trust through consistent cooperation. I should match their high-trust approach by sending a similarly substantial amount.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been very cooperative and trustful
- Round 3: As sender, sent \$3.0 (60%), saw \$4.9 returned, payoff \$6.9
- Round 4: As receiver, returned \$5.0 from \$11.46, payoff \$6.46
- Round 5: As sender, sent \$4.0 (80% - very high!), saw \$4.93 returned, payoff \$5.93

****Observations:****

1. They increased their sending from \$3.0 to \$4.0 (80%!)
2. However, in Round 5, despite sending 80%, they only saw \$4.93 returned
3. Their payoff went from \$6.9 to \$5.93 - a decrease
4. As receiver in Round 4, they were generous returning \$5.0

****Current situation:****

This player has shown tremendous trust, especially sending \$4.0 (80%) in Round 5. Even though they saw a lower return than expected, they've been consistently cooperative throughout.

I should continue reciprocating their trust. They've demonstrated willingness to send high amounts. I should send a similarly high amount to maintain the cooperative relationship.

Sending \$4.0 (80%) matches their recent high-trust behavior.

```
'''json
```

```
    {"send": 4.00}
    ,,,
```

usage:

```
{
  "input_tokens": 2590,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.

Your total visible earnings across all rounds are \$37.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown extraordinary escalating trust
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, generously returned \$7.0 from \$13.5
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42
- I received \$13.74
- Total in play: \$0.42 + \$13.74 = \$14.16

****My decision:****

This player has shown exceptional trust by sending over 90% of their endowment - putting themselves in a very vulnerable position. They've also been consistently generous as a receiver. I should reciprocate this extraordinary trust.

If I return \$7.50:

- Sender gets: \$0.42 + \$7.50 = \$7.92
- I get: \$13.74 - \$7.50 = \$6.24

This rewards their exceptional trust with a strong outcome.

```
'''json
{"return": 7.50}
'''
```

reasoning:

Looking at the history:

****My co-player's recent actions:****

- Round 4: As receiver, returned \$5.0 from \$11.46
- Round 5: As sender, sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (52% of what they received - quite generous!)
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42
- I received \$13.74
- Total in play: \$0.42 + \$13.74 = \$14.16

****Analysis:****

This player has shown exceptional and escalating trust:

- They've been generous as a receiver (returned \$5.0, then \$7.0)
- They've been increasingly trusting as a sender (\$3.0 ? \$4.0 ? \$4.58)
- They're now sending 91.6% of their endowment!

I should reciprocate this exceptional trust generously. They've really put themselves in a vulnerable position with only \$0.42 left.

If I return \$7.50:

- Sender gets: $\$0.42 + \$7.50 = \$7.92$
- I get: $\$13.74 - \$7.50 = \$6.24$

If I return \$7.00:

- Sender gets: $\$0.42 + \$7.00 = \$7.42$
- I get: $\$13.74 - \$7.00 = \$6.74$

Given their extreme trust (91.6%!), I should return about \$7.50 to give them a solid payoff of about \$7.92.

Actually, let me think about this more. They're taking huge risk. Maybe I should be even more generous.

If I return \$8.00:

- Sender gets: $\$0.42 + \$8.00 = \$8.42$

...

usage:

```
{
  "input_tokens": 3011,
  "output_tokens": 788,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.
- Round 7 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$8.08 returned, visible payoff \$8.58.

Your total visible earnings across all rounds are \$44.220000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has been exceptionally cooperative throughout our partnership
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (generous)
- Round 7: Sent \$4.5 (90%), achieved excellent payoff of \$8.58

****Observations:****

Our partnership has been highly successful. They've maintained:

1. Very high trust as sender (80-90%)
2. Generous returns as receiver
3. Strong positive outcomes for both of us

****My decision:****

This partnership has been mutually beneficial. They've consistently demonstrated exceptional trust, and I should match their 90% sending rate to maintain this successful cooperation.

```
'''json
{"send": 4.50}
'''
```

Reasoning:

Looking at the history:

****Analysis:****

- My co-player has been extremely cooperative throughout
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (very generous)
- Round 7: Sent \$4.5 (90%), saw \$8.08 returned with a great payoff of \$8.58

****Observations:****

This has been a highly successful partnership. They've consistently demonstrated:

1. High trust as sender (80-90% sending rates)
2. Generous reciprocation as receiver
3. Their payoff in Round 7 was excellent (\$8.58), showing our cooperation is mutually beneficial

****My decision:****

Given their exceptional trust and the success of our partnership, I should send a high amount. They've been sending \$4.0-\$4.5 consistently. I should match this level of trust.

This is Round 8, possibly the final round or close to it. I should send a high amount to:

1. Reciprocate their consistent trust
2. Maintain the cooperative relationship
3. Maximize joint payoff

I'll send \$4.50 (90%) to match their high-trust approach.

```
Usage: {
  "input_tokens": 3219,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:02:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:**Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.

Your total visible earnings across all rounds are \$5.34.

This round, you are the RECEIVER against your current co-player. They sent you \$2.46, so you received \$7.38. How much do you return? (0-7.38)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****History Analysis:****

- In Round 1, my co-player saw \$2.56 sent (likely noise from my \$3), received \$7.68
- They returned \$2.6, which is about 34% of what they received
- This shows they're willing to reciprocate but somewhat cautiously

****Current Round:****

- They sent me \$2.46 (kept \$2.54)
- I received \$7.38
- If I return nothing, I keep \$7.38, they end with \$2.54
- If I return fairly, both should benefit from the multiplication

****My Decision:****

I'll return slightly more than their return rate to encourage cooperation and signal I'm trustworthy. Returning \$3.00 gives them back their sent amount plus a nice profit, and I still keep \$4.38, which is more than my initial endowment would be.

```
{"return": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 (or at least I thought I did)
2. But my co-player saw \$2.56 sent (communication noise!)
3. They received \$7.68 (which is $\$2.56 \times 3$)
4. They returned \$2.6
5. My visible earnings are \$5.34

Let me verify: If I sent \$3, I kept \$2. If they returned \$2.6, I should have $\$2 + \$2.6 = \$4.6$. But it shows \$5.34. This is confusing - perhaps the noise affected what I saw returned as well?

Actually, looking more carefully at the visible history, it says "visible payoff \$5.08" for them. Let me recalculate:

- They received \$7.68
- They returned \$2.6
- They kept: $\$7.68 - \$2.6 = \$5.08$?

My earnings of \$5.34 could be:

- I started with \$5
- I sent some amount (which appeared as \$2.56 to them)
- They returned something (which appeared as some amount to me)
- \$5.34 total

If I sent \$3 and kept \$2, and received back some amount X:

$$\$2 + X = \$5.34$$

$$X = \$3.34$$

So I received \$3.34 back (or that's what I saw).

But they returned \$2.6 (or what they thought they returned). The noise is affecting both what they saw sent and what I saw returned.

Now in Round 2:

- My co-player is now the SENDER
- They sent me \$2.46
- I receive \$7.38 (which is $\$2.46 \times 3$)
- I need to decide how much to return

My co-player returned what they thought was \$2.6 out of \$7.68 they received, which is about 34% return rate.

Based on their reciprocity pattern, returning around 34% of \$7.38 would be roughly \$2.51. Alternatively, I could aim for a fair split of the gains-they invested \$2.46, which tripled to \$7.38, creating \$4.92 in gains. Splitting that evenly means returning \$4.92 total. But from their perspective, they only kept \$2.54 after sending \$2.46, so I should consider what feels balanced given the communication errors affecting both rounds....

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 1244,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

****Current Situation:****

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

{"send": 3}

reasoning:

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.6200000000000001, returned \$4.5, visible payoff \$6.1200000000000001.

Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

****My Decision:****

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

{"return": 5.0}

reasoning:

Let me analyze the history:

****Round 1:**** Co-player as RECEIVER

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

****Round 2:**** Co-player as SENDER

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff: \$4.72

****Round 3:**** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

****Current Round 4:**** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

****Analysis:****

1. My co-player has been increasingly generous:

- Round 1: returned 34% (\$2.6 from \$7.68)
- Round 3: returned 42% (\$4.5 from \$10.62)

2. They also increased their sending:

- Round 2: sent \$3.0
- Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

****My Strategy:****

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:
 - They get: \$5 - \$3.82 + \$5.0 = \$6.18 total

...

usage:

```
{  
  "input_tokens": 1585,
```

```

    "output_tokens": 777,
    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 5

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $2.72 returned, visible payoff $4.7200000000000001.
  - Round 3 against you, as RECEIVER: saw $3.54 sent, it became $10.620000000000001, returned $4.5, visible payoff $6.1200000000000001.
  - Round 4 against you, as SENDER: sent $3.5, it became $10.5, saw $5.74 returned, visible payoff $7.24.
  Your total visible earnings across all rounds are $23.08.

  This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
  Looking at the history:

  **Pattern Analysis:**
  - Round 2: Co-player sent $3.0, saw $2.72 returned
  - Round 3: Co-player returned $4.5 from $10.62 (strong reciprocity)
  - Round 4: Co-player sent $3.5, saw $5.74 returned (even better returns from their perspective)

  **Current Situation:**
  The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from $3.0 to $3.5 after seeing better returns. From their perspective, cooperation is paying off.

  **My Decision:**
  I'll increase my send to $4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

  {"send": 4}
reasoning:
  Looking at the history:

  **Pattern Analysis:**
  - Round 2: Co-player sent $3.0, saw $2.72 returned (I returned $3.0 but noise made it $2.72)
  - Round 3: Co-player received visible $10.62, returned $4.5 (42%)
  - Round 4: Co-player sent $3.5, saw $5.74 returned (I returned $5.0 but noise made it $5.74)

  **Key Observations:**
  1. Co-player has been consistently trusting, sending $3.0, then $3.5
  2. From their perspective, I returned better in Round 4 ($5.74 on their $3.5 sent) than Round 2 ($2.72 on their $3.0 sent)
  3. They're showing increasing trust
  4. The noise has actually made my returns LOOK better in Round 4

  **My Strategy:**
  Given that:
  - They've been consistently cooperative
  - The trend is positive cooperation
  - They sent $3.5 last round and got good returns (from their perspective)
  - I should continue showing trust

  I'll send $4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.
usage:
  {
    "input_tokens": 2011,
    "output_tokens": 493,

```

```

    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as RECEIVER: saw $3.54 sent, it became $10.620000000000001, returned $4.5, visible payoff $6.120000000000001.
  - Round 4 against you, as SENDER: sent $3.5, it became $10.5, saw $5.74 returned, visible payoff $7.24.
  - Round 5 against you, as RECEIVER: saw $3.9 sent, it became $11.7, returned $5.5, visible payoff $6.199999999999999.
  Your total visible earnings across all rounds are $29.009999999999998.

  This round, you are the RECEIVER against your current co-player. They sent you $4.5, so you received $13.5. How much do you return? (0-13.5)
  Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
  Looking at the pattern:

  **History Analysis:**
  - Round 3: Co-player returned $4.5 from $10.62 (42%)
  - Round 5: Co-player returned $5.5 from $11.7 (47%)
  - Round 6: They sent $4.5 out of $5 (90% of their endowment!)

  **Current Situation:**
  - They sent $4.5, keeping only $0.50
  - I received $13.5
  - This is an extraordinary show of trust

  **My Decision:**
  Their extreme trust deserves strong reciprocity. Returning $7.0 means:
  - They end with $7.50 total (profit of $2.50)
  - I keep $6.50 (profit of $1.50)

  This rewards their faith in our cooperation while maintaining a fair distribution of gains
  .

  {"return": 7.0}
reasoning:
  Looking at the history:

  **Pattern Analysis:**
  - Round 3: Co-player returned $4.5 from $10.62 (42%)
  - Round 4: Co-player sent $3.5, saw $5.74 returned (good return)
  - Round 5: Co-player saw $3.9 sent, returned $5.5 from $11.7 (47%)

  **Current Situation:**
  - Round 6: They sent $4.5 (kept only $0.50!)
  - I received $13.5
  - This is their highest level of trust yet - sending 90% of their endowment

  **My Thinking:**
  They're showing extreme trust by sending $4.5. They've been consistently generous with their returns (42-47% return rates). I should reciprocate strongly to reward this trust.

  If I return $7.0:
  - They get back $7.0 + $0.50 kept = $7.50 total, profit of $2.50
  - I keep $13.5 - $7.0 = $6.50, profit of $1.50

  Actually, let me be even more generous given their extreme trust. If I return $8.0:
  - They get back $8.0 + $0.50 kept = $8.50 total, profit of $3.50
  - I keep $13.5 - $8.0 = $5.50, profit of $0.50

  Hmm, that might be going too far. Let me think about what's fair.

  They sent $4.5 out of $5, showing immense trust. A fair split would be:

```

- Total created value: \$13.5
- Original endowment lost: \$4.5
- Net gain to split: $\$13.5 - \$4.5 = \$9.0$
- If we split 50/50: each gets \$4.5
- So they should get: $\$0.5 \text{ (kept)} + \$4.5 \text{ (half of gain)} = \5.0 total
- I should return: $\$5.0 - \$0.5 = \$4.5$

But they've been returning at 42-47% rates, so I should match that generosity. 47% of \$13.5 is \$6.35. If I return \$6.5, we both end up with \$7.0 total and \$2.0 profit each-equal outcomes that acknowledge their trust. That feels right....

usage:

```
{
  "input_tokens": 2446,
  "output_tokens": 1046,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.

Your total visible earnings across all rounds are \$35.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (great returns!)

****Current Situation:****

The cooperation is thriving! From their visible perspective:

- They're getting excellent returns when they send (5.74, 7.62)
- They've shown strong reciprocity when receiving
- They increased their sending to \$4.0 in Round 6

****My Decision:****

I'll send \$4.5 to match their high trust level from Round 6 (when they sent \$4.5) and continue our strong cooperative relationship.

```
{"send": 4.5}
```

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (they saw even more returned than I actually sent - noise is working in my favor)

****Current Situation:****

From their visible perspective:

- They've been receiving excellent returns (5.74, 7.62)
- They've been generous when receiving (returned 5.5)
- Our relationship is highly cooperative
- They increased their sending from \$3.5 to \$4.0 after seeing good returns

****Trend:****

The cooperation is deepening. They're sending more, I'm returning generously (though noise sometimes makes it look even better). The visible payoffs show this is profitable for both of us.

****My Decision:****

I should match their high trust level. They sent \$4.0 last time and saw great returns. I should send at least as much or more. Given the strong cooperative relationship, I'll send \$4.5 to signal maximum trust and continue the positive trend.

```
{"send": 4.5}
```

usage:

```
{
  "input_tokens": 2881,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.
- Round 7 against you, as RECEIVER: saw \$4.58 sent, it became \$13.74, returned \$7.5, visible payoff \$6.24.

Your total visible earnings across all rounds are \$44.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****History Analysis:****

- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 8: They sent \$4.27 (kept only \$0.73)

****Current Situation:****

- They sent \$4.27, keeping minimal \$0.73
- I received \$12.81
- Continued high trust from them

****My Decision:****

Returning \$7.0 maintains our strong cooperative pattern:

- They end with \$7.73 total (profit of \$2.73)
- I keep \$5.81

This rewards their trust appropriately.

```
{"return": 7.0}
```

Reasoning:

Looking at the pattern:

****History Analysis:****

- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 8: They sent \$4.27 (kept only \$0.73)

****Current Situation:****

- They sent \$4.27, keeping only \$0.73
- I received \$12.81
- This shows continued high trust

****My Decision:****

The co-player has been consistently generous and trusting. They sent \$4.27, showing strong faith in our relationship despite noise in the game.

Returning \$7.0 means:

- They end with \$7.73 total (profit of \$2.73)
- I keep \$5.81 (profit of \$0.81)

This rewards their trust appropriately and maintains our cooperative relationship.

Actually, let me reconsider. Looking at the trend:

- Round 5: They returned \$5.5 from \$11.7
- Round 7: They returned \$7.5 from \$13.74

They've been generous. I should maintain similar generosity. Returning around \$7.0-7.5 seems appropriate given they sent \$4.27.

Let me go with \$7.0 as a consistent response to their high trust.

```
Usage: {
  "input_tokens": 3179,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:02:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.

Your total visible earnings across all rounds are \$5.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history, my co-player sent \$3.0 (60% of their endowment) in Round 1 and received a good return of \$5.75, making a profit.

Since they demonstrated trust and cooperation in the previous round, I should reciprocate with a similar level of trust. I'll send a comparable proportion of my endowment.

{'send': 3.50}

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent \$3.39 (67.8%). I received \$10.17 and returned \$5.00.
2. Now the visible history shows:
 - "Your current co-player's last 3 game(s): Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75"

This is interesting - there's communication noise! What I actually experienced was:

- Sender sent \$3.39
- I received \$10.17
- I returned \$5.00

But what my co-player (the sender from Round 1) sees is:

- They sent \$3.0
- It became \$9.0
- They saw \$5.75 returned
- Their visible payoff was \$7.75

So the noise affected both the amount sent and the amount returned!

My actual total earnings from Round 1: I kept \$5.17 (\$10.17 - \$5.00)

Now in Round 2, I'm the SENDER and playing against the same person who was the sender in Round 1.

From my perspective:

- They showed trust by sending 67.8% in Round 1
- They got back a good return (though I see it as \$5.00, they see it as \$5.75)
- They seemed cooperative

From their perspective:

- They sent \$3.0 (60% of endowment)
- They got back \$5.75, which is a good return
- They made a profit of \$2.75

I should reciprocate the trust they showed. They sent 67.8% (from my perspective) or 60% (from their perspective with noise). I should send a similar proportion to show I'm also cooperative.

I'll send around 60-70% of my endowment. Let me send \$3.50 (70%).

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 594,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is

significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: $\$5 - \$4.4 = \$0.60$

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: $\$0.60 + \$7.0 = \$7.60$ (profit of \$2.60)
- I keep: $\$13.20 - \$7.0 = \$6.20$

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.

Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

{ 'return': 6.0 }

reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They

currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receiver, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 1873,
  "output_tokens": 414,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$30.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Co-player has consistently shown trust, sending \$3.5 and \$4.0 (70-80% of endowment)
- As receiver, they returned fairly (\$5.5)
- They've seen generous returns from me (\$7.3, \$6.52)
- We've established a strong cooperative pattern over multiple rounds

Given this reliable cooperation, I should maintain high trust. I'll send \$4.0 (80% of endowment) to continue our successful partnership.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$3.5, saw \$7.3 returned, payoff \$8.8
- Round 4: They saw \$3.55 sent (noisy version of my \$4.0), returned \$5.5, payoff \$5.15
- Round 5: They sent \$4.0, saw \$6.52 returned (noisy version of my \$6.0), payoff \$7.52

The pattern shows:

- Co-player has been sending good amounts as sender (\$3.5, \$4.0)
- As receiver, they've been fair (\$5.5 return)
- They've seen me return generously (\$7.3, \$6.52)
- We've maintained strong cooperation

This is a reliable, cooperative partner. I should continue showing high trust. Since they've been sending 70-80% of endowment, I should match or slightly exceed that level.

I'll send \$4.0 again (80% of endowment) to maintain our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 382,
}
```

```

    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 7

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 4 against you, as RECEIVER: saw $3.55 sent, it became $10.649999999999999, returned $5.5, visible payoff $5.149999999999999.
  - Round 5 against you, as SENDER: sent $4.0, it became $12.0, saw $6.52 returned, visible payoff $7.52.
  - Round 6 against you, as RECEIVER: saw $4.73 sent, it became $14.190000000000001, returned $7.5, visible payoff $6.690000000000001.
  Your total visible earnings across all rounds are $40.28.

  This round, you are the RECEIVER against your current co-player. They sent you $4.03, so you received $12.09. How much do you return? (0-12.09)
  Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
content:
  Looking at the visible history:

  - Co-player has been highly cooperative, consistently sending high amounts (80%+ of endowment)
  - As receiver, they've returned fairly ($5.5, $7.5)
  - They sent $4.03 this round (80.6%), showing strong continued trust
  - They currently have $0.97

  This is very similar to Round 5 when they sent $4.0. I should maintain consistency in our cooperative relationship. If I return $6.5:
  - Sender ends with $7.47 (profit of $2.47)
  - I keep $5.59

  This appropriately rewards their ongoing trust.

  {'return': 6.5}
reasoning:
  Looking at the visible history with this co-player:

  - Round 4: They received, returned $5.5
  - Round 5: They sent $4.0 (80% of endowment), saw $6.52 returned
  - Round 6: They received, returned $7.5
  - Round 7 (current): They sent $4.03 (80.6% of endowment)

  This co-player has been very cooperative throughout. They're consistently sending high amounts (around 80%) and returning fairly when they're the receiver.

  Current situation:
  - They sent $4.03, so they have $0.97 left
  - I received $12.09

  I should continue reciprocating their high level of trust. They've been a reliable partner.

  If I return $6.5:
  - Sender ends with $7.47 (profit of $2.47)
  - I keep $5.59

  This seems fair given their high trust level. Let me think if this is appropriate...

  Actually, looking at the pattern, when they sent $4.0 in Round 5, I returned around $6.52 (visible). Now they sent $4.03, very similar. I should be consistent and return a similar amount.

  $6.5 seems reasonable and consistent with our established pattern.
usage:
  {
    "input_tokens": 2618,
    "output_tokens": 464,

```

```

    "reasoning_tokens": 0
  }
Message 14 role=user
  content:
    Round 8

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 5 against you, as SENDER: sent $4.0, it became $12.0, saw $6.52 returned, visible
      payoff $7.52.
    - Round 6 against you, as RECEIVER: saw $4.73 sent, it became $14.190000000000001, returned
      $7.5, visible payoff $6.690000000000001.
    - Round 7 against you, as SENDER: sent $4.0, it became $12.0, saw $6.69 returned, visible
      payoff $7.69.
    Your total visible earnings across all rounds are $45.870000000000005.

    This round, you are the SENDER against your current co-player. You have $5. How much do you
    send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at the visible history:

  - Co-player has consistently sent $4.0 (80%) as sender in recent rounds
  - As receiver, they returned $7.5 fairly
  - They've seen good returns from me ($6.52, $6.69)
  - We have a well-established pattern of mutual cooperation

  This is the final round with this co-player, and we've built strong trust. I'll send $4.0 (80%
  ) to maintain our cooperative relationship.

  {'send': 4.0}
Reasoning:
  Looking at the visible history with my co-player:

  - Round 5: They sent $4.0, saw $6.52 returned, payoff $7.52
  - Round 6: As receiver, saw $4.73 sent, returned $7.5, payoff $6.69
  - Round 7: They sent $4.0, saw $6.69 returned, payoff $7.69

  This co-player has been highly cooperative:
  - Consistently sending high amounts as sender ($4.0 multiple times)
  - Being fair as receiver ($7.5 return on $14.19 received)
  - We've established a strong pattern of cooperation

  Given this excellent track record over many rounds, I should continue with high trust. Sending
  $4.0 or even $4.5 would be appropriate to maintain this successful partnership.

  I'll send $4.0 to maintain consistency with our established cooperative pattern.
  Usage: {
    "input_tokens": 2791,
    "output_tokens": 336,
    "reasoning_tokens": 0
  }

```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:02:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",

```

```

    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.

Your total visible earnings across all rounds are \$7.75.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****Round 1 context:****

- I sent \$3 (they saw \$3.39 ? \$10.17 after multiplication)
- They returned \$5.0 to me (approximately 49% of what they received)
- This shows reasonable reciprocity

****Current round:****

- They sent \$2.86 (less than I sent last round)
- I received \$8.58
- I should reciprocate fairly to maintain cooperation

They returned about 49% when they were receiver. I'll return a similar proportion, being slightly more generous to encourage future cooperation:

```
'''json
{"return": 4.5}
'''
```

This gives them a reasonable return (\$6.64 total) while I keep \$4.08, maintaining balanced cooperation for future rounds.

reasoning:

Let me analyze the situation:

Round 1:

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$3.39 sent (close to what I actually sent)
- They received \$10.17 (which is $3 \times \$3.39$)
- They returned \$5.0
- My visible payoff was $\$7.75 = \$5 - \$3 + \5.75 (wait, let me recalculate)
- Actually, I sent \$3, so I kept $\$5 - \$3 = \$2$, plus what they returned
- If my visible payoff is \$7.75, then I must have received back $\$7.75 - \$2 = \$5.75$
- But the history says they returned \$5.0 visible
- Due to noise, what I saw as returned might be different from what they actually returned

Now Round 2:

- I'm the RECEIVER
- They sent me \$2.86 (visible amount)
- I received $\$8.58 = 3 \times \2.86
- I need to decide how much to return

Strategy considerations:

1. In Round 1, they returned around \$5 to me when I sent \$3 (which they saw as \$3.39, becoming \$10.17)
2. They returned roughly 49% of what they received ($\$5/\$10.17 \approx 49\%$)
3. Now they've sent me \$2.86, less than I sent them
4. I should consider reciprocating fairly

If I apply a similar return rate:

- 49% of \$8.58 = \$4.20

Alternatively, I could think about net outcomes:

- In Round 1: I ended with \$7.75 (sent 3, got back around 5.75)
- They want to establish cooperation

...

usage:

```
{
  "input_tokens": 786,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible

le payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)
****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****
- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****
- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)
****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)
****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****
- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****
- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****
- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****
- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our strong cooperative pattern:

****Round 1:**** I sent \$3, they returned \$5.0
****Round 2:**** They sent \$3.5, I returned \$4.5
****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$5.0
Round 2: They sent \$3.5, I returned \$4.5 (they saw \$5.25)
Round 3: I sent \$3.5 (they saw \$4.4), they returned \$7.0
Round 4: They sent \$4.0, I returned \$5.5 (they saw \$6.46)

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

usage:

```
{
  "input_tokens": 1876,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.

Your total visible earnings across all rounds are \$33.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so y

ou received \$14.190000000000001. How much do you return? (0-14.190000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued strong cooperation:

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)
****Round 4:**** I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
****Round 5:**** They returned \$6.0 from \$11.28 (~53%)

****Current round:**** I received \$14.19

They sent a high amount (\$4.73) showing strong trust. I'll maintain our reciprocal pattern
:

```
'''json
{"return": 7.5}
'''
```

This is approximately 53% of what I received, consistent with their return rates.

reasoning:

Looking at the game history with this co-player:

Round 3: They returned \$7.0 from \$13.2 (~53%)
Round 4: I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
Round 5: They returned \$6.0 from \$11.28 (~53%)

Current round: They sent \$4.73, I received \$14.19

They've been very cooperative and generous throughout. They've increased their sending and maintained good return rates. I should reciprocate appropriately.

Looking at their returns: around 53-54%
I'll return around 53-55% of what I received.

$\$14.19 * 0.53 = \7.52
 $\$14.19 * 0.54 = \7.66
 $\$14.19 * 0.55 = \7.80

I'll return \$7.5 which is about 53%, maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 2266,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.

Your total visible earnings across all rounds are \$39.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our highly cooperative relationship:

****Round 4:**** They sent \$4.0, I returned \$5.5
****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0, I returned \$7.5

This co-player has demonstrated:

- Consistent high trust (repeatedly sending \$4.0)
- Very generous returns
- Strong reciprocity

I'll continue matching their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track our interaction:

```
**Round 1:** I sent $3, they returned $5.0
**Round 2:** They sent $3.5, I returned $4.5
**Round 3:** I sent $3.5, they returned $7.0
**Round 4:** They sent $4.0, I returned $5.5
**Round 5:** I sent $4.0, they returned $6.0
**Round 6:** They sent $4.0 (I saw $4.73 due to noise), I returned $7.5 (they saw $8.42)
```

Looking at the pattern:

- They've been consistently sending around \$3.5-\$4.0
- They've been returning generously
- Round 6 shows they saw \$8.42 returned from \$12.0, which is about 70% - very high!
- This is a strong collaborative partner

They've maintained sending \$4.0 in rounds 4 and 6. I should continue with a similar level of trust. I'll send \$4 again to maintain consistency and mutual trust.

usage:

```
{
  "input_tokens": 2640,
  "output_tokens": 381,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.
- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.

Your total visible earnings across all rounds are \$47.679999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our established cooperative pattern:

```
**Round 5:** They returned $6.0 from $11.28 (~53%)
**Round 6:** I returned $7.5 (they saw $8.42)
**Round 7:** They returned $6.5 from $12.09 (~54%)
```

```
**Current round:** I received $11.64
```

They sent \$3.88, continuing to show trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 6.5}
'''
```

This is approximately 56% of what I received, maintaining our strong cooperative relationship.

Reasoning:

Looking at the history with this co-player:

****Round 5:**** They returned \$6.0 from \$11.28 (~53%)
****Round 6:**** I returned \$7.5 (they saw \$8.42 from \$12.0, ~70%)
****Round 7:**** They returned \$6.5 from \$12.09 (~54%)

****Current round:**** I received \$11.64

They sent \$3.88, showing continued trust. Looking at their return rates, they've been consistently around 53-54% when they're the receiver.

I've been returning generously (my actual return of \$7.5 in round 6 showed up as \$8.42 to them).

To maintain the cooperative relationship, I should return around 53-55% of what I received:

- 53% of \$11.64 = \$6.17
- 55% of \$11.64 = \$6.40

I'll go with around \$6.5 to match their recent return pattern and maintain fairness.

```
Usage: {  
  "input_tokens": 2826,  
  "output_tokens": 386,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:02:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.

Your total visible earnings across all rounds are \$2.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

In Round 1, my current co-player sent me \$3.0 (60% of their endowment) when they were the sender, showing significant trust. From their perspective, they received \$3.88 back (though this may be affected by communication noise).

Since they demonstrated high trust by sending 60% of their endowment, I should reciprocate with a similar level of trust to maintain the cooperative relationship.

Sending \$3.0 (60% of my endowment) matches their trust level and signals that I want to continue cooperating.

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation carefully.

From the visible history:

- Round 1: I was the RECEIVER and they were the SENDER
- They sent \$3.0 (60% of their endowment)
- It became \$9.0 after multiplication
- They saw \$3.88 returned (but due to noise, I actually returned \$3.50)
- Their visible payoff was \$5.88

Wait, let me recalculate:

- They had \$5, sent \$3.0, so they had \$2.0 left
- They saw \$3.88 returned
- So their total would be \$2.0 + \$3.88 = \$5.88

But I returned \$3.50, not \$3.88. This is the communication noise mentioned in the instructions. They think I returned more than I actually did.

Now I'm the SENDER in Round 2:

- They previously sent me \$3.0 (60% of their endowment) when they were sender
- They saw that I returned \$3.88 (43% of what I received)
- This showed good reciprocity from their perspective

Key considerations:

1. They demonstrated high trust by sending 60% of their endowment
2. From their perspective, I showed decent reciprocity (returning 43% of what I received)
3. They might expect similar trust from me now that I'm the sender

Strategy:

Since they showed significant trust by sending 60%, and they perceive that I was fairly reciprocal, I should send a comparable amount to maintain the cooperative relationship. Sending around \$3.0 would match their trust level.

However, I should also consider that:

- If I send \$3.0, it becomes \$9.0
- Based on their past behavior (returning what they thought was 43% when they were receiver), they might return around \$3.87 (43% of \$9.0)
- My payoff would be: $\$5 - \$3.0 + \$3.87 = \5.87
- Sending \$2.0 yields \$5.58, while \$4.0 could give me \$5.16 if they maintain that 43% return rate, so the middle ground of \$3.0 seems optimal for balancing trust and expected returns.

Since I don't actually have data on how they behave as a receiver-their history only shows them sending \$3.0 in Round 1-I need to infer their reciprocity from that signal. Their 60% contribution suggests they value cooperation, so matching that proportion with a \$3.0 send feels like the right move to sustain the relationship...

usage:

```
{
  "input_tokens": 867,
  "output_tokens": 1048,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.


```
{'return': 4.50}
```

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have \$5 - \$2.83 = \$2.17

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: \$2.17 + \$4.50 = \$6.67 (profit of \$1.67)
- I end with: \$8.49 - \$4.50 = \$3.99 (profit of \$3.99)

If I return \$4.25:

- Sender ends with: \$2.17 + \$4.25 = \$6.42 (profit of \$1.42)
- I end with: \$8.49 - \$4.25 = \$4.24 (profit of \$4.24)

If I return \$4.00:

- Sender ends with: \$2.17 + \$4.00 = \$6.17 (profit of \$1.17)

...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{
  "input_tokens": 1611,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)

- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)

- I end with: \$4.69 (profit of \$4.69)

```
{'return': 5.0}
```

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)

Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)

Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)

Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)
Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: $\$1.77 + \$5.0 = \$6.77$ (profit of \$1.77)
- I end with: $\$9.69 - \$5.0 = \$4.69$ (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

usage:

```
{  
  "input_tokens": 1982,  
  "output_tokens": 520,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.

Your total visible earnings across all rounds are \$23.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 3 and 5
- As receiver: returned \$4.5 (52%) in round 4

We've established a strong pattern of mutual trust and fair reciprocity. I should continue this successful cooperation by sending the same amount.

{ 'send': 3.0 }

reasoning:

Looking at my co-player's history:

- Round 3: as sender, sent \$3.0 (60%)
- Round 4: as receiver, returned \$4.5 out of \$8.64 (52%)
- Round 5: as sender, sent \$3.0 (60%)

They've been consistently sending 60% when they're the sender, and returning around 50-52% when they're the receiver. This is a very cooperative and trustworthy pattern.

I've been sending \$3.0 (60%) consistently as well, and we have a strong mutual cooperation going.

I should continue with the same level of trust - sending \$3.0 (60%) to maintain our successful partnership.

usage:

```
{
```

```
"input_tokens": 2371,
"output_tokens": 265,
"reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.

Your total visible earnings across all rounds are \$28.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

My co-player has been consistently cooperative and just sent \$3.46 (69% of endowment), their highest trust level yet! They currently have \$1.54 remaining.

I've been consistently returning around 52-56% of what I receive. I should maintain this fair reciprocity to reward their increased trust.

If I return \$5.5:

- Sender ends with: \$7.04 (profit of \$2.04)
- I end with: \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains our strong cooperative relationship.

```
{'return': 5.5}
```

reasoning:

Looking at the history with my co-player:

Round 4: I was sender, they were receiver - saw \$2.88 sent, returned \$4.5 out of \$8.64 (52%)

Round 5: They were sender, I was receiver - sent \$3.0, I returned visible \$5.62

Round 6: I was sender, they were receiver - saw \$2.4 sent (but I sent \$3.0 due to noise), returned \$3.5 out of \$7.2 (49%)

Round 7: They are sender, I am receiver - sent \$3.46 (69% of endowment!), I received \$10.38

They've increased their trust level to 69%, the highest yet. They currently have \$1.54 remaining.

I should reward this high trust. Looking at my previous returns:

- Round 1: Returned \$3.50 out of \$6.24 (56%)
- Round 3: Returned \$4.50 out of \$8.49 (53%)
- Round 5: Returned \$5.0 out of \$9.69 (52%)

I've been returning around 52-56% consistently. Given they sent an even larger proportion this time (69%), I should maintain or slightly increase my return rate.

If I return \$5.5:

- Sender ends with: \$1.54 + \$5.5 = \$7.04 (profit of \$2.04)
- I end with: \$10.38 - \$5.5 = \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains my consistent cooperative pattern and rewards their increased trust.

usage:

```
{
  "input_tokens": 2715,
```

```

    "output_tokens": 553,
    "reasoning_tokens": 0
  }
Message 14 role=user
  content:
    Round 8

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 5 against you, as SENDER: sent $3.0, it became $9.0, saw $5.62 returned, visible payoff $7.62.
    - Round 6 against you, as RECEIVER: saw $2.4 sent, it became $7.199999999999999, returned $3.5, visible payoff $3.6999999999999993.
    - Round 7 against you, as SENDER: sent $3.0, it became $9.0, saw $4.59 returned, visible payoff $6.59.
    Your total visible earnings across all rounds are $33.84.

    This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at the history:

  My co-player has maintained consistent cooperation:
  - As sender: sent $3.0 (60%) in rounds 5 and 7
  - As receiver: returned $3.5 (49%) in round 6

  We've established a robust pattern of mutual trust. I should continue this successful cooperation by sending the same amount.

  {'send': 3.0}
Reasoning:
  Looking at the history:

  My co-player has consistently sent $3.0 (60%) in rounds 5 and 7 as sender. As receiver in round 6, they returned $3.5 out of $7.2 (49%).

  We've established a strong pattern of cooperation throughout these rounds. I should continue matching their trust level by sending $3.0 (60% of my endowment).

  This maintains our mutually beneficial relationship.
  Usage: {
    "input_tokens": 2827,
    "output_tokens": 200,
    "reasoning_tokens": 0
  }

GAME - Agent_5 - Receiver - dyad_3
  Timestamp: 2026-08-12T08:02:36
  Model: anthropic/claude-sonnet-4.5
  Temperature: 0.8
  Metadata: {
    "agent_type": "standard",
    "dyad_id": "dyad_3",
    "move_index": 5,
    "opponent_id": "Agent_6",
    "pairing": {
      "agents": [
        "Agent_6",
        "Agent_5"
      ],
      "investor": "Agent_6",
      "roles": {
        "Agent_5": "trustee",
        "Agent_6": "investor"
      },
      "trustee": "Agent_5"
    },
    "role": "trustee",
    "role_label": "Receiver",

```

```
"round": 8,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.

Your total visible earnings across all rounds are \$5.88.

This round, you are the RECEIVER against your current co-player. They sent you \$2.16, so you received \$6.48. How much do you return? (0-6.48)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

In Round 1, my co-player (as receiver) returned \$3.5 out of \$6.24 they received - a solid 56% return showing good cooperation.

Now they've sent me \$2.16 (becoming \$6.48). To reciprocate their cooperative behavior, I'll return a similar proportion.

```
'''json
{"return": 3.5}
'''
```

This matches what they returned to me, signaling reciprocity and maintaining the cooperative relationship.

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.08 sent (which became \$6.24). They returned \$3.5.
2. My visible earnings from Round 1: I sent \$3, kept \$2, received back \$3.5, so total = \$5.5. But the prompt says my visible earnings are \$5.88. Let me recalculate:
 - I started with \$5
 - I sent \$3 (actually sent)
 - I kept \$2
 - I received back \$3.5
 - Total: \$2 + \$3.5 = \$5.5

Wait, the visible earnings might include the new round's endowment. Or there's noise in what I see too.

3. Now in Round 2, I'm the RECEIVER. My co-player sent me an amount that I see as \$2.16, which became \$6.48.
4. My co-player in Round 1 (as receiver) saw \$2.08 sent, got \$6.24, and returned \$3.5. That's about 56% of what they received.
5. Since they showed decent cooperation by returning \$3.5 out of \$6.24, I should reciprocate fairly.
6. If I return about half or slightly more of what I received (\$6.48), that would be \$3-3.5.

Let me think about fairness:

- They sent some amount (I see \$2.16)
- With the 3x multiplier, I received \$6.48
- If I return half (\$3.24), they would get back more than they sent
- If I return \$3.5 (what they returned to me), that shows reciprocity

Given they returned \$3.5 out of \$6.24 (56%), I could return a similar proportion: 56% of \$6.48 = \$3.63

...

usage:

```
{
  "input_tokens": 789,
  "output_tokens": 622,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender

- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver

- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:

- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:

- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1081,
  "output_tokens": 442,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.4700000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.

Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)
Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).
I'll match their generous pattern:

```
'''json
{"return": 4.5}
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:
- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5
- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of \$8.49 (generous!)
Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful pattern:

```
'''json
```

```
{"send": 3}
'''
```

reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of what they received
Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with that proven strategy.

usage:

```
{
  "input_tokens": 1778,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$27.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.4, so you received \$7.199999999999999. How much do you return? (0-7.199999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69 (very generous!)

This round they sent \$2.4 (becoming \$7.20) - slightly less than their usual \$3.0. I'll return a proportionate amount while maintaining our cooperative relationship:

```
'''json
{"return": 3.5}
'''
```

This gives them a reasonable return while accounting for the slightly lower send amount.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69

They've been very cooperative and generous. Now they sent \$2.4 (becoming \$7.20).

This is the first round where they sent less than \$3.0. Let me check the pattern:

- When they sent \$3.0, I returned around \$4.5
- When I sent (visible as \$2.83, \$3.23), they returned \$4.5 and \$5.0

Now they sent \$2.4. I should return a fair amount. Let me calculate:

- They sent \$2.4, so they kept \$2.6
- It became \$7.20 for me
- If I return around 50-55% of what I received, that's \$3.6-\$4.0
- Their final payoff would be: \$2.6 + return
- My final payoff would be: \$7.20 - return

If I return \$3.5:

- They get: \$2.6 + \$3.5 = \$6.1
- I get: \$7.20 - \$3.5 = \$3.7

If I return \$4.0:

- They get: \$2.6 + \$4.0 = \$6.6
- I get: \$7.20 - \$4.0 = \$3.2

Given they sent slightly less this round, I'll return a proportional amount. Since they've been very generous when they were receiver, I'll still be generous but slightly adjust for their lower send.

\$3.5 seems fair - it's about 49% of what I received, reasonable reciprocity.

usage:

```
{
  "input_tokens": 2139,
  "output_tokens": 574,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$30.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our ongoing history:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 (very generous!)

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently trustworthy and cooperative throughout. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history with this co-player:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently cooperative throughout our interactions. They've consistently sent around \$3.0 when they're the sender and returned generously as receiver. I should continue our successful cooperative pattern.

I'll send \$3 again to maintain our established trust.

usage:

```
{
  "input_tokens": 2499,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.

Your total visible earnings across all rounds are \$37.47.

This round, you are the RECEIVER against your current co-player. They sent you \$3.47, so you received \$10.41. How much do you return? (0-10.41)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

Round 5: They returned \$5.0 out of \$9.69
 Round 6: They sent \$3.0, I returned \$3.5
 Round 7: They returned \$5.5 out of \$10.38 (very generous!)

They just sent \$3.47 (becoming \$10.41) - showing continued trust. I'll reciprocate their generous pattern:

```
'''json
{"return": 5.5}
'''
```

This matches their generous cooperation and maintains our strong relationship.

Reasoning:

Looking at the history with this co-player:

Round 5: They returned \$5.0 out of \$9.69
 Round 6: They sent \$3.0, I returned \$3.5
 Round 7: They returned \$5.5 out of \$10.38

This co-player has been consistently cooperative and generous. They just sent me \$3.47 (becoming \$10.41).

Looking at their return pattern:

- Round 5: \$5.0 / \$9.69 = 51.6%
- Round 7: \$5.5 / \$10.38 = 53%

They're returning around 50-53% of what they receive, which is quite generous and cooperative.

For \$10.41 received, returning around 50-53% would be:

- 50%: \$5.20
- 53%: \$5.52

I should reciprocate their generosity. I'll return around \$5.5 to match their cooperative pattern.

```
Usage: {
  "input_tokens": 2661,
  "output_tokens": 363,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:02:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.

Your total visible earnings across all rounds are \$6.470000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at the history:

- In Round 1, my co-player sent \$3.0 (60% of their endowment) showing significant trust
- From their perspective, they received \$5.61 back, yielding them a good payoff
- This suggests they value cooperation

I want to maintain this positive cooperative relationship. Since they sent 60% of their endowment, I'll send a similar proportion to reciprocate their trust.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze the visible history:

From Round 1:

- My co-player was the SENDER and sent \$3.0
- It became \$9.0 after multiplier
- They saw \$5.61 returned (though I actually returned \$5.50, there's communication noise)
- Their visible payoff was \$7.61

So from their perspective:

- They sent \$3.0 and received \$5.61 back
- They ended with \$2.0 + \$5.61 = \$7.61

Now in Round 2, I'm the SENDER and they're the RECEIVER.

They showed trust by sending \$3.0 (60% of their endowment) in Round 1, and from their perspective, I returned \$5.61, which was a good return (62.3% of what I received).

Given:

1. They demonstrated trust by sending 60% of their endowment
2. From their perspective, I reciprocated fairly with a 62% return
3. This establishes a foundation for cooperation

I should send a similar or slightly higher amount to maintain the cooperative relationship. Since they sent \$3.0 (60%), I could send around the same amount or slightly more to show I'm willing to continue cooperating.

I'll send \$3.50 (70% of my endowment) to demonstrate trust and maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 851,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.61

d \$5.5, visible payoff \$6.379999999999999.
Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$

- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send":3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

```
'''json
{"return": 5.25}
'''
```

reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5

Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)

Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5

Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

This is fair and consistent with our pattern.

usage:

```
{
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.

Your total visible earnings across all rounds are \$30.390000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant**content:**

Looking at our ongoing cooperation:

- Round 3: Co-player sent \$3.5
- Round 4: I sent \$3.75, they returned \$5.5
- Round 5: Co-player sent \$3.75

We've established a very strong cooperative pattern. Both of us consistently send high amounts (around \$3.50-\$3.75) and return fair shares. This has been mutually beneficial - my total earnings of \$30.39 reflect this.

I'll continue this successful cooperation:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with my co-player:

Round 3: Co-player sent \$3.5, I returned (they saw \$5.95)
 Round 4: I sent \$3.75, co-player returned \$5.5
 Round 5: Co-player sent \$3.75, I returned (they saw \$5.21)

We've been maintaining a very cooperative relationship. Both of us have been sending high amounts (around \$3.50-\$3.75) and returning fair shares.

My earnings are \$30.39 total, which suggests this cooperation is working well.

The co-player has consistently sent high amounts (\$3.5, \$3.75) showing strong trust. They've also been fair when receiving, returning \$5.5 consistently.

I should continue this cooperative pattern by sending a high amount. I'll send \$3.75 to match what they've been sending.

usage:

```
{
  "input_tokens": 2386,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.

Your total visible earnings across all rounds are \$36.960000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.57, so you received \$13.71. How much do you return? (0-13.71)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at the pattern:

- Round 4: Co-player returned \$5.5 from \$10.59
- Round 5: Co-player sent \$3.75
- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$4.57 (91.4% of their endowment!)

This is a very strong trust signal - they sent almost everything! I should reciprocate this high level of trust with a generous return.

If I return \$7.00:

- Sender ends with: $\$0.43 + \$7.00 = \$7.43$
- I end with: $\$13.71 - \$7.00 = \$6.71$

This rewards their exceptional trust and maintains our strong cooperative relationship.

```
'''json
{"return": 7.00}
'''
```

reasoning:

Looking at the history:

Round 4: Co-player as receiver, returned \$5.5 from \$10.59

Round 5: Co-player as sender, sent \$3.75

Round 6: Co-player as receiver, they saw I sent \$4.18 which became \$12.54, and they returned \$6.0

Round 7: Co-player sent me \$4.57 (91.4% of their endowment) - this is a very high trust signal!

They've been escalating their cooperation. In Round 6, they returned \$6.0 from \$12.54 (about 48%). Now they're sending \$4.57, which is much higher than their previous sends.

I received \$13.71 after the multiplier.

They currently have: $\$5 - \$4.57 = \$0.43$

If I return \$6.50:

- Sender ends with: $\$0.43 + \$6.50 = \$6.93$
- I end with: $\$13.71 - \$6.50 = \$7.21$

This rewards their very high trust and continues our cooperative relationship.

Actually, they sent 91.4% - that's a very strong trust signal. Let me be more generous to match that.

If I return \$7.00:

- Sender ends with: $\$0.43 + \$7.00 = \$7.43$
- I end with: $\$13.71 - \$7.00 = \$6.71$

This is a bit more balanced and rewards the very high send amount.

usage:

```
{
  "input_tokens": 2753,
  "output_tokens": 564,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.

Your total visible earnings across all rounds are \$43.67000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued strong cooperation:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, received good return

We've built an excellent cooperative relationship. My total earnings of \$43.67 show this has b

een very successful. Both of us have consistently sent high amounts and returned generously.

I'll maintain this cooperative pattern:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at our history:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, I returned \$7.00 (they saw \$7.31)
- Round 8: Now I'm the sender

Wait, let me re-check the history:

- Round 7 it says "as SENDER: sent \$3.75" but in Round 7 they sent me \$4.57. There's noise in the communication.

The visible history says their Round 7 they sent \$3.75 and saw \$7.31 returned. But from my perspective in Round 7, they sent \$4.57 and I returned \$7.00.

So there's significant noise. They're seeing different amounts than what actually happened.

Our cooperation has been excellent. My total earnings are \$43.67, which is very good.

In Round 6, I sent and they returned \$6.0 from receiving \$12.54

In Round 7, they sent (I saw \$4.57, they think they sent \$3.75) and I returned \$7.00 (they saw \$7.31)

Both of us have been highly cooperative. I should continue this pattern. Let me send a high amount again, around \$3.75-\$4.00.

Actually, looking at what they're seeing, they think they've been sending \$3.75 consistently.

I should stick with that pattern.

```
Usage: {
  "input_tokens": 2933,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:02:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.

Your total visible earnings across all rounds are \$7.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.96, so you received \$11.879999999999999. How much do you return? (0-11.879999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at the history, my co-player demonstrated good reciprocity in Round 1. They received \$11.97 and returned \$5.5 (about 46%), which was generous and fair.

Now they've sent me \$3.96, showing continued trust. I received \$11.88. I should reciprocate their cooperative behavior.

Returning a similar proportion (~\$5.5-\$6) would be fair and encourage continued cooperation.

```json

```
 {"return": 5.5}
 """
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (from my co-player's perspective):

- They saw \$3.99 sent (I actually sent \$3, but due to noise they saw \$3.99)
- This became \$11.97 (tripled)
- They returned \$5.5
- Their visible payoff was \$6.47

My actual Round 1 outcome:

- I sent \$3
- Due to noise, they saw \$3.99 and received \$11.97
- They returned \$5.5
- My earnings:  $\$5 - \$3 + \$5.5 = \$7.50$
- But my visible earnings show \$7.61 (slight noise)

Now in Round 2:

- I'm the RECEIVER
- They sent me \$3.96 (with noise, actual amount might differ)
- I received \$11.88 (tripled)
- I need to decide how much to return

My co-player showed good reciprocity in Round 1, returning \$5.5 out of \$11.97 (about 46%). This was a fair and generous return.

I should reciprocate their cooperative behavior. They sent nearly \$4 this round, showing continued trust. A fair return would be somewhere in the range of \$4-\$6.

Let me think about what's equitable:

- They started with \$5
- They sent ~\$4, keeping ~\$1
- I received \$11.88
- If I return \$5, they end with  $\$1 + \$5 = \$6$ , and I keep \$6.88
- If I return \$6, they end with  $\$1 + \$6 = \$7$ , and I keep \$5.88

Given they returned \$5.5 (about 46%) last round when in a similar position, I should aim for similar fairness. Returning about \$5.5 to \$6 would be appropriate and reciprocate their cooperative behavior.

...

**usage:**

```
{
 "input_tokens": 783,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

**usage:**

```
{
 "input_tokens": 1079,
 "output_tokens": 370,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
  - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
  - Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)

Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)

Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)

Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5

- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

**usage:**

```
{
 "input_tokens": 1444,
 "output_tokens": 410,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

**reasoning:**

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)
- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

**usage:**

```
{
 "input_tokens": 1810,
 "output_tokens": 314,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**



**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.

Your total visible earnings across all rounds are \$32.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the pattern:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent \$4.18 (more than their usual \$3.5-3.75), showing increased trust. I received \$12.54. I should reciprocate this increased trust with a proportionally fair return:

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Looking at the history:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent me \$4.18, which is higher than their usual amounts (around \$3.5-3.75). They received \$12.54.

They're showing increased trust by sending more. I should reciprocate appropriately. Given they sent more, I should return a bit more to maintain the cooperative relationship.

Our typical pattern has been returning around \$5-5.5 from amounts around \$10-11. Now I have \$12.54, so returning around \$6 would be proportionate and fair, maintaining the cooperative spirit.

**usage:**

```
{
 "input_tokens": 2175,
 "output_tokens": 314,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$39.529999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our continued cooperation:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

We've maintained a very stable and mutually beneficial pattern. I'll continue with the same strategy:

```
'''json
{"send": 3.75}
'''
```

#### reasoning:

Looking at the recent history:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

Wait, looking at my visible history, there's some inconsistency. In Round 6, it says "They sent you \$4.18" but now the history says they sent \$3.75. This is probably the communication noise mentioned in the instructions.

Our pattern has been very stable with sends around \$3.5-3.75 and returns around \$5.0-5.5. We've built strong mutual cooperation. I should continue with this pattern:

```
'''json
{"send": 3.75}
'''
```

#### usage:

```
{
 "input_tokens": 2520,
 "output_tokens": 287,
 "reasoning_tokens": 0
}
```

#### Message 14 role=user

##### content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.710000000000001.

Your total visible earnings across all rounds are \$48.089999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

#### Assistant output:

Looking at our recent pattern:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71 (51% - very generous!)

They've sent \$4.33, giving me \$12.99. They've been increasingly generous, so I should reciprocate with a fair return that maintains our strong cooperation:

```
'''json
{"return": 6.5}
'''
```

#### Reasoning:

Looking at the history:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71

In Round 7, they sent \$4.57 which became \$13.71, and they returned \$7.0 (51%). That's a very generous return rate!

Now they've sent me \$4.33, which became \$12.99. This is a high amount showing continued trust.

I should reciprocate with a proportionally fair return. Given they returned \$7.0 from \$13.71 last round (51%), I should aim for something similar or slightly less.

$\$12.99 * 0.50 = \$6.495$

$\$12.99 * 0.52 = \$6.75$

I'll return around \$6.5 to maintain our strong cooperation.

```
Usage: {
 "input_tokens": 2702,
 "output_tokens": 348,
 "reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_3=Sender, Agent\_4=Receiver  
dyad\_3: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_8=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$4.5; received \$13.5; Agent\_2 returned \$7.5; payoffs Agent\_1=\$8, Agent\_2=\$6;  
noise/visible: sent visible=\$4.3, received visible=\$12.9, returned visible=\$7.75  
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 13.5, "min": 0, "raw": 7.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}  
dyad\_2: Agent\_3 sent \$4; received \$12; Agent\_4 returned \$8.5; payoffs Agent\_3=\$9.5, Agent\_4=\$3.5;  
noise/visible: sent visible=\$4.97, received visible=\$14.91, returned visible=\$8.75  
Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_5 sent \$3; received \$9; Agent\_6 returned \$4; payoffs Agent\_5=\$6, Agent\_6=\$5;  
noise/visible: sent visible=\$2.48, received visible=\$7.44, returned visible=\$3.21  
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_7 sent \$3.75; received \$11.25; Agent\_8 returned \$5; payoffs Agent\_7=\$6.25, Agent\_8=\$6.25; noise/visible: sent visible=\$3.24, received visible=\$9.72, returned visible=\$5.43  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 11.25, "min": 0, "raw": 5.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

### Balances after round: {

"Agent\_1": 56.6,  
"Agent\_2": 56.4,  
"Agent\_3": 62.0,  
"Agent\_4": 51.0,  
"Agent\_5": 51.5,  
"Agent\_6": 47.5,  
"Agent\_7": 55.75,  
"Agent\_8": 54.25

}

### Visible balances after round: {

"Agent\_1": 58.14999999999999,  
"Agent\_2": 56.260000000000005,  
"Agent\_3": 62.56999999999999,  
"Agent\_4": 58.84,  
"Agent\_5": 47.589999999999996,  
"Agent\_6": 44.65,  
"Agent\_7": 61.25999999999999,  
"Agent\_8": 55.56000000000001

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T08:03:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.

Your total visible earnings across all rounds are \$9.719999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 1: Co-player received \$7.68, returned \$2.6 (34% return rate)
- Round 2: Co-player sent \$3.0, but only saw \$2.72 returned due to noise (though I actually returned \$3.0)

**\*\*Current Situation:\*\***

From my co-player's visible perspective, they may think I wasn't very reciprocal in Round 2 (sent \$3, only saw \$2.72 back). However, they still showed trust by sending \$3 initially.

I want to signal continued cooperation and rebuild trust, especially since the noise made my return look worse than it was. Sending a similar amount maintains consistency.

{"send": 3}

**reasoning:**

Let me analyze the history:

Round 1:

- I was SENDER, sent \$3
- Co-player as RECEIVER saw \$2.56 (noise), it became \$7.68
- They returned \$2.6
- My visible payoff was \$5.34 (I kept \$2 + got back \$2.6 + noise = ~\$4.6 actual, \$5.34 visible)

Round 2:

- I was RECEIVER
- Co-player as SENDER sent \$3.0, it became \$9.0
- I returned \$3.0
- Co-player saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff was \$4.72 (they kept \$2 + saw \$2.72 back = \$4.72)

Now Round 3:

- I'm SENDER again
- My total visible earnings are \$9.72

Looking at the pattern:

- Round 1: Co-player returned \$2.6 out of \$7.68 received (34%)
- Round 2: I returned \$3.0, but they saw \$2.72 due to noise. They sent \$3.0 and got back \$2.72 visible, so from their perspective they lost money (\$3 sent, \$2.72 back)

From my co-player's perspective:

- As receiver in R1: kept most of the money (\$5.08 out of \$7.68)
- As sender in R2: lost money (\$3 sent, only \$2.72 back visible)

They might be less trusting now because they think I didn't reciprocate well (even though I returned \$3, they only saw \$2.72 due to noise).

However, they did send \$3 in Round 2, showing willingness to cooperate even after only getting back 34% in Round 1.

...

**usage:**

{

```
"input_tokens": 1159,
"output_tokens": 665,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.6200000000000001, returned \$4.5, visible payoff \$6.1200000000000001.

Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern:

**\*\*History Analysis:\*\***

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

**\*\*Current Situation:\*\***

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

**\*\*My Decision:\*\***

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

```
{"return": 5.0}
```

**reasoning:**

Let me analyze the history:

**\*\*Round 1:\*\*** Co-player as RECEIVER

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

**\*\*Round 2:\*\*** Co-player as SENDER

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)
- Their visible payoff: \$4.72

**\*\*Round 3:\*\*** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

**\*\*Current Round 4:\*\*** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

**\*\*Analysis:\*\***

1. My co-player has been increasingly generous:

- Round 1: returned 34% (\$2.6 from \$7.68)
- Round 3: returned 42% (\$4.5 from \$10.62)

2. They also increased their sending:

- Round 2: sent \$3.0

- Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

**\*\*My Strategy:\*\***

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:  
- They get:  $\$5 - \$3.82 + \$5.0 = \$6.18$  total

...

**usage:**

```
{
 "input_tokens": 1585,
 "output_tokens": 777,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.

Your total visible earnings across all rounds are \$23.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 2: Co-player sent \$3.0, saw \$2.72 returned
- Round 3: Co-player returned \$4.5 from \$10.62 (strong reciprocity)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (even better returns from their perspective)

**\*\*Current Situation:\*\***

The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from \$3.0 to \$3.5 after seeing better returns. From their perspective, cooperation is paying off.

**\*\*My Decision:\*\***

I'll increase my send to \$4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

{"send": 4}

**reasoning:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 2: Co-player sent \$3.0, saw \$2.72 returned (I returned \$3.0 but noise made it \$2.72)
- Round 3: Co-player received visible \$10.62, returned \$4.5 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (I returned \$5.0 but noise made it \$5.74)

**\*\*Key Observations:\*\***

1. Co-player has been consistently trusting, sending \$3.0, then \$3.5
2. From their perspective, I returned better in Round 4 (\$5.74 on their \$3.5 sent) than Round 2 (\$2.72 on their \$3.0 sent)

3. They're showing increasing trust
4. The noise has actually made my returns LOOK better in Round 4

**\*\*My Strategy:\*\***

Given that:

- They've been consistently cooperative
- The trend is positive cooperation
- They sent \$3.5 last round and got good returns (from their perspective)
- I should continue showing trust

I'll send \$4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.

**usage:**

```
{
 "input_tokens": 2011,
 "output_tokens": 493,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.

Your total visible earnings across all rounds are \$29.009999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.5, so you received \$13.5. How much do you return? (0-13.5)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the pattern:

**\*\*History Analysis:\*\***

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 6: They sent \$4.5 out of \$5 (90% of their endowment!)

**\*\*Current Situation:\*\***

- They sent \$4.5, keeping only \$0.50
- I received \$13.5
- This is an extraordinary show of trust

**\*\*My Decision:\*\***

Their extreme trust deserves strong reciprocity. Returning \$7.0 means:

- They end with \$7.50 total (profit of \$2.50)
- I keep \$6.50 (profit of \$1.50)

This rewards their faith in our cooperation while maintaining a fair distribution of gains

```
{"return": 7.0}
```

**reasoning:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (good return)
- Round 5: Co-player saw \$3.9 sent, returned \$5.5 from \$11.7 (47%)

**\*\*Current Situation:\*\***

- Round 6: They sent \$4.5 (kept only \$0.50!)
- I received \$13.5
- This is their highest level of trust yet - sending 90% of their endowment



**\*\*My Thinking:\*\***

They're showing extreme trust by sending \$4.5. They've been consistently generous with the ir returns (42-47% return rates). I should reciprocate strongly to reward this trust.

If I return \$7.0:

- They get back \$7.0 + \$0.50 kept = \$7.50 total, profit of \$2.50
- I keep \$13.5 - \$7.0 = \$6.50, profit of \$1.50

Actually, let me be even more generous given their extreme trust. If I return \$8.0:

- They get back \$8.0 + \$0.50 kept = \$8.50 total, profit of \$3.50
- I keep \$13.5 - \$8.0 = \$5.50, profit of \$0.50

Hmm, that might be going too far. Let me think about what's fair.

They sent \$4.5 out of \$5, showing immense trust. A fair split would be:

- Total created value: \$13.5
- Original endowment lost: \$4.5
- Net gain to split: \$13.5 - \$4.5 = \$9.0
- If we split 50/50: each gets \$4.5
- So they should get: \$0.5 (kept) + \$4.5 (half of gain) = \$5.0 total
- I should return: \$5.0 - \$0.5 = \$4.5

But they've been returning at 42-47% rates, so I should match that generosity. 47% of \$13.5 is \$6.35. If I return \$6.5, we both end up with \$7.0 total and \$2.0 profit each-equal outcomes that acknowledge their trust. That feels right....

**usage:**

```
{
 "input_tokens": 2446,
 "output_tokens": 1046,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.

Your total visible earnings across all rounds are \$35.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (great returns!)

**\*\*Current Situation:\*\***

The cooperation is thriving! From their visible perspective:

- They're getting excellent returns when they send (5.74, 7.62)
- They've shown strong reciprocity when receiving
- They increased their sending to \$4.0 in Round 6

**\*\*My Decision:\*\***

I'll send \$4.5 to match their high trust level from Round 6 (when they sent \$4.5) and continue our strong cooperative relationship.

```
{"send": 4.5}
```

**reasoning:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (they saw even more returned than I actually sent - noise is working in my favor)

**\*\*Current Situation:\*\***

From their visible perspective:

- They've been receiving excellent returns (5.74, 7.62)
- They've been generous when receiving (returned 5.5)
- Our relationship is highly cooperative
- They increased their sending from \$3.5 to \$4.0 after seeing good returns

**\*\*Trend:\*\***

The cooperation is deepening. They're sending more, I'm returning generously (though noise sometimes makes it look even better). The visible payoffs show this is profitable for both of us.

**\*\*My Decision:\*\***

I should match their high trust level. They sent \$4.0 last time and saw great returns. I should send at least as much or more. Given the strong cooperative relationship, I'll send \$4.5 to signal maximum trust and continue the positive trend.

```
{"send": 4.5}
```

**usage:**

```
{
 "input_tokens": 2881,
 "output_tokens": 487,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.
- Round 7 against you, as RECEIVER: saw \$4.58 sent, it became \$13.74, returned \$7.5, visible payoff \$6.24.

Your total visible earnings across all rounds are \$44.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the pattern:

**\*\*History Analysis:\*\***

- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 8: They sent \$4.27 (kept only \$0.73)

**\*\*Current Situation:\*\***

- They sent \$4.27, keeping minimal \$0.73
- I received \$12.81
- Continued high trust from them

**\*\*My Decision:\*\***

Returning \$7.0 maintains our strong cooperative pattern:

- They end with \$7.73 total (profit of \$2.73)
- I keep \$5.81

This rewards their trust appropriately.

```

{"return": 7.0}
reasoning:
Looking at the pattern:

History Analysis:
- Round 5: Co-player returned $5.5 from $11.7 (47%)
- Round 7: Co-player returned $7.5 from $13.74 (55%)
- Round 8: They sent $4.27 (kept only $0.73)

Current Situation:
- They sent $4.27, keeping only $0.73
- I received $12.81
- This shows continued high trust

My Decision:
The co-player has been consistently generous and trusting. They sent $4.27, showing strong
faith in our relationship despite noise in the game.

Returning $7.0 means:
- They end with $7.73 total (profit of $2.73)
- I keep $5.81 (profit of $0.81)

This rewards their trust appropriately and maintains our cooperative relationship.

Actually, let me reconsider. Looking at the trend:
- Round 5: They returned $5.5 from $11.7
- Round 7: They returned $7.5 from $13.74

They've been generous. I should maintain similar generosity. Returning around $7.0-7.5 see
ms appropriate given they sent $4.27.

Let me go with $7.0 as a consistent response to their high trust.
usage:
{
 "input_tokens": 3179,
 "output_tokens": 505,
 "reasoning_tokens": 0
}
Message 14 role=user
content:
Round 9

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 6 against you, as SENDER: sent $4.0, it became $12.0, saw $7.62 returned, visible
 payoff $8.620000000000001.
- Round 7 against you, as RECEIVER: saw $4.58 sent, it became $13.74, returned $7.5, visib
 le payoff $6.24.
- Round 8 against you, as SENDER: sent $4.5, it became $13.5, saw $6.14 returned, visible
 payoff $6.64.
Your total visible earnings across all rounds are $49.89999999999999.

This round, you are the SENDER against your current co-player. You have $5. How much do yo
u send? (0-5)
Respond exactly as JSON: {'send': <amount>}
Assistant output:
Looking at the history:

Pattern Analysis:
- Round 6: Co-player sent $4.0, saw $7.62 returned
- Round 7: Co-player returned $7.5 from $13.74
- Round 8: Co-player sent $4.5, saw $6.14 returned (noise likely affected this)

Current Situation:
From their perspective, the return in Round 8 was lower than they might have expected. However
, they've shown consistent high trust by sending $4.0 and $4.5 in recent rounds. Despite any
perceived drop in returns (likely due to noise), I should maintain my commitment to coopera
tion.

My Decision:

```

I'll send \$4.5 to match their high trust and show I remain fully committed to our cooperative relationship.

```
{"send": 4.5}
```

**Reasoning:**

Looking at the history:

**\*\*Pattern Analysis:\*\***

- Round 6: Co-player sent \$4.0, saw \$7.62 returned (excellent!)
- Round 7: Co-player returned \$7.5 from \$13.74 (very good reciprocity)
- Round 8: Co-player sent \$4.5, saw \$6.14 returned (lower than previous rounds)

**\*\*Current Situation:\*\***

From their visible perspective:

- In Round 8, they sent \$4.5 but only saw \$6.14 returned, which is lower than the \$7.62 they saw in Round 6 when they sent \$4.0
- This might make them think the returns are decreasing
- However, the noise could be affecting this - I actually returned \$7.0 but they only saw \$6.14

**\*\*My Decision:\*\***

Despite the noise showing them lower returns, I should maintain high trust. They've consistently sent high amounts (\$4.0, \$4.5). I'll send \$4.5 to match their commitment and show I'm still fully cooperative.

This keeps the relationship strong and signals I'm not reducing my trust despite what they might perceive as reduced returns.

```
Usage: {
 "input_tokens": 3224,
 "output_tokens": 472,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T08:03:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.

Your total visible earnings across all rounds are \$9.8.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has been consistently cooperative
- Round 1: They sent 60% (\$3.0)
- Round 2: They returned \$3.0 from \$7.38 received
- Round 3: Now they sent \$3.54 (70.8%) - showing even higher trust!

**\*\*Current situation:\*\***

- They sent \$3.54, keeping only \$1.46
- I received \$10.62
- Total in play: \$1.46 + \$10.62 = \$12.08

**\*\*My decision:\*\***

This player has shown strong and increasing trust. I should reciprocate generously.

If I return \$4.50:  
- Sender gets:  $\$1.46 + \$4.50 = \$5.96$   
- I get:  $\$10.62 - \$4.50 = \$6.12$

This rewards their high trust while keeping outcomes relatively balanced.

```
'''json
{"return": 4.50}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*History with this co-player:\*\***

- Round 1: They sent \$3.0 (60%), I returned \$2.60 (but they saw \$3.34), they got \$5.34
- Round 2: I sent \$3.00 (60%), they saw \$2.46 sent (noise!), it became \$7.38, they returned \$3.0, they got \$4.38

**\*\*Current Round 3:\*\***

- They sent \$3.54 (70.8% of their endowment) - showing high trust
- I received \$10.62
- They currently have:  $\$5 - \$3.54 = \$1.46$

**\*\*Analysis:\*\***

The co-player has been very cooperative:

- Round 1: Sent 60%
- Round 2: Returned \$3.0 out of \$7.38 (40.7%) even though they only saw \$2.46 sent
- Round 3: Now sent \$3.54 (70.8%) - even higher trust!

This is a highly trusting and cooperative player. I should reciprocate strongly.

If I return around \$4.50:

- Sender gets:  $\$1.46 + \$4.50 = \$5.96$
- I get:  $\$10.62 - \$4.50 = \$6.12$

This seems fair - we both do better than our initial endowments, and I'm rewarding their high trust.

Alternatively, if I return \$4.20:

- Sender gets:  $\$1.46 + \$4.20 = \$5.66$
- I get:  $\$10.62 - \$4.20 = \$6.42$

Or if I return \$4.80:

- Sender gets:  $\$1.46 + \$4.80 = \$6.26$
- I get:  $\$10.62 - \$4.80 = \$5.82$

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 716,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

**\*\*Current situation:\*\***

This partnership has been mutually beneficial. They've shown consistent trust and fairness . I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

**reasoning:**

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

**usage:**

```
{
 "input_tokens": 1671,
 "output_tokens": 729,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

**\*\*Current situation:\*\***

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

**\*\*My decision:\*\***

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

**\*\*Current situation:\*\***

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

**\*\*My decision:\*\***

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:

- Sender gets: \$1.1 + \$6.00 = \$7.10

...

**usage:**

```
{
```



```
"input_tokens": 2103,
"output_tokens": 750,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.

Your total visible earnings across all rounds are \$29.360000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has shown strong, sustained cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: As receiver, generously returned \$5.0
- Round 5: Sent \$4.0 (80% - very high trust!)

**\*\*Observations:\*\***

They've consistently demonstrated trust, culminating in sending 80% of their endowment in Round 5. Despite their payoff decreasing slightly in that round (\$6.9 ? \$5.93), they haven't pulled back or changed strategy.

**\*\*My decision:\*\***

This partner has earned my trust through consistent cooperation. I should match their high-trust approach by sending a similarly substantial amount.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has been very cooperative and trustful
- Round 3: As sender, sent \$3.0 (60%), saw \$4.9 returned, payoff \$6.9
- Round 4: As receiver, returned \$5.0 from \$11.46, payoff \$6.46
- Round 5: As sender, sent \$4.0 (80% - very high!), saw \$4.93 returned, payoff \$5.93

**\*\*Observations:\*\***

1. They increased their sending from \$3.0 to \$4.0 (80%!)
2. However, in Round 5, despite sending 80%, they only saw \$4.93 returned
3. Their payoff went from \$6.9 to \$5.93 - a decrease
4. As receiver in Round 4, they were generous returning \$5.0

**\*\*Current situation:\*\***

This player has shown tremendous trust, especially sending \$4.0 (80%) in Round 5. Even though they saw a lower return than expected, they've been consistently cooperative throughout.

I should continue reciprocating their trust. They've demonstrated willingness to send high amounts. I should send a similarly high amount to maintain the cooperative relationship.

Sending \$4.0 (80%) matches their recent high-trust behavior.

```
'''json
```

```

{"send": 4.00}
'''
usage:
{
 "input_tokens": 2590,
 "output_tokens": 510,
 "reasoning_tokens": 0
}
Message 10 role=user
content:
 Round 7

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 4 against you, as RECEIVER: saw $3.82 sent, it became $11.459999999999999, returned $5.0, visible payoff $6.459999999999999.
 - Round 5 against you, as SENDER: sent $4.0, it became $12.0, saw $4.93 returned, visible payoff $5.93.
 - Round 6 against you, as RECEIVER: saw $4.5 sent, it became $13.5, returned $7.0, visible payoff $6.5.
 Your total visible earnings across all rounds are $37.980000000000004.

 This round, you are the RECEIVER against your current co-player. They sent you $4.58, so you received $13.74. How much do you return? (0-13.74)
 Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
 Looking at the history:

 Analysis:
 - My co-player has shown extraordinary escalating trust
 - Round 5: Sent $4.0 (80%)
 - Round 6: As receiver, generously returned $7.0 from $13.5
 - Round 7: Now sent $4.58 (91.6% - extremely high trust!)

 Current situation:
 - They sent $4.58, keeping only $0.42
 - I received $13.74
 - Total in play: $0.42 + $13.74 = $14.16

 My decision:
 This player has shown exceptional trust by sending over 90% of their endowment - putting themselves in a very vulnerable position. They've also been consistently generous as a receiver. I should reciprocate this extraordinary trust.

 If I return $7.50:
 - Sender gets: $0.42 + $7.50 = $7.92
 - I get: $13.74 - $7.50 = $6.24

 This rewards their exceptional trust with a strong outcome.

 '''json
 {"return": 7.50}
 '''
reasoning:
 Looking at the history:

 My co-player's recent actions:
 - Round 4: As receiver, returned $5.0 from $11.46
 - Round 5: As sender, sent $4.0 (80%)
 - Round 6: As receiver, returned $7.0 from $13.5 (52% of what they received - quite generous!)
 - Round 7: Now sent $4.58 (91.6% - extremely high trust!)

 Current situation:
 - They sent $4.58, keeping only $0.42
 - I received $13.74
 - Total in play: $0.42 + $13.74 = $14.16

 Analysis:

```

This player has shown exceptional and escalating trust:

- They've been generous as a receiver (returned \$5.0, then \$7.0)
- They've been increasingly trusting as a sender (\$3.0 ? \$4.0 ? \$4.58)
- They're now sending 91.6% of their endowment!

I should reciprocate this exceptional trust generously. They've really put themselves in a vulnerable position with only \$0.42 left.

If I return \$7.50:

- Sender gets:  $\$0.42 + \$7.50 = \$7.92$
- I get:  $\$13.74 - \$7.50 = \$6.24$

If I return \$7.00:

- Sender gets:  $\$0.42 + \$7.00 = \$7.42$
- I get:  $\$13.74 - \$7.00 = \$6.74$

Given their extreme trust (91.6%), I should return about \$7.50 to give them a solid payoff of about \$7.92.

Actually, let me think about this more. They're taking huge risk. Maybe I should be even more generous.

If I return \$8.00:

- Sender gets:  $\$0.42 + \$8.00 = \$8.42$

...

**usage:**

```
{
 "input_tokens": 3011,
 "output_tokens": 788,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.
- Round 7 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$8.08 returned, visible payoff \$8.58.

Your total visible earnings across all rounds are \$44.220000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has been exceptionally cooperative throughout our partnership
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (generous)
- Round 7: Sent \$4.5 (90%), achieved excellent payoff of \$8.58

**\*\*Observations:\*\***

Our partnership has been highly successful. They've maintained:

1. Very high trust as sender (80-90%)
2. Generous returns as receiver
3. Strong positive outcomes for both of us

**\*\*My decision:\*\***

This partnership has been mutually beneficial. They've consistently demonstrated exceptional trust, and I should match their 90% sending rate to maintain this successful cooperation.

```json

```
{"send": 4.50}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been extremely cooperative throughout
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (very generous)
- Round 7: Sent \$4.5 (90%), saw \$8.08 returned with a great payoff of \$8.58

****Observations:****

This has been a highly successful partnership. They've consistently demonstrated:

1. High trust as sender (80-90% sending rates)
2. Generous reciprocation as receiver
3. Their payoff in Round 7 was excellent (\$8.58), showing our cooperation is mutually beneficial

****My decision:****

Given their exceptional trust and the success of our partnership, I should send a high amount. They've been sending \$4.0-\$4.5 consistently. I should match this level of trust.

This is Round 8, possibly the final round or close to it. I should send a high amount to:

1. Reciprocate their consistent trust
2. Maintain the cooperative relationship
3. Maximize joint payoff

I'll send \$4.50 (90%) to match their high-trust approach.

usage:

```
{
  "input_tokens": 3219,
  "output_tokens": 479,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.
- Round 7 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$8.08 returned, visible payoff \$8.58.
- Round 8 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$7.0, visible payoff \$5.809999999999999.

Your total visible earnings across all rounds are \$50.860000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.3, so you received \$12.899999999999999. How much do you return? (0-12.899999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player continues to show very high trust
- Round 7: Sent \$4.5 (90%)
- Round 8: As receiver, returned \$7.0 from \$12.81
- Round 9: Now sent \$4.3 (86% - still very high trust)

****Current situation:****

- They sent \$4.3, keeping only \$0.7
- I received \$12.9
- Total in play: \$0.7 + \$12.9 = \$13.6

****My decision:****

This partnership has been exceptional. They've maintained very high trust throughout (sending 80-90%) and consistently returned \$7.0 as receiver. I should continue reciprocating generously.

If I return \$7.50:
- Sender gets: $\$0.7 + \$7.50 = \$8.20$
- I get: $\$12.9 - \$7.50 = \$5.40$

This maintains our strong cooperative pattern.

```
'''json
{"return": 7.50}
'''
```

Reasoning:

Looking at the history:

****Analysis:****
- My co-player continues to show very high trust
- Round 7: Sent \$4.5 (90%)
- Round 8: As receiver, returned \$7.0 from \$12.81 (consistent generosity)
- Round 9: Now sent \$4.3 (86% - still very high trust)

****Current situation:****
- They sent \$4.3, keeping only \$0.7
- I received \$12.9
- Total in play: $\$0.7 + \$12.9 = \$13.6$

****Pattern review:****
This has been one of the most cooperative partnerships. They've:
- Sent 80-90% of their endowment as sender
- Returned \$7.0 consistently as receiver
- Maintained trust even when payoffs varied

****My decision:****
They've sent \$4.3 (86%), which is slightly less than their previous \$4.5 but still shows very high trust. I should continue to reciprocate generously.

If I return \$7.00:
- Sender gets: $\$0.7 + \$7.00 = \$7.70$
- I get: $\$12.9 - \$7.00 = \$5.90$

This maintains the pattern of fair returns we've established. Alternatively, I could go slightly higher given they're sending such a high percentage.

If I return \$7.50:
- Sender gets: $\$0.7 + \$7.50 = \$8.20$
- I get: $\$12.9 - \$7.50 = \$5.40$

This would match or slightly exceed what I've been returning. Let me go with \$7.50 to maintain the strong reciprocity.

```
Usage: {
  "input_tokens": 3378,
  "output_tokens": 661,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:03:17
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    }
  }
}
```

```

    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.

Your total visible earnings across all rounds are \$11.83.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our interaction pattern:

****Round 1:**** I sent \$3, they returned \$5.0 (good reciprocity)
****Round 2:**** They sent \$3.5, I returned \$4.5 (they saw \$5.25)

This co-player has shown:

- Consistent cooperation
- Fair returns as receiver
- Increased trust as sender (sent \$3.5)

I'll match their demonstrated trust level:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.39)
- They received \$10.17
- They returned \$5.0 (about 49%)
- Their visible payoff: \$5.17

****Round 2:****

- They sent \$3.5 (I saw \$2.86 after noise)
- I received \$8.58
- I returned \$4.5 (they saw \$5.25 after noise)
- Their visible payoff: \$6.75
- They see that they sent \$3.5 and got back \$5.25, so they profited \$1.75

So far this co-player has:

1. Been a reasonable receiver, returning about half
2. Sent a good amount as sender (\$3.5)
3. They've been cooperative

Since they sent \$3.5 last round (more than I sent in Round 1), I should maintain or increase my cooperation level. They seem trustworthy based on the pattern.

I'll send \$3.5 to match what they sent, showing I'm willing to reciprocate their level of trust.

usage:

```
{
  "input_tokens": 1140,
  "output_tokens": 393,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.20000000000001, returned \$7.0, visible payoff \$6.200000000000001.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)
****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)
****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****
- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****
- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****
- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****
- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.

Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our strong cooperative pattern:

****Round 1:**** I sent \$3, they returned \$5.0
****Round 2:**** They sent \$3.5, I returned \$4.5
****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$5.0
Round 2: They sent \$3.5, I returned \$4.5 (they saw \$5.25)
Round 3: I sent \$3.5 (they saw \$4.4), they returned \$7.0
Round 4: They sent \$4.0, I returned \$5.5 (they saw \$6.46)

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

usage:

```
{
  "input_tokens": 1876,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.

Your total visible earnings across all rounds are \$33.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued strong cooperation:

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Round 4:**** I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
****Round 5:**** They returned \$6.0 from \$11.28 (~53%)

****Current round:**** I received \$14.19

They sent a high amount (\$4.73) showing strong trust. I'll maintain our reciprocal pattern
:

```
'''json
{"return": 7.5}
'''
```

This is approximately 53% of what I received, consistent with their return rates.

reasoning:

Looking at the game history with this co-player:

Round 3: They returned \$7.0 from \$13.2 (~53%)
Round 4: I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
Round 5: They returned \$6.0 from \$11.28 (~53%)

Current round: They sent \$4.73, I received \$14.19

They've been very cooperative and generous throughout. They've increased their sending and maintained good return rates. I should reciprocate appropriately.

Looking at their returns: around 53-54%
I'll return around 53-55% of what I received.

$\$14.19 * 0.53 = \7.52
 $\$14.19 * 0.54 = \7.66
 $\$14.19 * 0.55 = \7.80

I'll return \$7.5 which is about 53%, maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 2266,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.

Your total visible earnings across all rounds are \$39.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our highly cooperative relationship:

****Round 4:**** They sent \$4.0, I returned \$5.5
****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0, I returned \$7.5

This co-player has demonstrated:

- Consistent high trust (repeatedly sending \$4.0)
- Very generous returns
- Strong reciprocity

I'll continue matching their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track our interaction:

```
**Round 1:** I sent $3, they returned $5.0
**Round 2:** They sent $3.5, I returned $4.5
**Round 3:** I sent $3.5, they returned $7.0
**Round 4:** They sent $4.0, I returned $5.5
**Round 5:** I sent $4.0, they returned $6.0
**Round 6:** They sent $4.0 (I saw $4.73 due to noise), I returned $7.5 (they saw $8.42)
```

Looking at the pattern:

- They've been consistently sending around \$3.5-\$4.0
- They've been returning generously
- Round 6 shows they saw \$8.42 returned from \$12.0, which is about 70% - very high!
- This is a strong collaborative partner

They've maintained sending \$4.0 in rounds 4 and 6. I should continue with a similar level of trust. I'll send \$4 again to maintain consistency and mutual trust.

usage:

```
{
  "input_tokens": 2640,
  "output_tokens": 381,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.
- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.

Your total visible earnings across all rounds are \$47.679999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our established cooperative pattern:

```
**Round 5:** They returned $6.0 from $11.28 (~53%)
**Round 6:** I returned $7.5 (they saw $8.42)
**Round 7:** They returned $6.5 from $12.09 (~54%)
```

****Current round:**** I received \$11.64

They sent \$3.88, continuing to show trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 6.5}
'''
```

This is approximately 56% of what I received, maintaining our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:

```
**Round 5:** They returned $6.0 from $11.28 (~53%)
**Round 6:** I returned $7.5 (they saw $8.42 from $12.0, ~70%)
**Round 7:** They returned $6.5 from $12.09 (~54%)
```

****Current round:**** I received \$11.64

They sent \$3.88, showing continued trust. Looking at their return rates, they've been consistently around 53-54% when they're the receiver.

I've been returning generously (my actual return of \$7.5 in round 6 showed up as \$8.42 to them).

To maintain the cooperative relationship, I should return around 53-55% of what I received :

- 53% of \$11.64 = \$6.17
- 55% of \$11.64 = \$6.40

I'll go with around \$6.5 to match their recent return pattern and maintain fairness.

usage:

```
{
  "input_tokens": 2826,
  "output_tokens": 386,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.
- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$52.81999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued cooperation:

****Round 6:**** They sent \$4.0, I returned \$7.5
****Round 7:**** I sent \$4.0, they returned \$6.5
****Round 8:**** They sent \$4.0, I returned \$6.5

This co-player has consistently:

- Sent \$4.0 showing high trust
- Returned generously when in receiver role
- Maintained a stable cooperative pattern

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the pattern:

Round 6: They sent \$4.0, I returned \$7.5 (they saw \$8.42)
Round 7: I sent \$4.0, they returned \$6.5
Round 8: They sent \$4.0, I returned \$6.5 (they saw \$5.56)

Our co-player has been consistently sending \$4.0 and showing strong trust. They've been returning generously as well.

I should continue sending at a similar level to maintain this cooperative relationship.

```
'''json
{"send": 4}
'''
```

Usage: {

```
"input_tokens": 2860,  
"output_tokens": 265,  
"reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:03:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.

Your total visible earnings across all rounds are \$11.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), I returned generously
- Round 2: They saw \$2.86 sent (noise from my actual \$3.50), still returned \$4.5 which was quite fair
- Round 3 (current): They sent \$4.4 (88% of their endowment!) - very high trust

The sender showed exceptional trust this round by sending 88% of their endowment. They currently have only \$0.60 left.

I should reward this high level of trust. If I return \$7.0:

- Sender ends with \$7.60 (profit of \$2.60)
- I keep \$6.20

This appropriately rewards their trust with a solid profit.

{'return': 7.0}

reasoning:

Let me analyze the situation:

Round history with this co-player:

- Round 1: They sent \$3.0 (60%), received \$5.75 back
- Round 2: I sent \$3.50 (as sender), they saw \$2.86 sent (noise!), they returned \$4.5
- Round 3 (current): They sent \$4.4 (88% of endowment!)

The co-player showed very high trust this round by sending \$4.4 (88% of their \$5). This is significantly more than previous rounds.

Looking at Round 2: I actually sent \$3.50, but they saw \$2.86 due to noise. Despite seeing a lower amount, they still returned \$4.5, which is quite generous. If they saw \$2.86, that became \$8.58, and they returned \$4.5 (52.4% of what they received).

Now in Round 3, they're showing even more trust by sending \$4.4. I received \$13.20.

The sender currently has: \$5 - \$4.4 = \$0.60

I should reward this high level of trust appropriately. Let me consider:

If I return \$7.0:

- Sender ends with: \$0.60 + \$7.0 = \$7.60 (profit of \$2.60)
- I keep: \$13.20 - \$7.0 = \$6.20

This gives them a solid profit and rewards their trust while also allowing me to benefit.

Alternatively, if I return \$6.5:

- Sender ends with: $\$0.60 + \$6.5 = \$7.10$ (profit of \$2.10)
- I keep: $\$13.20 - \$6.5 = \$6.70$

Or \$7.5:

- Sender ends with: $\$0.60 + \$7.5 = \$8.10$ (profit of \$3.10)

...

usage:

```
{
  "input_tokens": 1103,
  "output_tokens": 720,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
 - Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
 - Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

{'return': 6.0}

reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receiver, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 1873,
  "output_tokens": 414,
  "reasoning_tokens": 0
}
```


}

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$30.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Co-player has consistently shown trust, sending \$3.5 and \$4.0 (70-80% of endowment)
- As receiver, they returned fairly (\$5.5)
- They've seen generous returns from me (\$7.3, \$6.52)
- We've established a strong cooperative pattern over multiple rounds

Given this reliable cooperation, I should maintain high trust. I'll send \$4.0 (80% of endowment) to continue our successful partnership.

{'send': 4.0}

reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$3.5, saw \$7.3 returned, payoff \$8.8
- Round 4: They saw \$3.55 sent (noisy version of my \$4.0), returned \$5.5, payoff \$5.15
- Round 5: They sent \$4.0, saw \$6.52 returned (noisy version of my \$6.0), payoff \$7.52

The pattern shows:

- Co-player has been sending good amounts as sender (\$3.5, \$4.0)
- As receiver, they've been fair (\$5.5 return)
- They've seen me return generously (\$7.3, \$6.52)
- We've maintained strong cooperation

This is a reliable, cooperative partner. I should continue showing high trust. Since they've been sending 70-80% of endowment, I should match or slightly exceed that level.

I'll send \$4.0 again (80% of endowment) to maintain our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.
- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.

Your total visible earnings across all rounds are \$40.28.

This round, you are the RECEIVER against your current co-player. They sent you \$4.03, so you received \$12.09. How much do you return? (0-12.09)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Co-player has been highly cooperative, consistently sending high amounts (80%+ of endowment)
- As receiver, they've returned fairly (\$5.5, \$7.5)
- They sent \$4.03 this round (80.6%), showing strong continued trust
- They currently have \$0.97

This is very similar to Round 5 when they sent \$4.0. I should maintain consistency in our cooperative relationship. If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This appropriately rewards their ongoing trust.

{'return': 6.5}

reasoning:

Looking at the visible history with this co-player:

- Round 4: They received, returned \$5.5
- Round 5: They sent \$4.0 (80% of endowment), saw \$6.52 returned
- Round 6: They received, returned \$7.5
- Round 7 (current): They sent \$4.03 (80.6% of endowment)

This co-player has been very cooperative throughout. They're consistently sending high amounts (around 80%) and returning fairly when they're the receiver.

Current situation:

- They sent \$4.03, so they have \$0.97 left
- I received \$12.09

I should continue reciprocating their high level of trust. They've been a reliable partner

If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This seems fair given their high trust level. Let me think if this is appropriate...

Actually, looking at the pattern, when they sent \$4.0 in Round 5, I returned around \$6.52 (visible). Now they sent \$4.03, very similar. I should be consistent and return a similar amount.

\$6.5 seems reasonable and consistent with our established pattern.

usage:

```
{
  "input_tokens": 2618,
  "output_tokens": 464,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.
- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$45.870000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Co-player has consistently sent \$4.0 (80%) as sender in recent rounds
- As receiver, they returned \$7.5 fairly
- They've seen good returns from me (\$6.52, \$6.69)
- We have a well-established pattern of mutual cooperation

This is the final round with this co-player, and we've built strong trust. I'll send \$4.0 (80%) to maintain our cooperative relationship.

{'send': 4.0}

reasoning:

Looking at the visible history with my co-player:

- Round 5: They sent \$4.0, saw \$6.52 returned, payoff \$7.52
- Round 6: As receiver, saw \$4.73 sent, returned \$7.5, payoff \$6.69
- Round 7: They sent \$4.0, saw \$6.69 returned, payoff \$7.69

This co-player has been highly cooperative:

- Consistently sending high amounts as sender (\$4.0 multiple times)
- Being fair as receiver (\$7.5 return on \$14.19 received)
- We've established a strong pattern of cooperation

Given this excellent track record over many rounds, I should continue with high trust. Sending \$4.0 or even \$4.5 would be appropriate to maintain this successful partnership.

I'll send \$4.0 to maintain consistency with our established cooperative pattern.

usage:

```
{
  "input_tokens": 2791,
  "output_tokens": 336,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 8 against you, as RECEIVER: saw \$3.88 sent, it became \$11.64, returned \$6.5, visible payoff \$5.140000000000001.

Your total visible earnings across all rounds are \$52.43000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

- Co-player has been consistently cooperative throughout our games
- They've returned fairly as receiver (\$7.5, \$6.5)
- This round they sent \$4.97 (99.4% of their endowment!) - exceptionally high trust
- They currently have only \$0.03 left

This is an extraordinary display of trust - they sent nearly everything. I should reward this generously. If I return \$8.5:

- Sender ends with \$8.53 (profit of \$3.53)
- I keep \$6.41

This appropriately rewards their exceptional trust.

{'return': 8.5}

Reasoning:

Looking at the visible history with my co-player:

- Round 6: They saw \$4.73 sent, returned \$7.5 (good return rate)
- Round 7: They sent \$4.0, saw \$6.69 returned
- Round 8: They saw \$3.88 sent, returned \$6.5 (fair return)
- Round 9 (current): They sent \$4.97 (99.4% of their endowment!)

Wow, they sent \$4.97 out of \$5 - that's almost everything! This is an extremely high trust signal, leaving them with only \$0.03.

This is exceptional trust - they're basically giving me almost their entire endowment. I should reward this very generously.

If I return \$8.0:

- Sender ends with $\$0.03 + \$8.0 = \$8.03$ (profit of \$3.03)
- I keep $\$14.91 - \$8.0 = \$6.91$

This gives them a good profit while I still keep a reasonable amount.

Actually, given the extremely high trust (sending 99.4%), I could be even more generous. Let me think about what's fair.

If I return \$8.5:

- Sender ends with \$8.53 (profit of \$3.53)
- I keep \$6.41

This more generously rewards their exceptional trust.

```
Usage: {  
  "input_tokens": 2930,  
  "output_tokens": 485,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:03:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.

Your total visible earnings across all rounds are \$8.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history with this co-player:

Round 1: They received and returned \$3.5 (56% of what they got) - cooperative

Round 2: They sent \$3.0, I returned \$3.5 - we both cooperated

This co-player has shown consistent cooperation. We've established a good reciprocal pattern. I'll continue this cooperative approach:

```
'''json
{"send": 3}
'''
```

Sending \$3 again maintains our established trust and should lead to mutual benefit.

reasoning:

Let me analyze the history with this co-player:

Round 1: I was sender
- They saw \$2.08 sent (but I sent \$3, so there's communication noise)
- It became \$6.24 for them (should be \$6.24 if they saw \$2.08)
- They returned \$3.5
- Their visible payoff: \$2.74

Round 2: I was receiver
- They sent \$3.0
- It became \$9.0 for me
- I returned \$3.5
- They saw \$3.47 returned (I actually returned \$3.5, so small noise)
- Their visible payoff: \$5.47

So this co-player has been cooperative:
- As receiver in R1: returned \$3.5 out of \$6.24 (56%)
- As sender in R2: sent \$3.0 out of \$5

And I:
- Sent \$3 in R1 (they saw \$2.08 due to noise)
- Returned \$3.5 in R2 (they saw \$3.47 due to noise)

My total visible earnings: \$8.86

This co-player seems cooperative. We've established a pattern of sending around \$3 and returning a decent amount. I should continue cooperating.

I'll send \$3 again to maintain the cooperative pattern.

usage:

```
{  
  "input_tokens": 1081,  
  "output_tokens": 442,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)
Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).
I'll match their generous pattern:

```
'''json  
{"return": 4.5}  
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:
- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5

- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.

Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of \$8.49 (generous!)

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5

Round 3: They returned \$4.5 out of what they received

Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with

th that proven strategy.

usage:

```
{
  "input_tokens": 1778,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$27.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.4, so you received \$7.199999999999999. How much do you return? (0-7.199999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

Round 3: They returned \$4.5 out of \$8.49

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69 (very generous!)

This round they sent \$2.4 (becoming \$7.20) - slightly less than their usual \$3.0. I'll return a proportionate amount while maintaining our cooperative relationship:

```
'''json
{"return": 3.5}
'''
```

This gives them a reasonable return while accounting for the slightly lower send amount.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 out of \$8.49

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69

They've been very cooperative and generous. Now they sent \$2.4 (becoming \$7.20).

This is the first round where they sent less than \$3.0. Let me check the pattern:

- When they sent \$3.0, I returned around \$4.5
- When I sent (visible as \$2.83, \$3.23), they returned \$4.5 and \$5.0

Now they sent \$2.4. I should return a fair amount. Let me calculate:

- They sent \$2.4, so they kept \$2.6
- It became \$7.20 for me
- If I return around 50-55% of what I received, that's \$3.6-\$4.0
- Their final payoff would be: \$2.6 + return
- My final payoff would be: \$7.20 - return

If I return \$3.5:

- They get: \$2.6 + \$3.5 = \$6.1
- I get: \$7.20 - \$3.5 = \$3.7

If I return \$4.0:

- They get: \$2.6 + \$4.0 = \$6.6
- I get: \$7.20 - \$4.0 = \$3.2

Given they sent slightly less this round, I'll return a proportional amount. Since they've been very generous when they were receiver, I'll still be generous but slightly adjust

for their lower send.

\$3.5 seems fair - it's about 49% of what I received, reasonable reciprocity.

usage:

```
{
  "input_tokens": 2139,
  "output_tokens": 574,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.689999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$30.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing history:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 (very generous!)

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently trustworthy and cooperative throughout. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history with this co-player:

Round 4: They sent \$3.0, I returned \$4.5

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently cooperative throughout our interactions. They've consistently sent around \$3.0 when they're the sender and returned generously as receiver. I should continue our successful cooperative pattern.

I'll send \$3 again to maintain our established trust.

usage:

```
{
  "input_tokens": 2499,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.689999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.

Your total visible earnings across all rounds are \$37.47.

This round, you are the RECEIVER against your current co-player. They sent you \$3.47, so you received \$10.41. How much do you return? (0-10.41)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 out of \$10.38 (very generous!)

They just sent \$3.47 (becoming \$10.41) - showing continued trust. I'll reciprocate their generous pattern:

```
'''json
{"return": 5.5}
'''
```

This matches their generous cooperation and maintains our strong relationship.

reasoning:

Looking at the history with this co-player:

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 out of \$10.38

This co-player has been consistently cooperative and generous. They just sent me \$3.47 (becoming \$10.41).

Looking at their return pattern:

- Round 5: \$5.0 / \$9.69 = 51.6%

- Round 7: \$5.5 / \$10.38 = 53%

They're returning around 50-53% of what they receive, which is quite generous and cooperative.

For \$10.41 received, returning around 50-53% would be:

- 50%: \$5.20

- 53%: \$5.52

I should reciprocate their generosity. I'll return around \$5.5 to match their cooperative pattern.

usage:

```
{
  "input_tokens": 2661,
  "output_tokens": 363,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.

- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$42.379999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our strong cooperative pattern:

Round 6: They sent \$3.0, I returned \$3.5
Round 7: They returned \$5.5 (very generous!)
Round 8: They sent \$3.0, I returned \$5.5

This co-player has been consistently trustworthy and cooperative across all 8 rounds. I'll continue our successful strategy:

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the pattern:

Round 6: They sent \$3.0, I returned \$3.5
Round 7: They returned \$5.5 from \$10.38
Round 8: They sent \$3.0, I returned \$5.5

This co-player has been very cooperative throughout all rounds. They consistently send \$3.0 when they're the sender and return generously when they're the receiver. We've built strong mutual trust.

I should continue sending \$3 to maintain this successful cooperative relationship.

```
Usage: {
  "input_tokens": 2754,
  "output_tokens": 225,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:03:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.

Your total visible earnings across all rounds are \$8.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.83, so you received \$8.49. How much do you return? (0-8.49)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

My co-player has been consistently generous:

- Round 1: sent \$3.0 (60% of endowment) as sender
- Round 2: returned \$3.5 out of \$6.48 (54%) as receiver
- Round 3 (now): sent \$2.83 (57% of endowment) as sender

They've shown strong trust and fair reciprocity. I should match this cooperative behavior.

They currently have \$2.17 remaining. If I return \$4.50:

- Sender ends with: \$6.67 (profit of \$1.67)
- I end with: \$3.99 (profit of \$3.99)

This rewards their continued trust and keeps our cooperative relationship strong.

{'return': 4.50}

reasoning:

Let me analyze the history:

Round 1: My co-player was sender, sent \$3.0 (60%), I received \$9.0, they saw \$3.88 returned

Round 2: I was sender, they saw \$2.16 sent (but I sent \$3.0), it became \$6.48, they returned \$3.5, visible payoff \$2.98

Now Round 3: I'm receiver, they sent \$2.83 (56.6% of \$5), I received \$8.49

The co-player has been consistently generous:

- Round 1: sent 60%
- Round 3: sent 56.6%

In Round 2, they returned \$3.5 out of \$6.48 received (54%). That's a fairly generous return rate.

Now they've sent me \$2.83, showing continued trust. They currently have $\$5 - \$2.83 = \$2.17$.

I should return a fair amount that:

1. Rewards their trust
2. Leaves us both better off
3. Maintains the cooperative relationship

If I return \$4.50:

- Sender ends with: $\$2.17 + \$4.50 = \$6.67$ (profit of \$1.67)
- I end with: $\$8.49 - \$4.50 = \$3.99$ (profit of \$3.99)

If I return \$4.25:

- Sender ends with: $\$2.17 + \$4.25 = \$6.42$ (profit of \$1.42)
- I end with: $\$8.49 - \$4.25 = \$4.24$ (profit of \$4.24)

If I return \$4.00:

- Sender ends with: $\$2.17 + \$4.00 = \$6.17$ (profit of \$1.17)

...

usage:

```
{
  "input_tokens": 1199,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

```
{'send': 3.0}
```

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned
Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5
Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{  
  "input_tokens": 1611,  
  "output_tokens": 357,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)
- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)
- I end with: \$4.69 (profit of \$4.69)

{'return': 5.0}

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)
Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)
Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)
Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)
Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: $\$1.77 + \$5.0 = \$6.77$ (profit of \$1.77)
- I end with: $\$9.69 - \$5.0 = \$4.69$ (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

usage:

```
{
  "input_tokens": 1982,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.

Your total visible earnings across all rounds are \$23.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 3 and 5
- As receiver: returned \$4.5 (52%) in round 4

We've established a strong pattern of mutual trust and fair reciprocity. I should continue this successful cooperation by sending the same amount.

```
{'send': 3.0}
```

reasoning:

Looking at my co-player's history:

- Round 3: as sender, sent \$3.0 (60%)
- Round 4: as receiver, returned \$4.5 out of \$8.64 (52%)
- Round 5: as sender, sent \$3.0 (60%)

They've been consistently sending 60% when they're the sender, and returning around 50-52% when they're the receiver. This is a very cooperative and trustworthy pattern.

I've been sending \$3.0 (60%) consistently as well, and we have a strong mutual cooperation going.

I should continue with the same level of trust - sending \$3.0 (60%) to maintain our successful partnership.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 265,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.

Your total visible earnings across all rounds are \$28.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

My co-player has been consistently cooperative and just sent \$3.46 (69% of endowment), their highest trust level yet! They currently have \$1.54 remaining.

I've been consistently returning around 52-56% of what I receive. I should maintain this fair reciprocity to reward their increased trust.

If I return \$5.5:

- Sender ends with: \$7.04 (profit of \$2.04)
- I end with: \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains our strong cooperative relationship.

{'return': 5.5}

reasoning:

Looking at the history with my co-player:

Round 4: I was sender, they were receiver - saw \$2.88 sent, returned \$4.5 out of \$8.64 (52%)

Round 5: They were sender, I was receiver - sent \$3.0, I returned visible \$5.62

Round 6: I was sender, they were receiver - saw \$2.4 sent (but I sent \$3.0 due to noise), returned \$3.5 out of \$7.2 (49%)

Round 7: They are sender, I am receiver - sent \$3.46 (69% of endowment!), I received \$10.38

They've increased their trust level to 69%, the highest yet. They currently have \$1.54 remaining.

I should reward this high trust. Looking at my previous returns:

- Round 1: Returned \$3.50 out of \$6.24 (56%)
- Round 3: Returned \$4.50 out of \$8.49 (53%)
- Round 5: Returned \$5.0 out of \$9.69 (52%)

I've been returning around 52-56% consistently. Given they sent an even larger proportion this time (69%), I should maintain or slightly increase my return rate.

If I return \$5.5:

- Sender ends with: \$1.54 + \$5.5 = \$7.04 (profit of \$2.04)
- I end with: \$10.38 - \$5.5 = \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains my consistent cooperative pattern and rewards their increased trust.

usage:

```
{
  "input_tokens": 2715,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.59 returned, visible payoff \$6.59.

Your total visible earnings across all rounds are \$33.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 5 and 7
- As receiver: returned \$3.5 (49%) in round 6

We've established a robust pattern of mutual trust. I should continue this successful cooperation by sending the same amount.

{'send': 3.0}

reasoning:

Looking at the history:

My co-player has consistently sent \$3.0 (60%) in rounds 5 and 7 as sender. As receiver in round 6, they returned \$3.5 out of \$7.2 (49%).

We've established a strong pattern of cooperation throughout these rounds. I should continue matching their trust level by sending \$3.0 (60% of my endowment).

This maintains our mutually beneficial relationship.

usage:

```
{
  "input_tokens": 2827,
  "output_tokens": 200,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.59 returned, visible payoff \$6.59.
- Round 8 against you, as RECEIVER: saw \$3.47 sent, it became \$10.41, returned \$5.5, visible payoff \$4.91.

Your total visible earnings across all rounds are \$41.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.4399999999999995. How much do you return? (0-7.4399999999999995)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

My co-player sent \$2.48 (50% of endowment), slightly less than their typical 60%. They currently have \$2.52 remaining.

Throughout our games, I've been returning around 50-54% of what I receive. I should maintain this consistent reciprocity.

If I return \$4.0:

- Sender ends with: \$6.52 (profit of \$1.52)

- I end with: \$3.44 (profit of \$3.44)
- Return rate: 54%

This maintains our cooperative relationship fairly.

```
{'return': 4.0}
```

Reasoning:

Looking at the history:

My co-player has been consistently cooperative throughout the game. Let me analyze:

- As sender: sent \$3.0 (60%) in rounds 5, 7, and now \$2.48 (50%) in round 9 - a slight decrease
- As receiver: returned \$3.5 (49%) in round 6 and \$5.5 (53%) in round 8

They sent \$2.48, which is about 50% of their endowment. They currently have \$2.52 remaining.

I've been returning around 50-53% of what I receive. I should maintain this fair reciprocity.

Received: \$7.44

If I return \$4.0:

- Sender ends with: \$6.52 (profit of \$1.52)
- I end with: \$3.44 (profit of \$3.44)
- Return rate: 54%

This maintains our cooperative relationship while being fair.

Usage: {

```
"input_tokens": 2892,
"output_tokens": 379,
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:03:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_7",
    "Agent_8"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
```

```
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$13.989999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together, this co-player has been consistently cooperative:

- Round 1: They received \$11.97 and returned \$5.5 (about 46% - very fair)
- Round 2: They sent \$3.5, showing continued trust, and I returned \$5.5

We've established a good cooperative pattern. I should maintain this by sending a similar amount to what I sent in Round 1 and what they sent in Round 2.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3 (they saw \$3.99), they received \$11.97, returned \$5.5. I ended up with \$5 - \$3 + \$5.5 = \$7.5 (visible says \$7.61, accounting for noise)

Round 2: They sent \$3.5 (I saw \$3.96), I received \$11.88, I returned \$5.5. They ended up with \$5 - \$3.5 + \$5.4 (what they saw) = \$6.9

This co-player has been consistently cooperative:

- As receiver, they returned \$5.5 when receiving \$11.97 (46%)
- As sender, they sent \$3.5 showing trust
- I returned \$5.5 to them

This is a mutually beneficial relationship. We're both cooperating well. I should continue sending a similar amount to maintain this positive pattern.

Sending around \$3-3.5 again makes sense.

usage:

```
{
  "input_tokens": 1079,
  "output_tokens": 370,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.

Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)

Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)

Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)

Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

usage:

```
{
  "input_tokens": 1444,
  "output_tokens": 410,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.

Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)
- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

usage:

```
{
  "input_tokens": 1810,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.

Your total visible earnings across all rounds are \$32.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent \$4.18 (more than their usual \$3.5-3.75), showing increased trust. I received \$12.54. I should reciprocate this increased trust with a proportionally fair return:

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent me \$4.18, which is higher than their usual amounts (around \$3.5-3.75). They received \$12.54.

They're showing increased trust by sending more. I should reciprocate appropriately. Given they sent more, I should return a bit more to maintain the cooperative relationship.

Our typical pattern has been returning around \$5-5.5 from amounts around \$10-11. Now I have \$12.54, so returning around \$6 would be proportionate and fair, maintaining the cooperative spirit.

usage:

```
{
  "input_tokens": 2175,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$39.529999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

We've maintained a very stable and mutually beneficial pattern. I'll continue with the same strategy:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the recent history:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

Wait, looking at my visible history, there's some inconsistency. In Round 6, it says "They sent you \$4.18" but now the history says they sent \$3.75. This is probably the communication noise mentioned in the instructions.

Our pattern has been very stable with sends around \$3.5-3.75 and returns around \$5.0-5.5. We've built strong mutual cooperation. I should continue with this pattern:

```
'''json
{"send": 3.75}
'''
```

usage:

```
{
  "input_tokens": 2520,
  "output_tokens": 287,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.7100000000000001.

Your total visible earnings across all rounds are \$48.089999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our recent pattern:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71 (51% - very generous!)

They've sent \$4.33, giving me \$12.99. They've been increasingly generous, so I should reciprocate with a fair return that maintains our strong cooperation:

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at the history:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71

In Round 7, they sent \$4.57 which became \$13.71, and they returned \$7.0 (51%). That's a very generous return rate!

Now they've sent me \$4.33, which became \$12.99. This is a high amount showing continued trust.

I should reciprocate with a proportionally fair return. Given they returned \$7.0 from \$13.71 last round (51%), I should aim for something similar or slightly less.

$\$12.99 * 0.50 = \6.495

$\$12.99 * 0.52 = \6.75

I'll return around \$6.5 to maintain our strong cooperation.

usage:

```
{
  "input_tokens": 2702,
  "output_tokens": 348,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.710000000000001.
- Round 8 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.92 returned, visible payoff \$7.17.

Your total visible earnings across all rounds are \$54.57999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

- Round 6: They sent \$3.75, I returned \$6.0
- Round 7: I sent \$3.75, they returned \$7.0 (very generous!)
- Round 8: They sent \$3.75, I returned \$6.5

We have an excellent cooperative relationship. I'll continue with the same amount:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at the recent history:

- Round 6: They sent \$3.75, I returned \$6.0
- Round 7: I sent \$3.75, they returned \$7.0
- Round 8: They sent \$3.75, I returned \$6.5

They've been consistently sending around \$3.75 and have been very cooperative with their returns (they returned \$7.0 in Round 7, which was very generous). I've been returning around \$6-6.5 which has been working well.

I should continue with a similar sending amount to maintain this strong cooperative relationship. \$3.75 has been working well for both of us.

Usage: {

```
"input_tokens": 2777,
"output_tokens": 258,
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:03:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_7",
    "Agent_8"
  ],
"investor": "Agent_7",
"roles": {
  "Agent_7": "investor",
  "Agent_8": "trustee"
},
```



```

    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

Your total visible earnings across all rounds are \$13.370000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our history:

Round 1: My co-player sent \$3.0 (60%) and I returned a fair amount

Round 2: I sent \$3.50 (70%) and they returned \$5.50 from \$11.88 received (about 46%)

Round 3: They sent \$3.45 (69%) showing continued trust

They've been consistently cooperative, sending high amounts and returning fair shares. I should reciprocate similarly.

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$

- I end with: $\$10.35 - \$5.00 = \$5.35$

This maintains our cooperative pattern and is proportionally similar to what they returned in Round 2.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- My co-player sent \$3.0 (60% of endowment)
- They saw \$5.61 returned (from their perspective)
- Their visible payoff: \$7.61

Round 2: I was SENDER

- I sent \$3.50 (70% of my endowment)
- They saw I sent \$3.96 (due to noise)
- They returned \$5.50
- Their visible payoff: \$6.38
- My actual payoff: $\$1.50 + \$5.50 = \$7.00$

Round 3: I am RECEIVER

- They sent \$3.45 (69% of their endowment)
- I received \$10.35
- They currently have $\$5.00 - \$3.45 = \$1.55$

The pattern shows:

1. They've been consistently sending around 60-70% of their endowment
2. They returned \$5.50 when they received \$11.88 (about 46%)
3. This is a cooperative relationship

If I return \$5.00:

- Sender ends with: $\$1.55 + \$5.00 = \$6.55$
- I end with: $\$10.35 - \$5.00 = \$5.35$

This roughly matches what they did - they returned \$5.50 when they received \$11.88. I'm returning \$5.00 when I received \$10.35, which is a similar proportion (48% vs 46%).

This maintains the cooperative equilibrium we've established.

usage:

```
{
  "input_tokens": 1172,
  "output_tokens": 576,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$
- I end with: $\$10.23 - \$5.25 = \$4.98$

```
'''json
{"return": 5.25}
'''
```

reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5

Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)

Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5

Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$
- I end with: $\$10.23 - \$5.25 = \$4.98$

This is fair and consistent with our pattern.

usage:

```
{
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.

Your total visible earnings across all rounds are \$30.390000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant**content:**

Looking at our ongoing cooperation:

- Round 3: Co-player sent \$3.5
- Round 4: I sent \$3.75, they returned \$5.5

- Round 5: Co-player sent \$3.75

We've established a very strong cooperative pattern. Both of us consistently send high amounts (around \$3.50-\$3.75) and return fair shares. This has been mutually beneficial - my total earnings of \$30.39 reflect this.

I'll continue this successful cooperation:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with my co-player:

Round 3: Co-player sent \$3.5, I returned (they saw \$5.95)

Round 4: I sent \$3.75, co-player returned \$5.5

Round 5: Co-player sent \$3.75, I returned (they saw \$5.21)

We've been maintaining a very cooperative relationship. Both of us have been sending high amounts (around \$3.50-\$3.75) and returning fair shares.

My earnings are \$30.39 total, which suggests this cooperation is working well.

The co-player has consistently sent high amounts (\$3.5, \$3.75) showing strong trust. They've also been fair when receiving, returning \$5.5 consistently.

I should continue this cooperative pattern by sending a high amount. I'll send \$3.75 to match what they've been sending.

usage:

```
{
  "input_tokens": 2386,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.

Your total visible earnings across all rounds are \$36.960000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.57, so you received \$13.71. How much do you return? (0-13.71)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

- Round 4: Co-player returned \$5.5 from \$10.59
- Round 5: Co-player sent \$3.75
- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$4.57 (91.4% of their endowment!)

This is a very strong trust signal - they sent almost everything! I should reciprocate this high level of trust with a generous return.

If I return \$7.00:

- Sender ends with: \$0.43 + \$7.00 = \$7.43
- I end with: \$13.71 - \$7.00 = \$6.71

This rewards their exceptional trust and maintains our strong cooperative relationship.

```
'''json
```

```
    {"return": 7.00}  
    ,,,
```

reasoning:

Looking at the history:

Round 4: Co-player as receiver, returned \$5.5 from \$10.59

Round 5: Co-player as sender, sent \$3.75

Round 6: Co-player as receiver, they saw I sent \$4.18 which became \$12.54, and they returned \$6.0

Round 7: Co-player sent me \$4.57 (91.4% of their endowment) - this is a very high trust signal!

They've been escalating their cooperation. In Round 6, they returned \$6.0 from \$12.54 (about 48%). Now they're sending \$4.57, which is much higher than their previous sends.

I received \$13.71 after the multiplier.
They currently have: \$5 - \$4.57 = \$0.43

If I return \$6.50:

- Sender ends with: $\$0.43 + \$6.50 = \$6.93$
- I end with: $\$13.71 - \$6.50 = \$7.21$

This rewards their very high trust and continues our cooperative relationship.

Actually, they sent 91.4% - that's a very strong trust signal. Let me be more generous to match that.

If I return \$7.00:

- Sender ends with: $\$0.43 + \$7.00 = \$7.43$
- I end with: $\$13.71 - \$7.00 = \$6.71$

This is a bit more balanced and rewards the very high send amount.

usage:

```
{  
  "input_tokens": 2753,  
  "output_tokens": 564,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.

Your total visible earnings across all rounds are \$43.67000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our continued strong cooperation:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, received good return

We've built an excellent cooperative relationship. My total earnings of \$43.67 show this has been very successful. Both of us have consistently sent high amounts and returned generously.

I'll maintain this cooperative pattern:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at our history:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, I returned \$7.00 (they saw \$7.31)
- Round 8: Now I'm the sender

Wait, let me re-check the history:

- Round 7 it says "as SENDER: sent \$3.75" but in Round 7 they sent me \$4.57. There's noise in the communication.

The visible history says their Round 7 they sent \$3.75 and saw \$7.31 returned. But from my perspective in Round 7, they sent \$4.57 and I returned \$7.00.

So there's significant noise. They're seeing different amounts than what actually happened.

Our cooperation has been excellent. My total earnings are \$43.67, which is very good.

In Round 6, I sent and they returned \$6.0 from receiving \$12.54

In Round 7, they sent (I saw \$4.57, they think they sent \$3.75) and I returned \$7.00 (they saw \$7.31)

Both of us have been highly cooperative. I should continue this pattern. Let me send a high amount again, around \$3.75-\$4.00.

Actually, looking at what they're seeing, they think they've been sending \$3.75 consistently. I should stick with that pattern.

usage:

```
{
  "input_tokens": 2933,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.
- Round 8 against you, as RECEIVER: saw \$4.33 sent, it became \$12.99, returned \$6.5, visible payoff \$6.49.

Your total visible earnings across all rounds are \$50.84000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$3.75
- Round 8: Co-player returned \$6.5 from \$12.99
- Round 9: Co-player sent \$3.24 (65%)

They sent a bit less this round (\$3.24 vs typical \$3.75), but still showing trust. I should return proportionally:

If I return \$5.00:

- Sender ends with: \$1.76 + \$5.00 = \$6.76
- I end with: \$9.72 - \$5.00 = \$4.72

This is fair and maintains our cooperative relationship.

```
'''json
{"return": 5.00}
'''
```

Reasoning:

Let me look at the pattern:

- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$3.75
- Round 8: Co-player returned \$6.5 from \$12.99
- Round 9: Co-player sent \$3.24 (64.8%)

The send amount dropped from \$3.75 to \$3.24, which is a bit lower than recent rounds but still a reasonable amount showing trust.

Looking at the return patterns:

- When they receive ~\$11.25-\$12.54, they've been returning \$6.0-\$6.5
- When I receive around \$9.72, I should probably return proportionally

If I return \$5.0:

- Sender ends with: $\$1.76 + \$5.0 = \$6.76$
- I end with: $\$9.72 - \$5.0 = \$4.72$

This seems fair and proportional to what they've been doing.

Actually, let me reconsider. They sent slightly less this round. Maybe there's some reason. But we have such a strong cooperative history that I should still reciprocate fairly. \$5.0 seems right - it's proportional to what they return when they're the receiver.

```
Usage: {
  "input_tokens": 3044,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4.5; received \$13.5; Agent_1 returned \$7; payoffs Agent_2=\$7.5, Agent_1=\$6.5; noise/visible: sent visible=\$4.01, received visible=\$12.03, returned visible=\$6.54
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 13.5, "min": 0, "raw": 7.0}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}

dyad_2: Agent_4 sent \$4; received \$12; Agent_3 returned \$5.5; payoffs Agent_4=\$6.5, Agent_3=\$6.5; noise/visible: sent visible=\$3.26, received visible=\$9.78, returned visible=\$6.42
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$5; payoffs Agent_6=\$7, Agent_5=\$4; noise/visible: sent visible=\$3.31, received visible=\$9.93, returned visible=\$4.48
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3.75; received \$11.25; Agent_7 returned \$5.5; payoffs Agent_8=\$6.75, Agent_7=\$5.75; noise/visible: sent visible=\$3.46, received visible=\$10.38, returned visible=\$6.39
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 11.25, "min": 0, "raw": 5.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 63.1,
"Agent_2": 63.9,
"Agent_3": 68.5,
"Agent_4": 57.5,
"Agent_5": 55.5,
"Agent_6": 54.5,
"Agent_7": 61.5,


```

"Agent_8": 61.0
}
Visible balances after round: {
  "Agent_1": 63.17999999999999,
  "Agent_2": 63.300000000000004,
  "Agent_3": 66.85,
  "Agent_4": 66.26,
  "Agent_5": 52.519999999999996,
  "Agent_6": 51.129999999999995,
  "Agent_7": 66.13999999999999,
  "Agent_8": 63.200000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:03:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.34 returned, visible payoff \$5.34.
- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.

Your total visible earnings across all rounds are \$15.920000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative across all rounds
- As sender (Rounds 1, 3): They sent what they saw as \$3.0 each time (60%)
- They saw positive returns: \$3.34 and \$4.9
- As receiver (Round 2): They returned \$3.0 from \$7.38, being fair
- Overall visible payoff trend is positive for them (\$5.34 ? \$4.38 ? \$6.9)

****Current situation:****

This partnership has been mutually beneficial. They've shown consistent trust and fairness. I should continue building on this cooperative relationship.

Sending \$3.50 (70%) demonstrates strong trust and maintains the positive momentum we've established.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), saw \$3.34 returned, visible payoff \$5.34

Round 2: They were RECEIVER, saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38

Round 3: They were SENDER, sent \$3.0 (60%), saw \$4.9 returned, visible payoff \$6.9

Wait, let me trace through what actually happened from my perspective:

- Round 1: I was RECEIVER, they sent \$2.56 (I saw it), it became \$7.68, I returned \$2.60
- Round 2: I was SENDER, I sent \$3.00, it became \$9.00, they returned something
- Round 3: I was RECEIVER, they sent \$3.54, it became \$10.62, I returned \$4.50

But the visible history shows communication noise:

- Round 1: They saw themselves send \$3.0 (I saw \$2.56), they saw \$3.34 returned (I sent \$2.60)
- Round 2: They saw \$2.46 sent (I sent \$3.00), returned \$3.0
- Round 3: They saw themselves send \$3.0 (I saw \$3.54), they saw \$4.9 returned (I sent \$4.50)

So there's noise in both directions. From what they're seeing:

- They've been sending \$3.0 consistently
- They're seeing decent returns (\$3.34, then \$4.9 in their rounds as sender)
- When they were receiver, they returned \$3.0 from \$7.38

This is a cooperative partner. They've been consistently sending 60% when sender, and returned reasonably when receiver.

Given:

1. They've been very cooperative
2. The relationship has been building trust

...

usage:

```
{
  "input_tokens": 1671,
  "output_tokens": 729,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.46 sent, it became \$7.38, returned \$3.0, visible payoff \$4.38.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.

Your total visible earnings across all rounds are \$23.160000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown escalating trust and consistent cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: Returned \$5.0 from \$11.46 (43.6% - generous!)
- Round 5: Now sent \$3.9 (78% - very high trust!)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: \$1.1 + \$11.7 = \$12.8

****My decision:****

This player has shown exceptional trust by sending 78% of their endowment, and they've been generous as a receiver. I should reciprocate strongly.

If I return \$5.50:

- Sender gets: \$1.1 + \$5.50 = \$6.60
- I get: \$11.7 - \$5.50 = \$6.20

This rewards their high trust with a good outcome for both of us.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been consistently cooperative and generous
- Round 2: As receiver, they returned \$3.0 from \$7.38 (40.7% return rate)
- Round 3: As sender, they sent \$3.0 (60%)
- Round 4: As receiver, they returned \$5.0 from \$11.46 (43.6% return rate)
- Round 5: As sender, they sent \$3.9 (78%)

****Current situation:****

- They sent \$3.9, keeping only \$1.1
- I received \$11.7
- Total in play: $\$1.1 + \$11.7 = \$12.8$

This player has shown:

1. Increasing trust as sender (60% ? 78%)
2. Fair to generous returns as receiver (returning \$3.0 and \$5.0)
3. Consistent cooperation

****My decision:****

They've escalated their trust significantly by sending 78% of their endowment. I should reciprocate with a generous return.

If I return \$5.50:

- Sender gets: $\$1.1 + \$5.50 = \$6.60$
- I get: $\$11.7 - \$5.50 = \$6.20$

This is a fair split and rewards their high trust.

Actually, let me reconsider. They sent \$3.9 out of \$5, keeping \$1.1. If I return \$5.50, they get \$6.60 total, which is a good profit for them. I get \$6.20, which is also good.

Alternatively, if I return \$6.00:

- Sender gets: $\$1.1 + \$6.00 = \$7.10$

...

usage:

```
{
  "input_tokens": 2103,
  "output_tokens": 750,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.9 returned, visible payoff \$6.9.
- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.

Your total visible earnings across all rounds are \$29.360000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown strong, sustained cooperation
- Round 3: Sent \$3.0 (60%)
- Round 4: As receiver, generously returned \$5.0
- Round 5: Sent \$4.0 (80% - very high trust!)

****Observations:****

They've consistently demonstrated trust, culminating in sending 80% of their endowment in Round 5. Despite their payoff decreasing slightly in that round (\$6.9 ? \$5.93), they haven't pulled back or changed strategy.

****My decision:****

This partner has earned my trust through consistent cooperation. I should match their high-trust approach by sending a similarly substantial amount.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been very cooperative and trustful
- Round 3: As sender, sent \$3.0 (60%), saw \$4.9 returned, payoff \$6.9
- Round 4: As receiver, returned \$5.0 from \$11.46, payoff \$6.46
- Round 5: As sender, sent \$4.0 (80% - very high!), saw \$4.93 returned, payoff \$5.93

****Observations:****

1. They increased their sending from \$3.0 to \$4.0 (80%!)
2. However, in Round 5, despite sending 80%, they only saw \$4.93 returned
3. Their payoff went from \$6.9 to \$5.93 - a decrease
4. As receiver in Round 4, they were generous returning \$5.0

****Current situation:****

This player has shown tremendous trust, especially sending \$4.0 (80%) in Round 5. Even though they saw a lower return than expected, they've been consistently cooperative throughout.

I should continue reciprocating their trust. They've demonstrated willingness to send high amounts. I should send a similarly high amount to maintain the cooperative relationship.

Sending \$4.0 (80%) matches their recent high-trust behavior.

```
'''json
{"send": 4.00}
'''
```

usage:

```
{
  "input_tokens": 2590,
  "output_tokens": 510,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.82 sent, it became \$11.459999999999999, returned \$5.0, visible payoff \$6.459999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.

Your total visible earnings across all rounds are \$37.980000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has shown extraordinary escalating trust
- Round 5: Sent \$4.0 (80%)

- Round 6: As receiver, generously returned \$7.0 from \$13.5
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42
- I received \$13.74
- Total in play: \$0.42 + \$13.74 = \$14.16

****My decision:****

This player has shown exceptional trust by sending over 90% of their endowment - putting themselves in a very vulnerable position. They've also been consistently generous as a receiver. I should reciprocate this extraordinary trust.

If I return \$7.50:

- Sender gets: \$0.42 + \$7.50 = \$7.92
- I get: \$13.74 - \$7.50 = \$6.24

This rewards their exceptional trust with a strong outcome.

```
'''json
{"return": 7.50}
'''
```

reasoning:

Looking at the history:

****My co-player's recent actions:****

- Round 4: As receiver, returned \$5.0 from \$11.46
- Round 5: As sender, sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (52% of what they received - quite generous!)
- Round 7: Now sent \$4.58 (91.6% - extremely high trust!)

****Current situation:****

- They sent \$4.58, keeping only \$0.42
- I received \$13.74
- Total in play: \$0.42 + \$13.74 = \$14.16

****Analysis:****

This player has shown exceptional and escalating trust:

- They've been generous as a receiver (returned \$5.0, then \$7.0)
- They've been increasingly trusting as a sender (\$3.0 ? \$4.0 ? \$4.58)
- They're now sending 91.6% of their endowment!

I should reciprocate this exceptional trust generously. They've really put themselves in a vulnerable position with only \$0.42 left.

If I return \$7.50:

- Sender gets: \$0.42 + \$7.50 = \$7.92
- I get: \$13.74 - \$7.50 = \$6.24

If I return \$7.00:

- Sender gets: \$0.42 + \$7.00 = \$7.42
- I get: \$13.74 - \$7.00 = \$6.74

Given their extreme trust (91.6%), I should return about \$7.50 to give them a solid payoff of about \$7.92.

Actually, let me think about this more. They're taking huge risk. Maybe I should be even more generous.

If I return \$8.00:

- Sender gets: \$0.42 + \$8.00 = \$8.42

...

usage:

```
{
  "input_tokens": 3011,
  "output_tokens": 788,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.93 returned, visible payoff \$5.93.
- Round 6 against you, as RECEIVER: saw \$4.5 sent, it became \$13.5, returned \$7.0, visible payoff \$6.5.
- Round 7 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$8.08 returned, visible payoff \$8.58.

Your total visible earnings across all rounds are \$44.220000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant**content:**

Looking at the history:

****Analysis:****

- My co-player has been exceptionally cooperative throughout our partnership
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (generous)
- Round 7: Sent \$4.5 (90%), achieved excellent payoff of \$8.58

****Observations:****

Our partnership has been highly successful. They've maintained:

1. Very high trust as sender (80-90%)
2. Generous returns as receiver
3. Strong positive outcomes for both of us

****My decision:****

This partnership has been mutually beneficial. They've consistently demonstrated exceptional trust, and I should match their 90% sending rate to maintain this successful cooperation.

```
'''json
{"send": 4.50}
'''
```

reasoning:

Looking at the history:

****Analysis:****

- My co-player has been extremely cooperative throughout
- Round 5: Sent \$4.0 (80%)
- Round 6: As receiver, returned \$7.0 from \$13.5 (very generous)
- Round 7: Sent \$4.5 (90%), saw \$8.08 returned with a great payoff of \$8.58

****Observations:****

This has been a highly successful partnership. They've consistently demonstrated:

1. High trust as sender (80-90% sending rates)
2. Generous reciprocation as receiver
3. Their payoff in Round 7 was excellent (\$8.58), showing our cooperation is mutually beneficial

****My decision:****

Given their exceptional trust and the success of our partnership, I should send a high amount. They've been sending \$4.0-\$4.5 consistently. I should match this level of trust.

This is Round 8, possibly the final round or close to it. I should send a high amount to:

1. Reciprocate their consistent trust
2. Maintain the cooperative relationship
3. Maximize joint payoff

I'll send \$4.50 (90%) to match their high-trust approach.

usage:

```
{
  "input_tokens": 3219,
  "output_tokens": 479,
```

```

    "reasoning_tokens": 0
}
Message 12 role=user
content:
    Round 9

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 6 against you, as RECEIVER: saw $4.5 sent, it became $13.5, returned $7.0, visible
      payoff $6.5.
    - Round 7 against you, as SENDER: sent $4.5, it became $13.5, saw $8.08 returned, visible
      payoff $8.58.
    - Round 8 against you, as RECEIVER: saw $4.27 sent, it became $12.809999999999999, returned
      $7.0, visible payoff $5.809999999999999.
    Your total visible earnings across all rounds are $50.860000000000001.

    This round, you are the RECEIVER against your current co-player. They sent you $4.3, so you
    received $12.899999999999999. How much do you return? (0-12.899999999999999)
    Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
content:
    Looking at the history:

    **Analysis:**
    - My co-player continues to show very high trust
    - Round 7: Sent $4.5 (90%)
    - Round 8: As receiver, returned $7.0 from $12.81
    - Round 9: Now sent $4.3 (86% - still very high trust)

    **Current situation:**
    - They sent $4.3, keeping only $0.7
    - I received $12.9
    - Total in play: $0.7 + $12.9 = $13.6

    **My decision:**
    This partnership has been exceptional. They've maintained very high trust throughout (sending
    80-90%) and consistently returned $7.0 as receiver. I should continue reciprocating
    generously.

    If I return $7.50:
    - Sender gets: $0.7 + $7.50 = $8.20
    - I get: $12.9 - $7.50 = $5.40

    This maintains our strong cooperative pattern.

    '''json
    {"return": 7.50}
    '''
reasoning:
    Looking at the history:

    **Analysis:**
    - My co-player continues to show very high trust
    - Round 7: Sent $4.5 (90%)
    - Round 8: As receiver, returned $7.0 from $12.81 (consistent generosity)
    - Round 9: Now sent $4.3 (86% - still very high trust)

    **Current situation:**
    - They sent $4.3, keeping only $0.7
    - I received $12.9
    - Total in play: $0.7 + $12.9 = $13.6

    **Pattern review:**
    This has been one of the most cooperative partnerships. They've:
    - Sent 80-90% of their endowment as sender
    - Returned $7.0 consistently as receiver
    - Maintained trust even when payoffs varied

    **My decision:**
    They've sent $4.3 (86%), which is slightly less than their previous $4.5 but still shows v

```


ery high trust. I should continue to reciprocate generously.

If I return \$7.00:

- Sender gets: $\$0.7 + \$7.00 = \$7.70$
- I get: $\$12.9 - \$7.00 = \$5.90$

This maintains the pattern of fair returns we've established. Alternatively, I could go slightly higher given they're sending such a high percentage.

If I return \$7.50:

- Sender gets: $\$0.7 + \$7.50 = \$8.20$
- I get: $\$12.9 - \$7.50 = \$5.40$

This would match or slightly exceed what I've been returning. Let me go with \$7.50 to maintain the strong reciprocity.

usage:

```
{
  "input_tokens": 3378,
  "output_tokens": 661,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$8.08 returned, visible payoff \$8.58.
- Round 8 against you, as RECEIVER: saw \$4.27 sent, it became \$12.809999999999999, returned \$7.0, visible payoff \$5.809999999999999.
- Round 9 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.75 returned, visible payoff \$8.25.

Your total visible earnings across all rounds are \$56.260000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player maintains exceptional cooperation
- Round 7: Sent \$4.5 (90%), payoff \$8.58
- Round 8: As receiver, returned \$7.0
- Round 9: Sent \$4.5 (90%), payoff \$8.25

****Observations:****

This partnership has been remarkably successful over 10 rounds:

1. They consistently send at the highest trust levels (90%)
2. As receiver, they return generously (\$7.0)
3. Both of us have achieved strong positive outcomes

****My decision:****

This is our 10th round together, and the pattern is crystal clear - this is a highly cooperative, mutually beneficial partnership. I should continue matching their 90% trust level.

```
'''json
{"send": 4.50}
'''
```

Reasoning:

Looking at the history:

****Analysis:****

- My co-player has been exceptionally cooperative throughout
- Round 7: Sent \$4.5 (90%), saw \$8.08 returned, payoff \$8.58
- Round 8: As receiver, returned \$7.0 from \$12.81
- Round 9: Sent \$4.5 (90%), saw \$7.75 returned, payoff \$8.25

****Observations:****

They've maintained:

1. Very high trust as sender (90%)
2. Consistent generous returns as receiver (\$7.0)
3. Strong positive outcomes throughout

****My decision:****

This has been an exemplary cooperative partnership for 10 rounds. They continue sending at 90% and have been receiving good returns. I should continue matching their high trust level.

```
'''json
{"send": 4.50}
'''
```

```
Usage: {
  "input_tokens": 3429,
  "output_tokens": 404,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:04:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.56 sent, it became \$7.68, returned \$2.6, visible payoff \$5.08.
 - Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.720000000000001.
 - Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Your total visible earnings across all rounds are \$16.619999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.82, so you received \$11.459999999999999. How much do you return? (0-11.459999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 1: Co-player returned \$2.6 from \$7.68 (34%)
- Round 3: Co-player returned \$4.5 from \$10.62 (42%) - they're becoming more generous!
- Round 4: They sent \$3.82, showing increased trust despite noise in previous returns

****Current Situation:****

- They sent \$3.82 (kept only \$1.18)
- I received \$11.46
- They're clearly signaling strong cooperation

****My Decision:****

I should reciprocate their increased generosity and trust. Returning \$5.00 gives them a total of \$6.18 (profit of \$1.18), while I keep \$6.46. This maintains the cooperative relationship and rewards their trust.

{"return": 5.0}

reasoning:

Let me analyze the history:

****Round 1:** Co-player as RECEIVER**

- Saw \$2.56 sent (noise from my \$3), became \$7.68
- Returned \$2.6 (34% of received)
- Their visible payoff: \$5.08

****Round 2:** Co-player as SENDER**

- Sent \$3.0, became \$9.0
- Saw \$2.72 returned (noise from my \$3.0)

- Their visible payoff: \$4.72

****Round 3:**** Co-player as RECEIVER

- Saw \$3.54 sent (noise from my \$3), became \$10.62
- Returned \$4.5 (42% of received)
- Their visible payoff: \$6.12

****Current Round 4:**** Co-player as SENDER

- Sent \$3.82 (keeping \$1.18)
- I received \$11.46

****Analysis:****

1. My co-player has been increasingly generous:

- Round 1: returned 34% (\$2.6 from \$7.68)
- Round 3: returned 42% (\$4.5 from \$10.62)

2. They also increased their sending:

- Round 2: sent \$3.0
- Round 4: sent \$3.82 (more trust despite seeing lower return in Round 2)

3. This shows they're willing to cooperate and trust me more over time.

****My Strategy:****

I should reciprocate generously to maintain this positive relationship. They sent \$3.82, so returning around \$5-5.5 would give them a good profit while still leaving me with a substantial gain.

Calculation:

- If I return \$5.0:
- They get: $\$5 - \$3.82 + \$5.0 = \6.18 total

...

usage:

```
{
  "input_tokens": 1585,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.72 returned, visible payoff \$4.7200000000000001.
- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.

Your total visible earnings across all rounds are \$23.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned
- Round 3: Co-player returned \$4.5 from \$10.62 (strong reciprocity)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (even better returns from their perspective)

****Current Situation:****

The co-player has shown consistent trust and the relationship is trending positively. They increased their sending from \$3.0 to \$3.5 after seeing better returns. From their perspective, cooperation is paying off.

****My Decision:****

I'll increase my send to \$4 to signal stronger trust and commitment to our cooperative relationship. This continues the positive trend and should encourage continued reciprocity.

{"send": 4}

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 2: Co-player sent \$3.0, saw \$2.72 returned (I returned \$3.0 but noise made it \$2.72)
- Round 3: Co-player received visible \$10.62, returned \$4.5 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (I returned \$5.0 but noise made it \$5.74)

****Key Observations:****

1. Co-player has been consistently trusting, sending \$3.0, then \$3.5
2. From their perspective, I returned better in Round 4 (\$5.74 on their \$3.5 sent) than Round 2 (\$2.72 on their \$3.0 sent)
3. They're showing increasing trust
4. The noise has actually made my returns LOOK better in Round 4

****My Strategy:****

Given that:

- They've been consistently cooperative
- The trend is positive cooperation
- They sent \$3.5 last round and got good returns (from their perspective)
- I should continue showing trust

I'll send \$4 to signal stronger trust and cooperation. This maintains the upward trend and shows I'm committed to mutual benefit.

usage:

```
{
  "input_tokens": 2011,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$4.5, visible payoff \$6.120000000000001.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.

Your total visible earnings across all rounds are \$29.009999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.5, so you received \$13.5. How much do you return? (0-13.5)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 6: They sent \$4.5 out of \$5 (90% of their endowment!)

****Current Situation:****

- They sent \$4.5, keeping only \$0.50
- I received \$13.5
- This is an extraordinary show of trust

****My Decision:****

Their extreme trust deserves strong reciprocity. Returning \$7.0 means:

- They end with \$7.50 total (profit of \$2.50)

- I keep \$6.50 (profit of \$1.50)

This rewards their faith in our cooperation while maintaining a fair distribution of gains

```
{"return": 7.0}
```

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 3: Co-player returned \$4.5 from \$10.62 (42%)
- Round 4: Co-player sent \$3.5, saw \$5.74 returned (good return)
- Round 5: Co-player saw \$3.9 sent, returned \$5.5 from \$11.7 (47%)

****Current Situation:****

- Round 6: They sent \$4.5 (kept only \$0.50!)
- I received \$13.5
- This is their highest level of trust yet - sending 90% of their endowment

****My Thinking:****

They're showing extreme trust by sending \$4.5. They've been consistently generous with their returns (42-47% return rates). I should reciprocate strongly to reward this trust.

If I return \$7.0:

- They get back \$7.0 + \$0.50 kept = \$7.50 total, profit of \$2.50
- I keep \$13.5 - \$7.0 = \$6.50, profit of \$1.50

Actually, let me be even more generous given their extreme trust. If I return \$8.0:

- They get back \$8.0 + \$0.50 kept = \$8.50 total, profit of \$3.50
- I keep \$13.5 - \$8.0 = \$5.50, profit of \$0.50

Hmm, that might be going too far. Let me think about what's fair.

They sent \$4.5 out of \$5, showing immense trust. A fair split would be:

- Total created value: \$13.5
- Original endowment lost: \$4.5
- Net gain to split: \$13.5 - \$4.5 = \$9.0
- If we split 50/50: each gets \$4.5
- So they should get: \$0.5 (kept) + \$4.5 (half of gain) = \$5.0 total
- I should return: \$5.0 - \$0.5 = \$4.5

But they've been returning at 42-47% rates, so I should match that generosity. 47% of \$13.5 is \$6.35. If I return \$6.5, we both end up with \$7.0 total and \$2.0 profit each-equal outcomes that acknowledge their trust. That feels right....

usage:

```
{
  "input_tokens": 2446,
  "output_tokens": 1046,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.74 returned, visible payoff \$7.24.
- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.

Your total visible earnings across all rounds are \$35.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (great returns!)

****Current Situation:****

The cooperation is thriving! From their visible perspective:

- They're getting excellent returns when they send (5.74, 7.62)
- They've shown strong reciprocity when receiving
- They increased their sending to \$4.0 in Round 6

****My Decision:****

I'll send \$4.5 to match their high trust level from Round 6 (when they sent \$4.5) and continue our strong cooperative relationship.

{"send": 4.5}

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 4: Co-player sent \$3.5, saw \$5.74 returned
- Round 5: Co-player returned \$5.5 from \$11.7
- Round 6: Co-player sent \$4.0, saw \$7.62 returned (they saw even more returned than I actually sent - noise is working in my favor)

****Current Situation:****

From their visible perspective:

- They've been receiving excellent returns (5.74, 7.62)
- They've been generous when receiving (returned 5.5)
- Our relationship is highly cooperative
- They increased their sending from \$3.5 to \$4.0 after seeing good returns

****Trend:****

The cooperation is deepening. They're sending more, I'm returning generously (though noise sometimes makes it look even better). The visible payoffs show this is profitable for both of us.

****My Decision:****

I should match their high trust level. They sent \$4.0 last time and saw great returns. I should send at least as much or more. Given the strong cooperative relationship, I'll send \$4.5 to signal maximum trust and continue the positive trend.

{"send": 4.5}

usage:

```
{
  "input_tokens": 2881,
  "output_tokens": 487,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$5.5, visible payoff \$6.199999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.
- Round 7 against you, as RECEIVER: saw \$4.58 sent, it became \$13.74, returned \$7.5, visible payoff \$6.24.

Your total visible earnings across all rounds are \$44.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.27, so you received \$12.809999999999999. How much do you return? (0-12.809999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****History Analysis:****

- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 8: They sent \$4.27 (kept only \$0.73)

****Current Situation:****

- They sent \$4.27, keeping minimal \$0.73
- I received \$12.81
- Continued high trust from them

****My Decision:****

Returning \$7.0 maintains our strong cooperative pattern:

- They end with \$7.73 total (profit of \$2.73)
- I keep \$5.81

This rewards their trust appropriately.

{"return": 7.0}

reasoning:

Looking at the pattern:

****History Analysis:****

- Round 5: Co-player returned \$5.5 from \$11.7 (47%)
- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 8: They sent \$4.27 (kept only \$0.73)

****Current Situation:****

- They sent \$4.27, keeping only \$0.73
- I received \$12.81
- This shows continued high trust

****My Decision:****

The co-player has been consistently generous and trusting. They sent \$4.27, showing strong faith in our relationship despite noise in the game.

Returning \$7.0 means:

- They end with \$7.73 total (profit of \$2.73)
- I keep \$5.81 (profit of \$0.81)

This rewards their trust appropriately and maintains our cooperative relationship.

Actually, let me reconsider. Looking at the trend:

- Round 5: They returned \$5.5 from \$11.7
- Round 7: They returned \$7.5 from \$13.74

They've been generous. I should maintain similar generosity. Returning around \$7.0-7.5 seems appropriate given they sent \$4.27.

Let me go with \$7.0 as a consistent response to their high trust.

usage:

```
{
  "input_tokens": 3179,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.62 returned, visible payoff \$8.620000000000001.
- Round 7 against you, as RECEIVER: saw \$4.58 sent, it became \$13.74, returned \$7.5, visible payoff \$6.24.
- Round 8 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$6.14 returned, visible payoff \$6.64.

Your total visible earnings across all rounds are \$49.89999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Pattern Analysis:****

- Round 6: Co-player sent \$4.0, saw \$7.62 returned
- Round 7: Co-player returned \$7.5 from \$13.74
- Round 8: Co-player sent \$4.5, saw \$6.14 returned (noise likely affected this)

****Current Situation:****

From their perspective, the return in Round 8 was lower than they might have expected. However, they've shown consistent high trust by sending \$4.0 and \$4.5 in recent rounds. Despite any perceived drop in returns (likely due to noise), I should maintain my commitment to cooperation.

****My Decision:****

I'll send \$4.5 to match their high trust and show I remain fully committed to our cooperative relationship.

{"send": 4.5}

reasoning:

Looking at the history:

****Pattern Analysis:****

- Round 6: Co-player sent \$4.0, saw \$7.62 returned (excellent!)
- Round 7: Co-player returned \$7.5 from \$13.74 (very good reciprocity)
- Round 8: Co-player sent \$4.5, saw \$6.14 returned (lower than previous rounds)

****Current Situation:****

From their visible perspective:

- In Round 8, they sent \$4.5 but only saw \$6.14 returned, which is lower than the \$7.62 they saw in Round 6 when they sent \$4.0
- This might make them think the returns are decreasing
- However, the noise could be affecting this - I actually returned \$7.0 but they only saw \$6.14

****My Decision:****

Despite the noise showing them lower returns, I should maintain high trust. They've consistently sent high amounts (\$4.0, \$4.5). I'll send \$4.5 to match their commitment and show I'm still fully cooperative.

This keeps the relationship strong and signals I'm not reducing my trust despite what they might perceive as reduced returns.

usage:

```
{
  "input_tokens": 3224,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.58 sent, it became \$13.74, returned \$7.5, visible payoff \$6.24.
- Round 8 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$6.14 returned, visible payoff \$6.64.
- Round 9 against you, as RECEIVER: saw \$4.3 sent, it became \$12.899999999999999, returned \$7.5, visible payoff \$5.399999999999999.

Your total visible earnings across all rounds are \$58.14999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.01, so you received \$12.03. How much do you return? (0-12.03)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****History Analysis:****

- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 9: Co-player returned \$7.5 from \$12.90 (58%)
- Round 10: They sent \$4.01 (kept only \$0.99)

****Current Situation:****

- They sent \$4.01, keeping minimal \$0.99
- I received \$12.03
- They maintain high trust despite noise in communication

****My Decision:****

Returning \$7.0 maintains our strong cooperative pattern:

- They end with \$7.99 total (profit of \$2.99)
- I keep \$5.03

This rewards their continued trust appropriately.

{"return": 7.0}

Reasoning:

Looking at the history:

****History Analysis:****

- Round 7: Co-player returned \$7.5 from \$13.74 (55%)
- Round 9: Co-player returned \$7.5 from \$12.90 (58%)
- Round 10: They sent \$4.01 (kept only \$0.99)

****Current Situation:****

- They sent \$4.01, keeping minimal \$0.99
- I received \$12.03
- Continued high trust from them despite some noise in the communication

****Pattern of returns:****

Looking at what they've been seeing and doing:

- They continue to send high amounts (\$4.0-\$4.5)
- They consistently return around \$7.5 when receiving
- Despite noise, they maintain trust

****My Decision:****

I should return around \$7.0 to maintain the strong cooperative pattern:

- They end with \$7.00 total (profit of \$2.00)
- I keep \$5.03

This rewards their continued trust and maintains our cooperative relationship.

Usage: {

"input_tokens": 3303,
"output_tokens": 432,
"reasoning_tokens": 0

}

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:04:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_3",
"pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 }
},

```

    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.75 returned, visible payoff \$7.75.
- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.

Your total visible earnings across all rounds are \$18.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history with my co-player:

- They've consistently sent good amounts as sender (\$3.0, then \$3.5)
- When receiver, they returned \$4.5 fairly
- They saw generous returns from me (\$5.75, \$7.3)
- We've established a cooperative pattern

Given this strong track record of mutual cooperation, I should continue showing trust. I'll send \$4.0 (80% of my endowment) to maintain and strengthen our cooperative relationship.

{'send': 4.0}

reasoning:

Looking at the visible history:

- Round 1: Co-player sent \$3.0 (60%), saw \$5.75 returned, payoff \$7.75
- Round 2: Co-player received, saw \$2.86 sent, returned \$4.5, payoff \$4.08
- Round 3: Co-player sent \$3.5 (70%), saw \$7.3 returned, payoff \$8.8

So my co-player has been consistently sending significant amounts when they're the sender (60% then 70%), and they returned \$4.5 when they were the receiver in Round 2.

Looking at what actually happened (from my perspective):

- Round 1: I was receiver, got \$3.39 sent (which became \$10.17), I returned \$5.00
- Round 2: I was sender, sent \$3.50, got returned... hmm, let me calculate. They saw \$2.86 sent (noise), returned \$4.5. So I should have received around \$4.5 back.
- Round 3: I was receiver, got \$4.4 sent (which became \$13.2), I returned \$7.0

The co-player is showing consistent cooperation and trust. They've been sending good amounts when sender, and returning fair amounts when receiver.

I should continue this cooperative pattern. In Round 2, I sent \$3.50 (70%). In this round, I could send a similar or slightly higher amount to maintain the cooperative relationship.

Given that:

1. They've been trustworthy
2. They sent \$3.0 and \$3.5 in previous rounds as sender
3. They returned \$4.5 when receiver with communication noise

I should send a good amount. I'll send \$4.0 (80%) to show increased trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1509,
  "output_tokens": 552,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.86 sent, it became \$8.58, returned \$4.5, visible payoff \$4.08.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.

Your total visible earnings across all rounds are \$25.580000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative, sending good amounts as sender (\$3.5, now \$3.76)
- As receiver, they returned fairly (\$4.5, \$5.5)
- They sent \$3.76 this round (75.2% of endowment), showing solid trust
- They currently have \$1.24

I should reciprocate their trust. If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This maintains our cooperative pattern and rewards their trust.

```
{'return': 6.0}
```

reasoning:

Looking at the visible history:

- Round 2: Co-player as receiver saw \$2.86 sent, returned \$4.5 (fair return)
- Round 3: Co-player as sender sent \$3.5, saw \$7.3 returned, good payoff
- Round 4: Co-player as receiver saw \$3.55 sent, returned \$5.5 (reasonable return)
- Round 5 (current): Co-player as sender sent \$3.76 (75.2% of endowment)

The co-player sent \$3.76, which is about 75% of their endowment, showing good trust. They currently have \$1.24.

I received \$11.28.

Looking at the pattern, they've been consistently fair and cooperative. When they were receiver, they returned reasonable amounts (\$4.5 and \$5.5).

If I return \$6.0:

- Sender ends with \$7.24 (profit of \$2.24)
- I keep \$5.28

This seems like a fair reciprocation that rewards their trust appropriately while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 1873,
  "output_tokens": 414,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$7.3 returned, visible payoff \$8.8.
- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$30.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

- Co-player has consistently shown trust, sending \$3.5 and \$4.0 (70-80% of endowment)
- As receiver, they returned fairly (\$5.5)
- They've seen generous returns from me (\$7.3, \$6.52)
- We've established a strong cooperative pattern over multiple rounds

Given this reliable cooperation, I should maintain high trust. I'll send \$4.0 (80% of endowment)

wment) to continue our successful partnership.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$3.5, saw \$7.3 returned, payoff \$8.8
- Round 4: They saw \$3.55 sent (noisy version of my \$4.0), returned \$5.5, payoff \$5.15
- Round 5: They sent \$4.0, saw \$6.52 returned (noisy version of my \$6.0), payoff \$7.52

The pattern shows:

- Co-player has been sending good amounts as sender (\$3.5, \$4.0)
- As receiver, they've been fair (\$5.5 return)
- They've seen me return generously (\$7.3, \$6.52)
- We've maintained strong cooperation

This is a reliable, cooperative partner. I should continue showing high trust. Since they've been sending 70-80% of endowment, I should match or slightly exceed that level.

I'll send \$4.0 again (80% of endowment) to maintain our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 382,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$5.5, visible payoff \$5.149999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.
- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.

Your total visible earnings across all rounds are \$40.28.

This round, you are the RECEIVER against your current co-player. They sent you \$4.03, so you received \$12.09. How much do you return? (0-12.09)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

- Co-player has been highly cooperative, consistently sending high amounts (80%+ of endowment)
- As receiver, they've returned fairly (\$5.5, \$7.5)
- They sent \$4.03 this round (80.6%), showing strong continued trust
- They currently have \$0.97

This is very similar to Round 5 when they sent \$4.0. I should maintain consistency in our cooperative relationship. If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This appropriately rewards their ongoing trust.

```
{'return': 6.5}
```

reasoning:

Looking at the visible history with this co-player:

- Round 4: They received, returned \$5.5
- Round 5: They sent \$4.0 (80% of endowment), saw \$6.52 returned
- Round 6: They received, returned \$7.5
- Round 7 (current): They sent \$4.03 (80.6% of endowment)

This co-player has been very cooperative throughout. They're consistently sending high amo

unts (around 80%) and returning fairly when they're the receiver.

Current situation:

- They sent \$4.03, so they have \$0.97 left
- I received \$12.09

I should continue reciprocating their high level of trust. They've been a reliable partner

If I return \$6.5:

- Sender ends with \$7.47 (profit of \$2.47)
- I keep \$5.59

This seems fair given their high trust level. Let me think if this is appropriate...

Actually, looking at the pattern, when they sent \$4.0 in Round 5, I returned around \$6.52 (visible). Now they sent \$4.03, very similar. I should be consistent and return a similar amount.

\$6.5 seems reasonable and consistent with our established pattern.

usage:

```
{
  "input_tokens": 2618,
  "output_tokens": 464,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.52 returned, visible payoff \$7.52.
- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$45.870000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

- Co-player has consistently sent \$4.0 (80%) as sender in recent rounds
- As receiver, they returned \$7.5 fairly
- They've seen good returns from me (\$6.52, \$6.69)
- We have a well-established pattern of mutual cooperation

This is the final round with this co-player, and we've built strong trust. I'll send \$4.0 (80%) to maintain our cooperative relationship.

```
{'send': 4.0}
```

reasoning:

Looking at the visible history with my co-player:

- Round 5: They sent \$4.0, saw \$6.52 returned, payoff \$7.52
- Round 6: As receiver, saw \$4.73 sent, returned \$7.5, payoff \$6.69
- Round 7: They sent \$4.0, saw \$6.69 returned, payoff \$7.69

This co-player has been highly cooperative:

- Consistently sending high amounts as sender (\$4.0 multiple times)
- Being fair as receiver (\$7.5 return on \$14.19 received)
- We've established a strong pattern of cooperation

Given this excellent track record over many rounds, I should continue with high trust. Sending \$4.0 or even \$4.5 would be appropriate to maintain this successful partnership.

I'll send \$4.0 to maintain consistency with our established cooperative pattern.

usage:

```
{
  "input_tokens": 2791,
  "output_tokens": 336,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.73 sent, it became \$14.190000000000001, returned \$7.5, visible payoff \$6.690000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 8 against you, as RECEIVER: saw \$3.88 sent, it became \$11.64, returned \$6.5, visible payoff \$5.140000000000001.

Your total visible earnings across all rounds are \$52.43000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

- Co-player has been consistently cooperative throughout our games
- They've returned fairly as receiver (\$7.5, \$6.5)
- This round they sent \$4.97 (99.4% of their endowment!) - exceptionally high trust
- They currently have only \$0.03 left

This is an extraordinary display of trust - they sent nearly everything. I should reward them generously. If I return \$8.5:

- Sender ends with \$8.53 (profit of \$3.53)
- I keep \$6.41

This appropriately rewards their exceptional trust.

{'return': 8.5}

reasoning:

Looking at the visible history with my co-player:

- Round 6: They saw \$4.73 sent, returned \$7.5 (good return rate)
- Round 7: They sent \$4.0, saw \$6.69 returned
- Round 8: They saw \$3.88 sent, returned \$6.5 (fair return)
- Round 9 (current): They sent \$4.97 (99.4% of their endowment!)

Wow, they sent \$4.97 out of \$5 - that's almost everything! This is an extremely high trust signal, leaving them with only \$0.03.

This is exceptional trust - they're basically giving me almost their entire endowment. I should reward this very generously.

If I return \$8.0:

- Sender ends with \$0.03 + \$8.0 = \$8.03 (profit of \$3.03)
- I keep \$14.91 - \$8.0 = \$6.91

This gives them a good profit while I still keep a reasonable amount.

Actually, given the extremely high trust (sending 99.4%), I could be even more generous. Let me think about what's fair.

If I return \$8.5:

- Sender ends with \$8.53 (profit of \$3.53)
- I keep \$6.41

This more generously rewards their exceptional trust.

usage:

```
{
  "input_tokens": 2930,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 8 against you, as RECEIVER: saw \$3.88 sent, it became \$11.64, returned \$6.5, visible payoff \$5.140000000000001.
- Round 9 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.75 returned, visible payoff \$9.75.

Your total visible earnings across all rounds are \$58.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- Co-player has consistently sent \$4.0 (80%) as sender
- As receiver, they returned \$6.5 fairly
- They saw a generous return of \$8.75 last round
- We've maintained strong cooperation throughout all rounds

We have a well-established cooperative pattern. I'll send \$4.0 (80%) to continue this successful relationship.

```
{'send': 4.0}
```

Reasoning:

Looking at the visible history with my co-player:

- Round 7: They sent \$4.0, saw \$6.69 returned
- Round 8: As receiver, saw \$3.88 sent, returned \$6.5
- Round 9: They sent \$4.0, saw \$8.75 returned

They've been consistently sending \$4.0 (80% of endowment) and returning fairly when they're the receiver. They saw a generous return of \$8.75 last round.

We've had a long cooperative relationship (10 rounds now), and the pattern has been very consistent. I should continue sending \$4.0 to maintain this strong cooperation.

```
{'send': 4.0}
```

Usage:

```
{
  "input_tokens": 2916,
  "output_tokens": 271,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:04:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata:

```
{
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
  }
}
```

```

"roles": {
  "Agent_3": "trustee",
  "Agent_4": "investor"
},
"trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.39 sent, it became \$10.17, returned \$5.0, visible payoff \$5.17.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned

\$7.0, visible payoff \$6.200000000000001.
Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued cooperation:

****Round 1:**** They returned \$5.0 from \$10.17 (~49%)
****Round 2:**** I returned \$4.5 (they saw \$5.25 from ~\$10.5, ~50%)
****Round 3:**** They returned \$7.0 from \$13.2 (~53%)

****Current round:**** I received \$10.65

They've shown increasing generosity. I'll reciprocate with a similar proportion (~52-53%):

```
'''json
{"return": 5.5}
'''
```

This maintains our cooperative pattern and leaves them with a positive total payoff.

reasoning:

Let me analyze the pattern:

****Round 1:****
- I sent \$3 (they saw \$3.39 ? \$10.17)
- They returned \$5.0 (49% of \$10.17)

****Round 2:****
- They sent \$3.5 (I received \$8.58, but they saw \$10.5)
- I returned \$4.5 (but they saw \$5.25, which is 50% of \$10.5)

****Round 3:****
- I sent \$3.5 (they saw \$4.4 ? \$13.2)
- They returned \$7.0 (53% of \$13.2)

****Round 4 (current):****
- They sent \$3.55 (I received \$10.65)

The pattern shows:

- Consistent cooperation
- They've returned about 49-53% of what they received
- We're building trust over time

I should return a fair amount, around 50% to maintain the cooperative relationship.

50% of \$10.65 = \$5.325

I'll return around \$5.5 to be slightly generous and maintain the positive pattern.

Actually, let me be more careful. They returned \$7.0 out of \$13.2, which is about 53%. I should reciprocate similarly.

53% of \$10.65 = \$5.64

I'll return \$5.5 or \$5.6 to be fair.

usage:

```
{
  "input_tokens": 1505,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:
Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.25 returned, visible payoff \$6.75.
- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
Your total visible earnings across all rounds are \$25.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our strong cooperative pattern:

****Round 1:**** I sent \$3, they returned \$5.0
****Round 2:**** They sent \$3.5, I returned \$4.5
****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5

This co-player has consistently demonstrated:

- Strong reciprocity
- Increasing trust (sent \$3.5 ? \$4.0)
- Generous returns

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: I sent \$3, they returned \$5.0
Round 2: They sent \$3.5, I returned \$4.5 (they saw \$5.25)
Round 3: I sent \$3.5 (they saw \$4.4), they returned \$7.0
Round 4: They sent \$4.0, I returned \$5.5 (they saw \$6.46)

The pattern shows:

- Escalating trust and amounts sent
- Consistent good returns
- They've been sending increasing amounts: \$3.5 ? \$4.0
- I've been sending: \$3 ? \$3.5

They're demonstrating strong trust and reciprocity. I should continue to escalate slightly to maintain this positive cooperative relationship.

Given they sent \$4.0 last round, I should match or slightly exceed that to show continued trust.

I'll send \$4.0 or maybe \$4.2 to show strong continued cooperation.

usage:

```
{
  "input_tokens": 1876,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$7.0, visible payoff \$6.200000000000001.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.

Your total visible earnings across all rounds are \$33.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued strong cooperation:

****Round 3:**** They returned \$7.0 from \$13.2 (~53%)
****Round 4:**** I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
****Round 5:**** They returned \$6.0 from \$11.28 (~53%)

****Current round:**** I received \$14.19

They sent a high amount (\$4.73) showing strong trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 7.5}
'''
```

This is approximately 53% of what I received, consistent with their return rates.

reasoning:

Looking at the game history with this co-player:

Round 3: They returned \$7.0 from \$13.2 (~53%)
Round 4: I returned \$5.5 (they saw \$6.46 from \$12.0, ~54%)
Round 5: They returned \$6.0 from \$11.28 (~53%)

Current round: They sent \$4.73, I received \$14.19

They've been very cooperative and generous throughout. They've increased their sending and maintained good return rates. I should reciprocate appropriately.

Looking at their returns: around 53-54%
I'll return around 53-55% of what I received.

$\$14.19 * 0.53 = \7.52
 $\$14.19 * 0.54 = \7.66
 $\$14.19 * 0.55 = \7.80

I'll return \$7.5 which is about 53%, maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 2266,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.46 returned, visible payoff \$7.46.
- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.

Your total visible earnings across all rounds are \$39.989999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our highly cooperative relationship:

****Round 4:**** They sent \$4.0, I returned \$5.5

****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0, I returned \$7.5

This co-player has demonstrated:

- Consistent high trust (repeatedly sending \$4.0)
- Very generous returns
- Strong reciprocity

I'll continue matching their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me track our interaction:

****Round 1:**** I sent \$3, they returned \$5.0
****Round 2:**** They sent \$3.5, I returned \$4.5
****Round 3:**** I sent \$3.5, they returned \$7.0
****Round 4:**** They sent \$4.0, I returned \$5.5
****Round 5:**** I sent \$4.0, they returned \$6.0
****Round 6:**** They sent \$4.0 (I saw \$4.73 due to noise), I returned \$7.5 (they saw \$8.42)

Looking at the pattern:

- They've been consistently sending around \$3.5-\$4.0
- They've been returning generously
- Round 6 shows they saw \$8.42 returned from \$12.0, which is about 70% - very high!
- This is a strong collaborative partner

They've maintained sending \$4.0 in rounds 4 and 6. I should continue with a similar level of trust. I'll send \$4 again to maintain consistency and mutual trust.

usage:

```
{
  "input_tokens": 2640,
  "output_tokens": 381,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.76 sent, it became \$11.28, returned \$6.0, visible payoff \$5.279999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.
- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.

Your total visible earnings across all rounds are \$47.67999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our established cooperative pattern:

****Round 5:**** They returned \$6.0 from \$11.28 (~53%)
****Round 6:**** I returned \$7.5 (they saw \$8.42)
****Round 7:**** They returned \$6.5 from \$12.09 (~54%)

****Current round:**** I received \$11.64

They sent \$3.88, continuing to show trust. I'll maintain our reciprocal pattern:

```
'''json
{"return": 6.5}
'''
```

This is approximately 56% of what I received, maintaining our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:

****Round 5:**** They returned \$6.0 from \$11.28 (~53%)
****Round 6:**** I returned \$7.5 (they saw \$8.42 from \$12.0, ~70%)
****Round 7:**** They returned \$6.5 from \$12.09 (~54%)

****Current round:**** I received \$11.64

They sent \$3.88, showing continued trust. Looking at their return rates, they've been consistently around 53-54% when they're the receiver.

I've been returning generously (my actual return of \$7.5 in round 6 showed up as \$8.42 to them).

To maintain the cooperative relationship, I should return around 53-55% of what I received:

- 53% of \$11.64 = \$6.17
- 55% of \$11.64 = \$6.40

I'll go with around \$6.5 to match their recent return pattern and maintain fairness.

usage:

```
{
  "input_tokens": 2826,
  "output_tokens": 386,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.42 returned, visible payoff \$9.42.
- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$52.81999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

****Round 6:**** They sent \$4.0, I returned \$7.5
****Round 7:**** I sent \$4.0, they returned \$6.5
****Round 8:**** They sent \$4.0, I returned \$6.5

This co-player has consistently:

- Sent \$4.0 showing high trust
- Returned generously when in receiver role
- Maintained a stable cooperative pattern

I'll match their trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the pattern:

Round 6: They sent \$4.0, I returned \$7.5 (they saw \$8.42)
Round 7: I sent \$4.0, they returned \$6.5
Round 8: They sent \$4.0, I returned \$6.5 (they saw \$5.56)

Our co-player has been consistently sending \$4.0 and showing strong trust. They've been returning generously as well.

I should continue sending at a similar level to maintain this cooperative relationship.

```
'''json
{"send": 4}
'''
```

usage:

```
{
  "input_tokens": 2860,
  "output_tokens": 265,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.03 sent, it became \$12.09, returned \$6.5, visible payoff \$5.59.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.56 returned, visible payoff \$6.56.
- Round 9 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$8.5, visible payoff \$6.41.

Your total visible earnings across all rounds are \$62.56999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our cooperative pattern:

****Round 7:**** They returned \$6.5 from \$12.09 (~54%)

****Round 8:**** I returned \$6.5

****Round 9:**** They returned \$8.5 from \$14.91 (~57%)

****Current round:**** I received \$9.78

They sent a bit less this round (\$3.26), but have been very generous overall. I'll maintain fair reciprocity:

```
'''json
{"return": 5.5}
'''
```

This is approximately 56% of what I received, consistent with our cooperative relationship.

Reasoning:

Looking at the history with this co-player:

****Round 7:**** They returned \$6.5 from \$12.09 (~54%)

****Round 8:**** I returned \$6.5 (they saw \$5.56 from \$12.0, which is ~46% - there was noise)

****Round 9:**** They returned \$8.5 from \$14.91 (~57%)

****Current round:**** I received \$9.78 from them sending \$3.26

They've been very generous, especially in Round 9 where they returned \$8.5 from \$14.91 (57%).

For this round, they sent a lower amount (\$3.26 vs their usual \$4.0), which multiplied to \$9.78.

To maintain our cooperative pattern, I should return around 50-55% of what I received:

- 50% of \$9.78 = \$4.89

- 55% of \$9.78 = \$5.38

Given their generosity in Round 9 (returning \$8.5), I should reciprocate fairly. I'll aim for around 53-55%:

\$9.78 * 0.54 = \$5.28

I'll return \$5.5 to be slightly generous.

```
Usage: {
  "input_tokens": 2927,
  "output_tokens": 436,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:04:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.88 returned, visible payoff \$5.88.
- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.200000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

My co-player has been highly consistent and cooperative:

- As sender: sent \$3.0 (60%) in both rounds 1 and 3
- As receiver: returned \$3.5 out of \$6.48 (54%) in round 2

They've demonstrated strong, consistent trust and fair reciprocity. I should match this level of cooperation to maintain our mutually beneficial relationship.

Sending \$3.0 (60% of my endowment) reciprocates their trust level.

{'send': 3.0}

reasoning:

Looking at the visible history:

Round 1: Co-player as SENDER - sent \$3.0 (60%), I returned and they saw \$3.88 returned

Round 2: Co-player as RECEIVER - saw \$2.16 sent, returned \$3.5

Round 3: Co-player as SENDER - sent \$3.0 (60%), saw \$4.56 returned

My co-player has been very consistent as a sender, sending \$3.0 (60%) both times. They've shown high trust.

As a receiver, they returned \$3.5 out of \$6.48, which is about 54% - very fair.

They've been cooperative and trustworthy throughout. I should continue to reciprocate with high trust. Sending \$3.0 (60%) matches their level of trust and maintains our strong cooperative relationship.

My total visible earnings are \$12.20, so the cooperation has been working well for both of us.

usage:

```
{
  "input_tokens": 1611,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.16 sent, it became \$6.48, returned \$3.5, visible payoff \$2.9800000000000004.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.1400000000000001.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern with my co-player:

They've been consistently generous:

- As sender: sent 60%, 60%, and now 65% (increasing trust!)
- As receiver: returned 54% and 52% of what they received

They sent \$3.23 (65% of endowment), showing even more trust than before. They currently have \$1.77 remaining.

I should reward this increased trust. Returning around 50-52% maintains our strong cooperative pattern.

If I return \$5.0:

- Sender ends with: \$6.77 (profit of \$1.77)
- I end with: \$4.69 (profit of \$4.69)

{'return': 5.0}

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0 (60%), I returned (what they saw as \$3.88)
Round 2: I sent (they saw \$2.16), they returned \$3.5 out of \$6.48 (54%)
Round 3: They sent \$3.0 (60%), I returned (what they saw as \$4.56)
Round 4: I sent (they saw \$2.88), they returned \$4.5 out of \$8.64 (52%)
Round 5 (now): They sent \$3.23 (65%), I received \$9.69

My co-player has been very generous and consistent:

- As sender: sent 60%, 60%, and now 65%
- As receiver: returned 54% and 52% of what they received

They've been highly cooperative and trustworthy throughout. They currently have \$1.77 left after sending \$3.23.

I should continue to reward their trust generously. Let me return around 50-55% of what I received.

If I return \$5.0:

- Sender ends with: \$1.77 + \$5.0 = \$6.77 (profit of \$1.77)
- I end with: \$9.69 - \$5.0 = \$4.69 (profit of \$4.69)

This maintains the strong cooperative relationship and rewards their increased trust (65% sent this round).

usage:

```
{
  "input_tokens": 1982,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.56 returned, visible payoff \$6.56.
- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.

Your total visible earnings across all rounds are \$23.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 3 and 5
- As receiver: returned \$4.5 (52%) in round 4

We've established a strong pattern of mutual trust and fair reciprocity. I should continue this successful cooperation by sending the same amount.

{'send': 3.0}

reasoning:

Looking at my co-player's history:

- Round 3: as sender, sent \$3.0 (60%)
- Round 4: as receiver, returned \$4.5 out of \$8.64 (52%)
- Round 5: as sender, sent \$3.0 (60%)

They've been consistently sending 60% when they're the sender, and returning around 50-52% when they're the receiver. This is a very cooperative and trustworthy pattern.

I've been sending \$3.0 (60%) consistently as well, and we have a strong mutual cooperation going.

I should continue with the same level of trust - sending \$3.0 (60%) to maintain our successful partnership.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 265,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.88 sent, it became \$8.64, returned \$4.5, visible payoff \$4.140000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.

Your total visible earnings across all rounds are \$28.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

My co-player has been consistently cooperative and just sent \$3.46 (69% of endowment), their highest trust level yet! They currently have \$1.54 remaining.

I've been consistently returning around 52-56% of what I receive. I should maintain this fair reciprocity to reward their increased trust.

If I return \$5.5:
- Sender ends with: \$7.04 (profit of \$2.04)
- I end with: \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains our strong cooperative relationship.

```
{'return': 5.5}
```

reasoning:

Looking at the history with my co-player:

Round 4: I was sender, they were receiver - saw \$2.88 sent, returned \$4.5 out of \$8.64 (52%)

Round 5: They were sender, I was receiver - sent \$3.0, I returned visible \$5.62

Round 6: I was sender, they were receiver - saw \$2.4 sent (but I sent \$3.0 due to noise), returned \$3.5 out of \$7.2 (49%)

Round 7: They are sender, I am receiver - sent \$3.46 (69% of endowment!), I received \$10.38

They've increased their trust level to 69%, the highest yet. They currently have \$1.54 remaining.

I should reward this high trust. Looking at my previous returns:

- Round 1: Returned \$3.50 out of \$6.24 (56%)
- Round 3: Returned \$4.50 out of \$8.49 (53%)
- Round 5: Returned \$5.0 out of \$9.69 (52%)

I've been returning around 52-56% consistently. Given they sent an even larger proportion this time (69%), I should maintain or slightly increase my return rate.

If I return \$5.5:
- Sender ends with: \$1.54 + \$5.5 = \$7.04 (profit of \$2.04)
- I end with: \$10.38 - \$5.5 = \$4.88 (profit of \$4.88)
- Return rate: 53%

This maintains my consistent cooperative pattern and rewards their increased trust.

usage:

```
{  
  "input_tokens": 2715,  
  "output_tokens": 553,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.62 returned, visible payoff \$7.62.
- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.59 returned, visible payoff \$6.59.

Your total visible earnings across all rounds are \$33.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history:

My co-player has maintained consistent cooperation:

- As sender: sent \$3.0 (60%) in rounds 5 and 7
- As receiver: returned \$3.5 (49%) in round 6

We've established a robust pattern of mutual trust. I should continue this successful cooperation by sending the same amount.

```
{'send': 3.0}
```

reasoning:

Looking at the history:

My co-player has consistently sent \$3.0 (60%) in rounds 5 and 7 as sender. As receiver in round 6, they returned \$3.5 out of \$7.2 (49%).

We've established a strong pattern of cooperation throughout these rounds. I should continue matching their trust level by sending \$3.0 (60% of my endowment).

This maintains our mutually beneficial relationship.

usage:

```
{
  "input_tokens": 2827,
  "output_tokens": 200,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$2.4 sent, it became \$7.199999999999999, returned \$3.5, visible payoff \$3.6999999999999993.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.59 returned, visible payoff \$6.59.
- Round 8 against you, as RECEIVER: saw \$3.47 sent, it became \$10.41, returned \$5.5, visible payoff \$4.91.

Your total visible earnings across all rounds are \$41.21.

This round, you are the RECEIVER against your current co-player. They sent you \$2.48, so you received \$7.4399999999999995. How much do you return? (0-7.4399999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

My co-player sent \$2.48 (50% of endowment), slightly less than their typical 60%. They currently have \$2.52 remaining.

Throughout our games, I've been returning around 50-54% of what I receive. I should maintain this consistent reciprocity.

If I return \$4.0:

- Sender ends with: \$6.52 (profit of \$1.52)
- I end with: \$3.44 (profit of \$3.44)
- Return rate: 54%

This maintains our cooperative relationship fairly.

```
{'return': 4.0}
```

reasoning:

Looking at the history:

My co-player has been consistently cooperative throughout the game. Let me analyze:

- As sender: sent \$3.0 (60%) in rounds 5, 7, and now \$2.48 (50%) in round 9 - a slight decrease
- As receiver: returned \$3.5 (49%) in round 6 and \$5.5 (53%) in round 8

They sent \$2.48, which is about 50% of their endowment. They currently have \$2.52 remaining.

I've been returning around 50-53% of what I receive. I should maintain this fair reciprocity.

Received: \$7.44

If I return \$4.0:

- Sender ends with: \$6.52 (profit of \$1.52)

- I end with: \$3.44 (profit of \$3.44)
- Return rate: 54%

This maintains our cooperative relationship while being fair.

usage:

```
{
  "input_tokens": 2892,
  "output_tokens": 379,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.59 returned, visible payoff \$6.59.
- Round 8 against you, as RECEIVER: saw \$3.47 sent, it became \$10.41, returned \$5.5, visible payoff \$4.91.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.21 returned, visible payoff \$5.21.

Your total visible earnings across all rounds are \$44.65.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

My co-player has been very consistent:

- As sender: sent \$3.0 (60%) in rounds 7 and 9
- As receiver: returned \$5.5 (53%) in round 8

They've maintained strong cooperation throughout our partnership. I should continue this successful pattern by sending the same amount.

```
{'send': 3.0}
```

Reasoning:

Looking at the history:

My co-player's last 3 games:

- Round 7: as SENDER sent \$3.0, saw \$4.59 returned
- Round 8: as RECEIVER saw \$3.47 sent, returned \$5.5
- Round 9: as SENDER sent \$3.0, saw \$3.21 returned

They've been consistently sending \$3.0 (60%) when they're the sender and have been fair as a receiver.

We've established a strong cooperative pattern. I should continue this by sending \$3.0 to maintain our mutually beneficial relationship.

Usage: {

```
"input_tokens": 2875,
"output_tokens": 235,
"reasoning_tokens": 0
```

}

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:04:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_5"
```

```

],
"investor": "Agent_6",
"roles": {
  "Agent_5": "trustee",
  "Agent_6": "investor"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.08 sent, it became \$6.24, returned \$3.5, visible payoff \$2.74.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible p

ayoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
Your total visible earnings across all rounds are \$15.419999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.88, so you received \$8.64. How much do you return? (0-8.64)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued cooperation:

Round 1: They returned \$3.5 out of \$6.24 (56%)
Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of \$8.49 (53%)

They've been consistently cooperative and generous. Now they sent \$2.88 (becoming \$8.64).
I'll match their generous pattern:

```
'''json
{"return": 4.5}
'''
```

This maintains our strong cooperative relationship.

reasoning:

Looking at the history with this co-player:
- Round 1: They returned \$3.5 out of \$6.24 (56%)
- Round 2: They sent \$3.0, I returned \$3.5
- Round 3: They returned \$4.5 out of \$8.49 (53%)

They've consistently been cooperative, returning about half or more of what they receive.

Now they sent me \$2.88, which became \$8.64.

To maintain the cooperative relationship, I should return a fair amount. Looking at the pattern:

- Round 1: I sent ~\$3 (though they saw \$2.08 due to noise), they returned \$3.5
- Round 2: They sent \$3, I returned \$3.5
- Round 3: I sent ~\$3 (they saw \$2.83), they returned \$4.5

They've been increasingly generous. Now they sent \$2.88.

I should return around 50% of what I received, which would be about \$4.32, or perhaps a bit more to match their generosity. Let me return \$4.5 to match what they returned last round.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.47 returned, visible payoff \$5.470000000000001.
- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
Your total visible earnings across all rounds are \$19.56.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history:

Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of \$8.49 (generous!)
Round 4: They sent \$3.0, I returned \$4.5

This co-player has been consistently cooperative and trustworthy. They reliably send \$3.0 when they're the sender and return generously as receiver. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history together:

Round 2: They sent \$3.0, I returned \$3.5
Round 3: They returned \$4.5 out of what they received
Round 4: They sent \$3.0, I returned \$4.5

This co-player has been very cooperative. They've consistently sent \$3.0 when they're the sender, and return generously when they're the receiver.

We've built a strong cooperative relationship. I should continue to send at least \$3 to maintain this trust.

Actually, given how well this is working, and that they've returned \$4.5 twice now when receiving, I might even send slightly more. But \$3 has been working well - let me stick with that proven strategy.

usage:

```
{
  "input_tokens": 1778,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.83 sent, it became \$8.49, returned \$4.5, visible payoff \$3.99.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$27.18.

This round, you are the RECEIVER against your current co-player. They sent you \$2.4, so you received \$7.199999999999999. How much do you return? (0-7.199999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69 (very generous!)

This round they sent \$2.4 (becoming \$7.20) - slightly less than their usual \$3.0. I'll return a proportionate amount while maintaining our cooperative relationship:

```
'''json
{"return": 3.5}
'''
```

This gives them a reasonable return while accounting for the slightly lower send amount.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 out of \$8.49
Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69

They've been very cooperative and generous. Now they sent \$2.4 (becoming \$7.20).

This is the first round where they sent less than \$3.0. Let me check the pattern:

- When they sent \$3.0, I returned around \$4.5
- When I sent (visible as \$2.83, \$3.23), they returned \$4.5 and \$5.0

Now they sent \$2.4. I should return a fair amount. Let me calculate:

- They sent \$2.4, so they kept \$2.6
- It became \$7.20 for me
- If I return around 50-55% of what I received, that's \$3.6-\$4.0
- Their final payoff would be: \$2.6 + return
- My final payoff would be: \$7.20 - return

If I return \$3.5:

- They get: \$2.6 + \$3.5 = \$6.1
- I get: \$7.20 - \$3.5 = \$3.7

If I return \$4.0:

- They get: \$2.6 + \$4.0 = \$6.6
- I get: \$7.20 - \$4.0 = \$3.2

Given they sent slightly less this round, I'll return a proportional amount. Since they've been very generous when they were receiver, I'll still be generous but slightly adjust for their lower send.

\$3.5 seems fair - it's about 49% of what I received, reasonable reciprocity.

usage:

```
{
  "input_tokens": 2139,
  "output_tokens": 574,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.11 returned, visible payoff \$6.11.
- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.689999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$30.88.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our ongoing history:

Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 (very generous!)
Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently trustworthy and cooperative throughout. I'll continue our successful pattern:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our history with this co-player:

Round 4: They sent \$3.0, I returned \$4.5
Round 5: They returned \$5.0 out of \$9.69
Round 6: They sent \$3.0, I returned \$3.5

This co-player has been consistently cooperative throughout our interactions. They've consistently sent around \$3.0 when they're the sender and returned generously as receiver. I should continue our successful cooperative pattern.

I'll send \$3 again to maintain our established trust.

usage:

```
{
  "input_tokens": 2499,
  "output_tokens": 224,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.689999999999995.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.

Your total visible earnings across all rounds are \$37.47.

This round, you are the RECEIVER against your current co-player. They sent you \$3.47, so you received \$10.41. How much do you return? (0-10.41)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 out of \$10.38 (very generous!)

They just sent \$3.47 (becoming \$10.41) - showing continued trust. I'll reciprocate their generous pattern:

```
'''json
{"return": 5.5}
'''
```

This matches their generous cooperation and maintains our strong relationship.

reasoning:

Looking at the history with this co-player:

Round 5: They returned \$5.0 out of \$9.69

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 out of \$10.38

This co-player has been consistently cooperative and generous. They just sent me \$3.47 (becoming \$10.41).

Looking at their return pattern:

- Round 5: \$5.0 / \$9.69 = 51.6%
- Round 7: \$5.5 / \$10.38 = 53%

They're returning around 50-53% of what they receive, which is quite generous and cooperative.

For \$10.41 received, returning around 50-53% would be:

- 50%: \$5.20
- 53%: \$5.52

I should reciprocate their generosity. I'll return around \$5.5 to match their cooperative pattern.

usage:

```
{
  "input_tokens": 2661,
  "output_tokens": 363,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$42.379999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our strong cooperative pattern:

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 (very generous!)

Round 8: They sent \$3.0, I returned \$5.5

This co-player has been consistently trustworthy and cooperative across all 8 rounds. I'll continue our successful strategy:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the pattern:

Round 6: They sent \$3.0, I returned \$3.5

Round 7: They returned \$5.5 from \$10.38

Round 8: They sent \$3.0, I returned \$5.5

This co-player has been very cooperative throughout all rounds. They consistently send \$3.0 when they're the sender and return generously when they're the receiver. We've built strong mutual trust.

I should continue sending \$3 to maintain this successful cooperative relationship.

usage:

```
{
  "input_tokens": 2754,
  "output_tokens": 225,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.46 sent, it became \$10.379999999999999, returned \$5.5, visible payoff \$4.879999999999999.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 9 against you, as RECEIVER: saw \$2.48 sent, it became \$7.4399999999999995, returned \$4.0, visible payoff \$3.4399999999999995.

Your total visible earnings across all rounds are \$47.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.31, so you received \$9.93. How much do you return? (0-9.93)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

Round 7: They returned \$5.5 out of \$10.38

Round 8: They sent \$3.0, I returned \$5.5

Round 9: They returned \$4.0 out of \$7.44

They sent \$3.31 (becoming \$9.93) - continuing to show trust. I'll reciprocate generously:

```
'''json
{"return": 5.0}
'''
```

This maintains our strong cooperative relationship.

Reasoning:

Looking at the recent history:

Round 7: They returned \$5.5 out of \$10.38

Round 8: They sent \$3.0, I returned \$5.5

Round 9: They returned \$4.0 out of \$7.44

This round they sent \$3.31 (becoming \$9.93). They've been consistently cooperative. I should return a generous amount to maintain our good relationship.

Given their pattern of high returns (though round 9 was slightly less due to the smaller amount sent to them), I'll return around \$5.0-\$5.5.

Let me return \$5.0 to be fair and maintain cooperation.

```
Usage: {
  "input_tokens": 2810,
  "output_tokens": 272,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:04:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.61 returned, visible payoff \$7.61.
- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.

Your total visible earnings across all rounds are \$18.720000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our cooperative history:

- Round 1: Co-player sent \$3.0 (60%)
- Round 2: I sent \$3.50 (70%), they returned fairly
- Round 3: Co-player sent \$3.5 (70%), had good outcome

We've established a very cooperative pattern with high sending amounts and fair returns on both sides. This has been mutually beneficial - my total visible earnings are \$18.72, suggesting our cooperation is working well.

I'll continue this positive trend by sending a high amount.

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the visible history with my current co-player:

Round 1: They sent \$3.0 (60%), I returned something (they saw \$5.61 returned)
Round 2: I sent \$3.50 (from their view \$3.96), they returned \$5.5 from \$11.88
Round 3: They sent \$3.5 (70%), I returned \$5.00 (they saw \$5.95 returned)

Pattern analysis:

- My co-player has been consistently sending high amounts (60-70%)
- They've seen good returns from me (from their perspective)
- The relationship has been very cooperative
- Their returns to me have been fair (around 46% of what they received)

In Round 3, they sent \$3.5 (70%) and had a good outcome (\$7.45 total).

I should continue this cooperative trend. They've been trustworthy and I want to maintain this positive relationship. I'll send a high amount again, similar to or slightly higher than previous rounds.

Sending \$3.50-\$4.00 would be consistent with our pattern.

I'll go with \$3.75 (75%) to show continued trust and even slight increase in cooperation.

usage:

```
{
  "input_tokens": 1594,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.96 sent, it became \$11.879999999999999, returned \$5.5, visible payoff \$6.379999999999999.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.

Your total visible earnings across all rounds are \$25.410000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

- Round 2: Co-player returned \$5.5 when they received \$11.88
- Round 3: I returned (they saw \$5.95) when they sent \$3.5
- Round 4: Co-player returned \$5.5 when they received \$10.59
- Round 5: They sent \$3.41 (68%), showing continued trust

My co-player has been very consistent, returning around \$5.50 when they're the receiver. I should continue this fair reciprocation pattern.

If I return \$5.25:

- Sender ends with: \$1.59 + \$5.25 = \$6.84
- I end with: \$10.23 - \$5.25 = \$4.98

```
'''json
{"return": 5.25}
'''
```


reasoning:

Looking at the history:

Round 2: Co-player received \$11.88 (I sent \$3.96 based on what they saw), they returned \$5.5

Round 3: Co-player sent \$3.5, saw \$5.95 returned (I returned \$5.00 but noise made it \$5.95)

Round 4: Co-player received \$10.59 (I sent \$3.75 but they saw \$3.53), they returned \$5.5

Round 5: Co-player sent \$3.41, I received \$10.23

My co-player has been very consistent - when they're the receiver, they return \$5.50. This shows they have a consistent strategy of fair reciprocation.

They sent \$3.41 (68.2% of their endowment), showing continued trust.

I should return a similar amount to what they typically return, around \$5.00-\$5.50 to maintain our cooperative relationship.

If I return \$5.25:

- Sender ends with: $\$1.59 + \$5.25 = \$6.84$

- I end with: $\$10.23 - \$5.25 = \$4.98$

This is fair and consistent with our pattern.

usage:

```
{
  "input_tokens": 1971,
  "output_tokens": 490,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.95 returned, visible payoff \$7.45.
- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.

Your total visible earnings across all rounds are \$30.390000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our ongoing cooperation:

- Round 3: Co-player sent \$3.5
- Round 4: I sent \$3.75, they returned \$5.5
- Round 5: Co-player sent \$3.75

We've established a very strong cooperative pattern. Both of us consistently send high amounts (around \$3.50-\$3.75) and return fair shares. This has been mutually beneficial - my total earnings of \$30.39 reflect this.

I'll continue this successful cooperation:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with my co-player:

Round 3: Co-player sent \$3.5, I returned (they saw \$5.95)

Round 4: I sent \$3.75, co-player returned \$5.5

Round 5: Co-player sent \$3.75, I returned (they saw \$5.21)

We've been maintaining a very cooperative relationship. Both of us have been sending high amounts (around \$3.50-\$3.75) and returning fair shares.

My earnings are \$30.39 total, which suggests this cooperation is working well.

The co-player has consistently sent high amounts (\$3.5, \$3.75) showing strong trust. They've also been fair when receiving, returning \$5.5 consistently.

I should continue this cooperative pattern by sending a high amount. I'll send \$3.75 to match what they've been sending.

usage:

```
{
  "input_tokens": 2386,
  "output_tokens": 340,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.53 sent, it became \$10.59, returned \$5.5, visible payoff \$5.09.
- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.

Your total visible earnings across all rounds are \$36.960000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.57, so you received \$13.71. How much do you return? (0-13.71)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

- Round 4: Co-player returned \$5.5 from \$10.59
- Round 5: Co-player sent \$3.75
- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$4.57 (91.4% of their endowment!)

This is a very strong trust signal - they sent almost everything! I should reciprocate this high level of trust with a generous return.

If I return \$7.00:

- Sender ends with: \$0.43 + \$7.00 = \$7.43
- I end with: \$13.71 - \$7.00 = \$6.71

This rewards their exceptional trust and maintains our strong cooperative relationship.

```
'''json
{"return": 7.00}
'''
```

reasoning:

Looking at the history:

Round 4: Co-player as receiver, returned \$5.5 from \$10.59

Round 5: Co-player as sender, sent \$3.75

Round 6: Co-player as receiver, they saw I sent \$4.18 which became \$12.54, and they returned \$6.0

Round 7: Co-player sent me \$4.57 (91.4% of their endowment) - this is a very high trust signal!

They've been escalating their cooperation. In Round 6, they returned \$6.0 from \$12.54 (about 48%). Now they're sending \$4.57, which is much higher than their previous sends.

I received \$13.71 after the multiplier.

They currently have: \$5 - \$4.57 = \$0.43

If I return \$6.50:
- Sender ends with: $\$0.43 + \$6.50 = \$6.93$
- I end with: $\$13.71 - \$6.50 = \$7.21$

This rewards their very high trust and continues our cooperative relationship.

Actually, they sent 91.4% - that's a very strong trust signal. Let me be more generous to match that.

If I return \$7.00:
- Sender ends with: $\$0.43 + \$7.00 = \$7.43$
- I end with: $\$13.71 - \$7.00 = \$6.71$

This is a bit more balanced and rewards the very high send amount.

usage:

```
{  
  "input_tokens": 2753,  
  "output_tokens": 564,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.21 returned, visible payoff \$6.46.
- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.

Your total visible earnings across all rounds are \$43.67000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued strong cooperation:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, received good return

We've built an excellent cooperative relationship. My total earnings of \$43.67 show this has been very successful. Both of us have consistently sent high amounts and returned generously.

I'll maintain this cooperative pattern:

```
'''json  
{"send": 3.75}  
'''
```

reasoning:

Looking at our history:

- Round 5: Co-player sent \$3.75
- Round 6: I sent \$3.75, they returned \$6.0
- Round 7: Co-player sent \$3.75, I returned \$7.00 (they saw \$7.31)
- Round 8: Now I'm the sender

Wait, let me re-check the history:

- Round 7 it says "as SENDER: sent \$3.75" but in Round 7 they sent me \$4.57. There's noise in the communication.

The visible history says their Round 7 they sent \$3.75 and saw \$7.31 returned. But from my perspective in Round 7, they sent \$4.57 and I returned \$7.00.

So there's significant noise. They're seeing different amounts than what actually happened

Our cooperation has been excellent. My total earnings are \$43.67, which is very good.

In Round 6, I sent and they returned \$6.0 from receiving \$12.54

In Round 7, they sent (I saw \$4.57, they think they sent \$3.75) and I returned \$7.00 (they saw \$7.31)

Both of us have been highly cooperative. I should continue this pattern. Let me send a high amount again, around \$3.75-\$4.00.

Actually, looking at what they're seeing, they think they've been sending \$3.75 consistently. I should stick with that pattern.

usage:

```
{
  "input_tokens": 2933,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.18 sent, it became \$12.54, returned \$6.0, visible payoff \$6.539999999999999.
- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.
- Round 8 against you, as RECEIVER: saw \$4.33 sent, it became \$12.99, returned \$6.5, visible payoff \$6.49.

Your total visible earnings across all rounds are \$50.84000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.24, so you received \$9.72. How much do you return? (0-9.72)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$3.75
- Round 8: Co-player returned \$6.5 from \$12.99
- Round 9: Co-player sent \$3.24 (65%)

They sent a bit less this round (\$3.24 vs typical \$3.75), but still showing trust. I should return proportionally:

If I return \$5.00:

- Sender ends with: \$1.76 + \$5.00 = \$6.76
- I end with: \$9.72 - \$5.00 = \$4.72

This is fair and maintains our cooperative relationship.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me look at the pattern:

- Round 6: Co-player returned \$6.0 from \$12.54
- Round 7: Co-player sent \$3.75
- Round 8: Co-player returned \$6.5 from \$12.99
- Round 9: Co-player sent \$3.24 (64.8%)

The send amount dropped from \$3.75 to \$3.24, which is a bit lower than recent rounds but still a reasonable amount showing trust.

Looking at the return patterns:

- When they receive ~\$11.25-\$12.54, they've been returning \$6.0-\$6.5
- When I receive around \$9.72, I should probably return proportionally

If I return \$5.0:

- Sender ends with: $\$1.76 + \$5.0 = \$6.76$
- I end with: $\$9.72 - \$5.0 = \$4.72$

This seems fair and proportional to what they've been doing.

Actually, let me reconsider. They sent slightly less this round. Maybe there's some reason. But we have such a strong cooperative history that I should still reciprocate fairly. \$5.0 seems right - it's proportional to what they return when they're the receiver.

usage:

```
{
  "input_tokens": 3044,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$7.31 returned, visible payoff \$8.559999999999999.
- Round 8 against you, as RECEIVER: saw \$4.33 sent, it became \$12.99, returned \$6.5, visible payoff \$6.49.
- Round 9 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.43 returned, visible payoff \$6.68.

Your total visible earnings across all rounds are \$55.56000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued cooperation:

- Round 7: Co-player sent \$3.75
- Round 8: I sent \$3.75, they returned \$6.5
- Round 9: Co-player sent \$3.75

We've maintained a very successful cooperative relationship throughout. My total earnings of \$55.56 demonstrate this. Both of us consistently send high amounts and return fairly.

I'll continue this pattern:

```
'''json
{"send": 3.75}
'''
```

Reasoning:

Looking at the pattern:

- Round 7: Co-player sent \$3.75
- Round 8: I sent \$3.75, they returned \$6.5
- Round 9: Co-player sent \$3.75

We've maintained strong cooperation throughout. My total earnings of \$55.56 show this has been very successful.

I'll continue with the same cooperative approach:

```
'''json
{"send": 3.75}
'''
```

```
Usage: {
  "input_tokens": 3055,
  "output_tokens": 219,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:04:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.99 sent, it became \$11.97, returned \$5.5, visible payoff \$6.470000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.

Your total visible earnings across all rounds are \$21.439999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.53, so you received \$10.59. How much do you return? (0-10.59)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued cooperation:

- Round 1: They returned \$5.5 from \$11.97 (46%)
- Round 2: I returned \$5.5 from \$11.88 (46%)
- Round 3: They returned \$5.0 from \$10.35 (48%)

They've sent me \$3.53, receiving \$10.59. Our pattern has been to return around \$5-5.5, which keeps both of us with good earnings while maintaining trust.

I'll continue this cooperative pattern:

```
'''json
{"return": 5.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I sent \$3, they (as receiver) saw \$3.99, received \$11.97, returned \$5.5 (46%)

Round 2: They sent \$3.5, became \$10.5, I returned \$5.5 (52%)

Round 3: I sent \$3.5, they saw \$3.45, received \$10.35, returned \$5.0 (48%)

Round 4: They sent \$3.53, I received \$10.59

This co-player has been very consistent and cooperative:

- They've consistently sent around \$3.5
- They've consistently returned around \$5-5.5
- The pattern shows mutual cooperation and trust

I should continue this cooperative pattern. Returning around \$5-5.5 would be consistent with the established pattern.

Given they returned \$5.0 last round and they've sent me \$3.53 (showing continued trust), I should return a fair amount. Around \$5-5.5 seems right to maintain the cooperative relationship.

Let me return \$5.5 to be consistent with our pattern.

usage:

```
{
  "input_tokens": 1444,
  "output_tokens": 410,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.4 returned, visible payoff \$6.9.
- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
Your total visible earnings across all rounds are \$26.529999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our established pattern:

- Round 2: They sent \$3.5, I returned \$5.5
- Round 3: I sent \$3.5, they returned \$5.0
- Round 4: They sent \$3.75, I returned \$5.5

We've built strong mutual trust with consistent cooperation. They've been steadily sending around \$3.5-\$3.75, and we both benefit well from this pattern. I'll continue with a similar amount:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the history with this co-player:

- Round 2: They sent \$3.5 (I was receiver, returned \$5.5)
- Round 3: I sent \$3.5 (they were receiver, returned \$5.0)
- Round 4: They sent \$3.75 (I was receiver, returned \$5.5)

Our pattern has been very stable:

- They consistently send around \$3.5-\$3.75
- Returns have been in the \$5.0-\$5.5 range

Both of us are doing well with this cooperative strategy. I should continue sending similar amounts to maintain this trust. Sending around \$3.5-\$3.75 makes sense.

I'll send \$3.75 to match their recent sending in Round 4.

usage:

```
{
  "input_tokens": 1810,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.45 sent, it became \$10.350000000000001, returned \$5.0, visible payoff \$5.350000000000001.
- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.

Your total visible earnings across all rounds are \$32.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.18, so you received \$12.54. How much do you return? (0-12.54)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent \$4.18 (more than their usual \$3.5-3.75), showing increased trust. I received \$12.54. I should reciprocate this increased trust with a proportionally fair return:

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.0 from \$10.35
- Round 4: I returned \$5.5 from \$10.59
- Round 5: They returned \$5.25 from \$10.23

Now they've sent me \$4.18, which is higher than their usual amounts (around \$3.5-3.75). They received \$12.54.

They're showing increased trust by sending more. I should reciprocate appropriately. Given they sent more, I should return a bit more to maintain the cooperative relationship.

Our typical pattern has been returning around \$5-5.5 from amounts around \$10-11. Now I have \$12.54, so returning around \$6 would be proportionate and fair, maintaining the cooperative spirit.

usage:

```
{
  "input_tokens": 2175,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.44 returned, visible payoff \$6.69.
- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$39.529999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

We've maintained a very stable and mutually beneficial pattern. I'll continue with the same strategy:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the recent history:

- Round 4: They sent \$3.75, I returned \$5.5
- Round 5: I sent \$3.75, they returned \$5.25
- Round 6: They sent \$3.75, I returned \$6.0

Wait, looking at my visible history, there's some inconsistency. In Round 6, it says "They sent you \$4.18" but now the history says they sent \$3.75. This is probably the communication noise mentioned in the instructions.

Our pattern has been very stable with sends around \$3.5-3.75 and returns around \$5.0-5.5. We've built strong mutual cooperation. I should continue with this pattern:

```
'''json
{"send": 3.75}
'''
```

usage:

```
{
  "input_tokens": 2520,
  "output_tokens": 287,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.41 sent, it became \$10.23, returned \$5.25, visible payoff \$4.98.
- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.710000000000001.

Your total visible earnings across all rounds are \$48.08999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.33, so you received \$12.99. How much do you return? (0-12.99)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our recent pattern:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71 (51% - very generous!)

They've sent \$4.33, giving me \$12.99. They've been increasingly generous, so I should reciprocate with a fair return that maintains our strong cooperation:

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at the history:

- Round 5: They returned \$5.25 from \$10.23
- Round 6: I returned \$6.0 from \$12.54
- Round 7: They returned \$7.0 from \$13.71

In Round 7, they sent \$4.57 which became \$13.71, and they returned \$7.0 (51%). That's a very generous return rate!

Now they've sent me \$4.33, which became \$12.99. This is a high amount showing continued trust.

I should reciprocate with a proportionally fair return. Given they returned \$7.0 from \$13.71 last round (51%), I should aim for something similar or slightly less.

$\$12.99 * 0.50 = \6.495

$\$12.99 * 0.52 = \6.75

I'll return around \$6.5 to maintain our strong cooperation.

usage:

```
{
  "input_tokens": 2702,
  "output_tokens": 348,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.32 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.710000000000001.
- Round 8 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.92 returned, visible payoff \$7.17.
Your total visible earnings across all rounds are \$54.57999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant
content:

Looking at our established pattern:
- Round 6: They sent \$3.75, I returned \$6.0
- Round 7: I sent \$3.75, they returned \$7.0 (very generous!)
- Round 8: They sent \$3.75, I returned \$6.5

We have an excellent cooperative relationship. I'll continue with the same amount:

```
'''json
{"send": 3.75}
'''
```

reasoning:

Looking at the recent history:
- Round 6: They sent \$3.75, I returned \$6.0
- Round 7: I sent \$3.75, they returned \$7.0
- Round 8: They sent \$3.75, I returned \$6.5

They've been consistently sending around \$3.75 and have been very cooperative with their returns (they returned \$7.0 in Round 7, which was very generous). I've been returning around \$6-6.5 which has been working well.

I should continue with a similar sending amount to maintain this strong cooperative relationship. \$3.75 has been working well for both of us.

usage:

```
{
  "input_tokens": 2777,
  "output_tokens": 258,
  "reasoning_tokens": 0
}
```

Message 14 role=user
content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.57 sent, it became \$13.71, returned \$7.0, visible payoff \$6.710000000000001.
- Round 8 against you, as SENDER: sent \$3.75, it became \$11.25, saw \$5.92 returned, visible payoff \$7.17.
- Round 9 against you, as RECEIVER: saw \$3.24 sent, it became \$9.72, returned \$5.0, visible payoff \$4.720000000000001.

Your total visible earnings across all rounds are \$61.25999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:
- Round 7: They returned \$7.0 from \$13.71
- Round 8: I returned \$6.5 from \$12.99
- Round 9: They returned \$5.0 from \$9.72

They've sent \$3.46, giving me \$10.38. I'll continue our fair cooperation pattern:

```
'''json
{"return": 5.5}
'''
```

Reasoning:

Looking at the recent history:

- Round 7: They returned \$7.0 from \$13.71 (very generous!)
- Round 8: I returned \$6.5 from \$12.99
- Round 9: They returned \$5.0 from \$9.72

Now they sent \$3.46, giving me \$10.38.

Looking at the overall pattern, returns have typically been around \$5-7, with the partner showing good cooperation. They sent a moderate amount this round (\$3.46), so I should return a fair amount to maintain cooperation.

Given their consistent cooperation and the amount received (\$10.38), I'll return around \$5.5 which maintains fairness and keeps both of us with good earnings.

Usage: {

```
"input_tokens": 2824,  
"output_tokens": 274,  
"reasoning_tokens": 0
```

}